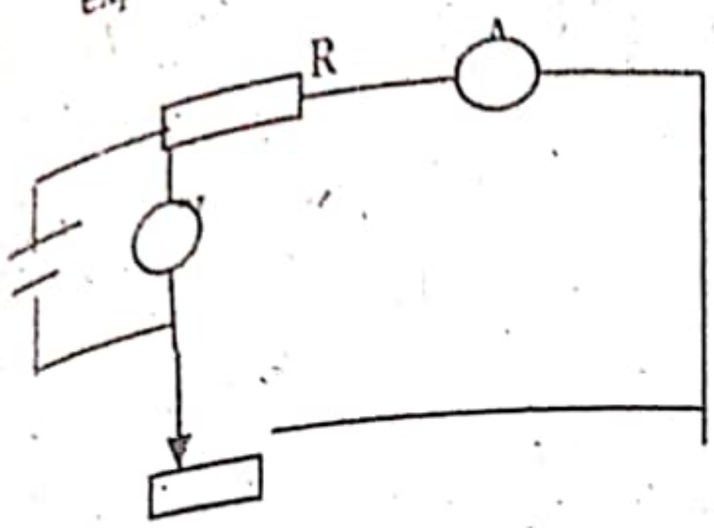
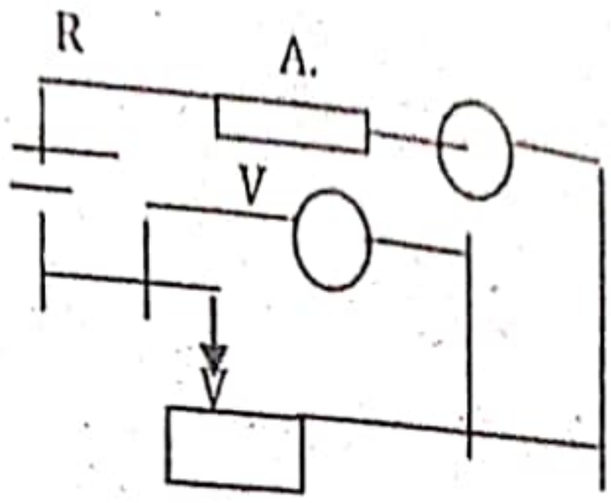
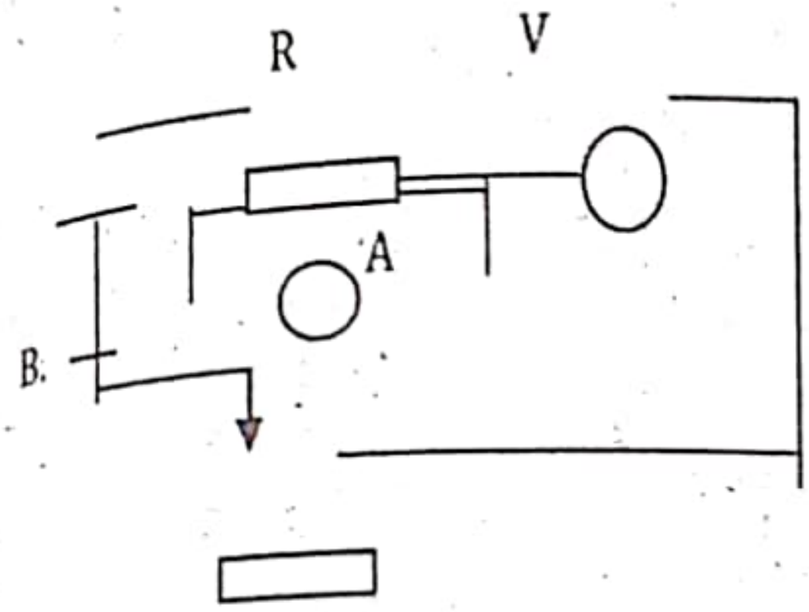
connections is the correct set up of the circuit to perform the experiment?



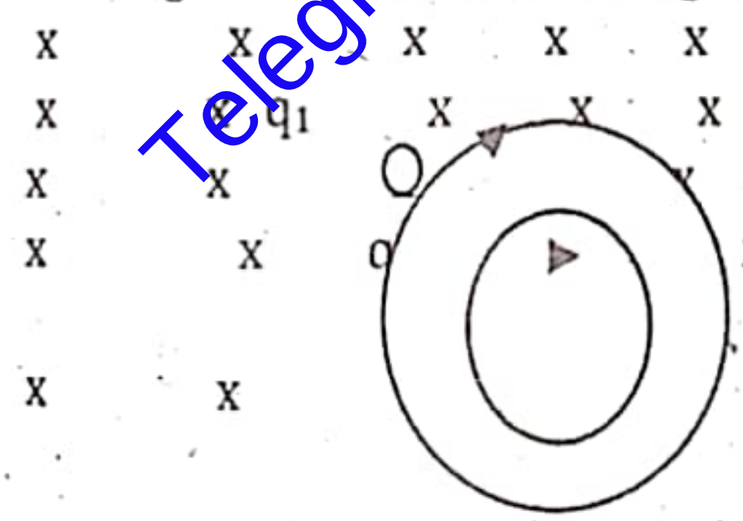
C.





D. DRAW THE DIAGRAM

38. Circular paths followed by two charges q_1 and q_2 moving with the same speed in uniform magnetic field directed into the page (x) are shown in the figure below. If the two charges have the same mass, which of the following is correct about the sign and the magnitude of charges?



q_2 is in the inner circle. draw the charge

A. q_1 is positive, q_2 is negative and $|q_1| < |q_2|$.

B. q_1 is positive, q_2 is negative and $|q_1| > |q_2|$.

C. q_1 is positive, q_2 is negative and $|q_1| < |q_2|$.

D. q_1 is negative, q_2 is positive and $|q_1| > |q_2|$.

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39. The primary coil of a transformer has 250 turns and its secondary coil consists of 1500 turns. If 220 is applied to its primary coil, the voltage induced in the secondary coil and the ratio of current in the primary coil to the current in the secondary coil, respectively, are.....
A. 1220 V and 60. B. 1320 V and 60. C. 1320 V and 6. D. 1330 V and 6.

40. A diode is an electronic component that
A. allows the flow of current when its p - type region is connected to the negative terminal of a cell.
B. allows current to pass through it either from cathode to anode or from anode to cathode.
C. is useful to convert a DC voltage to AC voltage.
D. allows electric current to pass through it in one direction.

41. Three forces, each having equal magnitudes of 10 N, are applied to an object on a horizontal surface. A force F_1 makes an angle of 30° above positive x - axis, F_2 makes 30° above the negative x - axis, and the third force F_3 is directed along the positive y - axis. The magnitude and direction of their resultant force are, respectively,
A. 20 N in the negative y - axis.
B. 14.1 N 45° above positive x - axis.
C. 20 N 45° above positive x - axis.
D. 14.1 N in the positive y - axis.

42. A student throws a ball vertically upward with a speed of 20 m/s. The maximum height reached by the ball and its velocity 3.0 s after it was thrown are, respectively
A. 105 m and 50 m/s upward.
B. 20 m and 10 m/s downward.
C. 20 m and 30 m/s upward.
D. 15 m and 10 m/s downward.

43. Which one of the following statements describes the difference between different types of friction?
A. If there is a constant force applied to a stationary object on a horizontal surface, the static friction becomes equal to the limiting friction.
B. Limiting and kinetic frictions exist at the same time between two surfaces sliding over each other.
C. For an object at rest on a horizontal surface, static friction increases with the increase of the applied force and reaches a maximum value just before sliding occurs.
D. Limiting friction is the minimum value of static friction that keeps an object stationary.

44. The apparent weight of a 60 kg student standing inside a lift is
A. 480 N when the lift accelerates down ward with 2 m/s^2.
B. 720 N when the lift is at rest.

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C. 600 N when the lift accelerates with acceleration equal to gravitational acceleration.

D. 600 N when the lift accelerates upward with 2 m/s².

45. Which of the following statements describe the use of energy resources?

A. Geothermal energy is the most accessible, environment friendly and inexpensive source of energy.

B. Wind energy, which is due to the pressure difference in space as the result of uneven sun heat,is a consistent and economical source of energy.

C. Solar energy produced by photovoltaic cells is free of greenhouse gases and has a very low running cost.

D. Hydroelectric power generates a large amount of energy using small dam with little environmental impact and cheap initial cost of construction.

46. A student wants to raise an object of mass m through a height 'h' using an inclined plane that makes an angle θ above the horizontal. If 'l' is the length of the inclined surface, which one of the following is correct about the actual mechanical advantage (AMA) and ideal mechanical advantage (IMA) of the inclined plane?

A. In the presence of friction, its AMA is $\frac{1}{\sin\theta}$.

B. In the absence of friction, AMA is $\sin\theta$ and its IMA is $\cos\theta$.

C. Neglecting friction, its AMA is $\frac{1}{\sin\theta}$.

D. Neglecting friction, its IMA is $\frac{1}{\sin\theta}$.

47. According to the law of floatation, an object must displace a volume of fluid

A. less than its weight.

C. equal to the weight of un displaced fluid.

B. equal to its own weight.

D. equal to its apparent weight.

48. Which of the following statements describes the unusual behavior of expansion of water?

A. At 0°C, water has larger density and smaller volume than its density and volume at 4°C.

B. As a temperature of water increases from 0°C to 4°C, its density increases since one molecule of water forms one hydrogen bond with another molecule.

C. The density of water at 0°C is less than its density at 4°C, since water molecules move further apart and form a second hydrogen bond with other water molecule.

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D. As a temperature of water increases above 4°C, its density increases since water molecules move further apart to from a second hydrogen bond with other molecules.

49. A 40 g of water at 50°C is poured into a well insulated 30 g of copper calorimeter at 20°C. The temperature water at equilibrium becomes.....

A. 35°C B. 30°C C. 52.4°C D. 47.9°C.

50. An FM radio signal is transmitted from a radio station at a frequency of 1.5×10^8 Hz. If the wavelength of the radio wave is 2.00 m, the speed of the FM signal is

A. $3 \times 10^8 m/s.$

B. $0.75 \times 10^8 m/s.$

C. $6 \times 10^8 m/s.$

D. $1.33 \times 10^{-8} m/s.$

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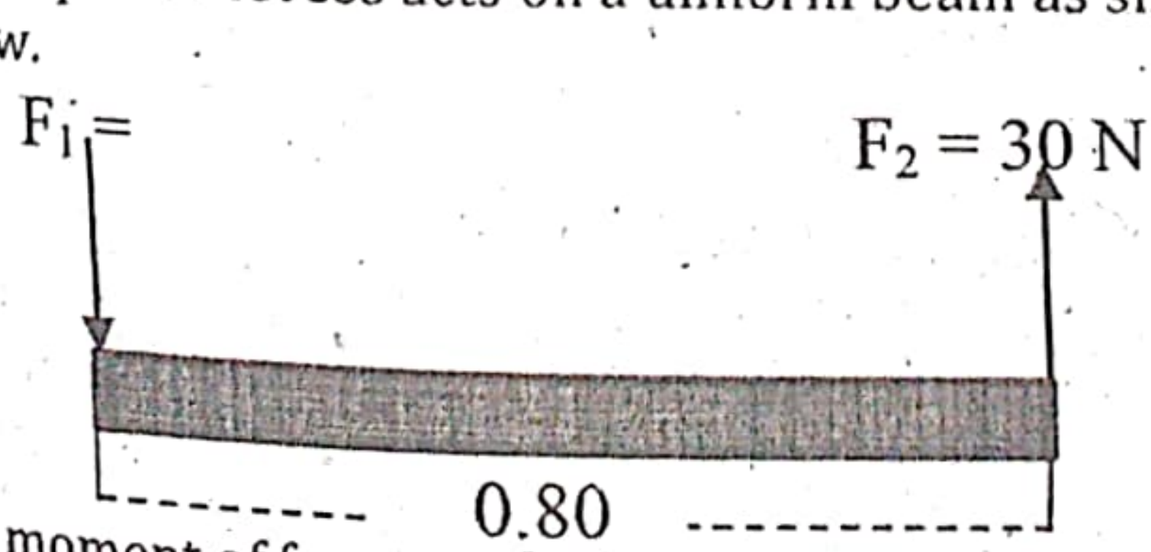
1. velling over an objects surface exerts less pressure than if the fluid was static. This is a statement of
   A. Bernoulli's principle.
   B. Principle of continuity.
   C. Pascal's principle.
   D. Archimedes' principle.
2. Which one of the following activities of experiment is NOT necessary to determine the specific latent heat of vaporization of water?
   A. Measuring temperature
   B. Measuring volume
   C. Measuring time
   D. Measuring mass
3. Which one of the following statements is correct about critical and boiling points substance?
   A. The critical point and the boiling point of a substance occur at the same specific temperature and pressure.
   B. Both critical temperature and boiling point of a substance can be increased or decreased based on the pressure of the surrounding.
   C. The boiling point of a substance can be different based on the pressure of the surrounding whereas critical point occurs at one specific pressure and temperature.
   D. The critical temperature of a substance can be increased or decrease based on the pressure of the surrounding whereas boiling point does not vary with the pressure of the surrounding.
4. A 30 kg rigid object has a moment of inertia 45 kg m^2 about its center of mass. What is the object's moment of inertia when it rotates through an axis 2m away and parallel to the axis through its center of mass?
   A. 90 kgm^2
   B. 165 kgm^2
   C. 180 kgm^2
   D. 105 kgm^2
5. Two coplanar forces acts on a uniform beam as shown in the figure below.



The moment of force on the beam is
A. 48.0 Nm.
C. 30.0 Nm.

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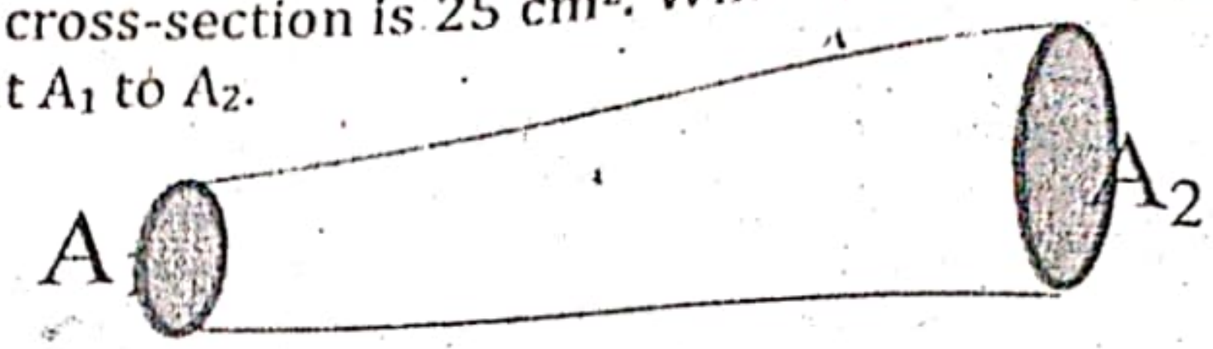
B. 12.0 Nm. D. 24.0 Nm.

6. Water flows at the rate of 500.0 cm^3/s at small cross-section 5 cm^2. The larger cross-section is 25 cm^2. What is the ratio of the speed of the water at A_1 to A_2.



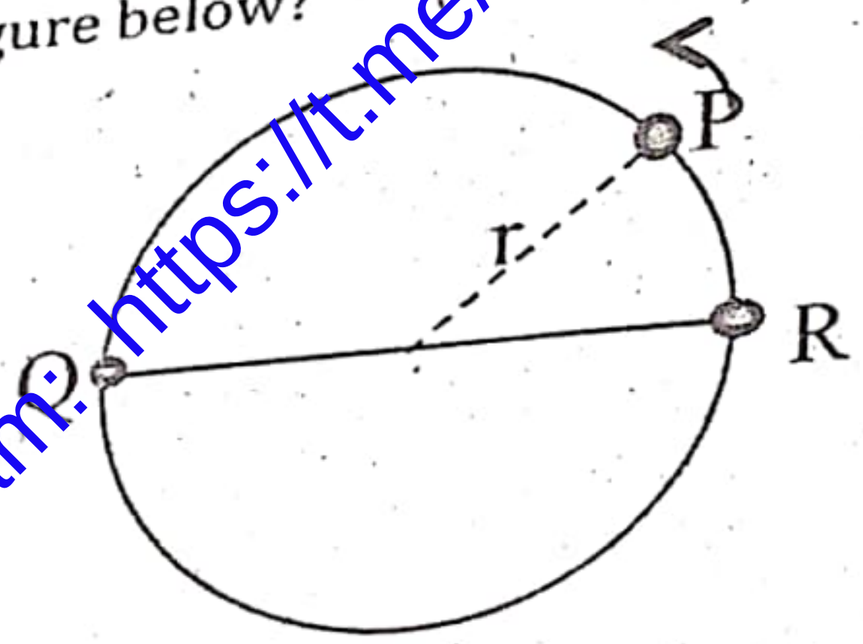
A. 5 B. 100 C. 0.2 D. 20

7. An ideal Monoatomic gas expands adiabatically from an initial volume of 2 Liters a t a temperature of 300 K to 16 liters of volume. What is the final temperature of the gas? (Use the ratio of specific heat capacity at constant pressure to that at constant volume to be 5/3)

A. 37.5 k C. 2400 k

B. 75 k D. 1200 k

8. Consider a particle performing a circular motion around a circle of radius r, with an angular velocity ω and the particle's projection on the horizontal diameter of the circle (between points Q and R) as shown in the figure below?



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Which one of the following statements is correct about the motion of the particle and its projection on $\overline{QR}$

A. The projection of the particle on $\overline{QR}$ swings between points Q and R with constant speed.

B. The circular motion of the particle can be taken as a simple harmonic motion.

C. The projection of the particle on $\overline{QR}$ demonstrates simple harmonic motion.

D. The projection of the particle on $\overline{QR}$ demonstrates motion with constant acceleration.

9. Air molecules vibrate in a pipe of length L closed at one end but open at the other. If the speed of sound through the pipe is V, which

one of the following statements is correct about the resonance
...ation of air molecules in the pipe?
... the fundamental resonance frequency is $\frac{v}{4L}$.
... the fundamental wave length of resonance is equal to 2L.
... the resonance frequency of the first harmonics is $\frac{v}{L}$.
... the wavelength of the third harmonic is equal to L.
9. A monochromatic light of wave length λ passes through double slits separated by a distance d. if the path difference between the lights is p and n being an integer (0, 1, 2, 3, ...). Which ons of the following would be observed on screen behind the slits?
   A. Constructive interference if p = (n+1/2) λ
   B. Constructive interference if p = n λ
   C. Destructive interference if p = 2 n λ
   D. Destructive interference if p = 0
10. Which one of the following statements is NOT correct about power of a lens?
   A. The power of a lens decreases as the focal length of the lens increases.
   B. The larger the focal length of the lens the greater is the power of the lens.
   C. The power of a lens is the measure of the strength of the lens.
   D. Optician gives the name dioptres for the unit of power of a lens.
11. Which one of the following groups of physical quantities contains only vectors?
   A. Work, Electric field, Displacement and Force
   B. Acceleration, Speed, Force and Electric field
   C. Momentum, Energy, Magnetic field and Force
   D. Displacement, Velocity, Magnetic field and Momentum
13. Which rule about the number of significant figures is correct? Zeros in the number
   A. 450.0 are significant figures.
   B. 0.0023 are significant figures.
   C. 1.1000 are not significant figures.
   D. 2.70018 are not significant figures.
14. Given two vectors $V_1 = 10$ units along the positive y-axis and $V_2 = 6$ units at an angle of 37^0 above the positive x-axis. What is the scalar product of the vectors in square of units?
   A. 45 B. 36 C. 48 D. 60
15. An airplane take-off at airport A and travels 500 km due east within one hour and then turning to south it flies for 20 minutes to land on airport B which is 100 km away from the turning point. The

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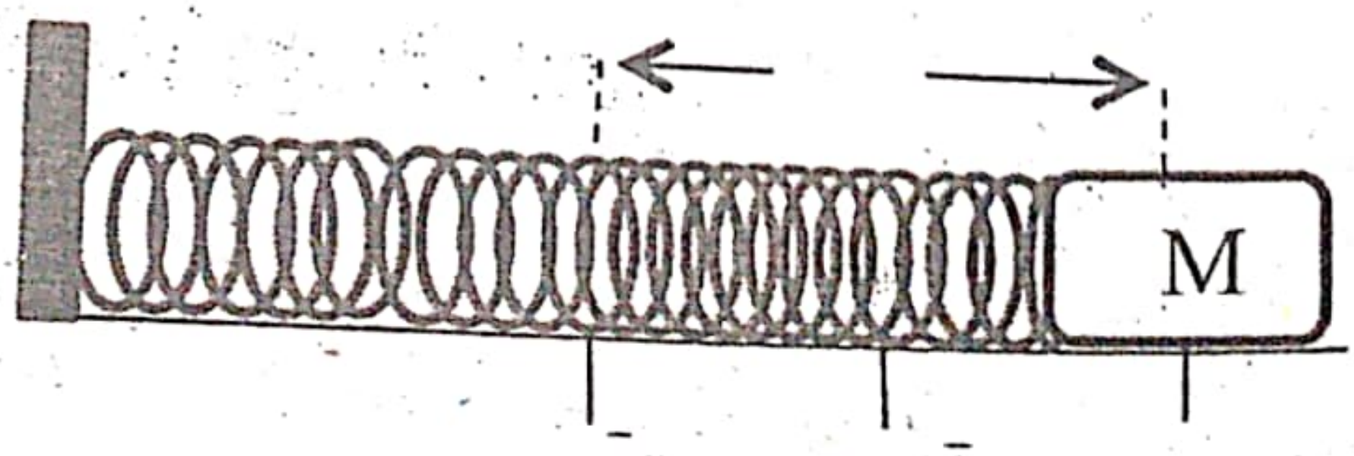
magnitude average velocity of the plane during its flight between the two airports is

A. $75\sqrt{24}$ Km/h.
B. $100\sqrt{34}$ Km/h.
C. 450 Km/h.
D. $75\sqrt{26}$ Km/h.

16. A bullet is fired with an initial speed v at an angle θ with the horizontal. It takes a time T to reach its maximum vertical displacement h_{max}. It hits the target point a distance R away and exactly in the horizontal level to the point where it was fired. Which one of the following statement is **NOT** correct about the motion of the bullet?
    A. The bullet hits the target at a time 2T after it was fired.
    B. h_{max} is directly proportional to the initial speed v.
    C. R will be maximum if $\cos\theta = \sqrt{2/2}$
    D. θ should be different from 90^0 to hit target at a distance R.
17. A man pushes a 25.0 kg object by 250 N force along an inclined plane inclined at an angel of 37^0 to the horizontal. If the object moves with constant speed, the friction force exerts on the block is
    A. 250.0 N, up the inclined plane.
    B. 100.0 N, up the inclined plane.
    C. 100.0 N, down the inclined plane.
    D. 250.0 N, down the inclined plane.
18. The following three collisions occur in different systems.
    I. A billiard ball A collides with an identical ball B and comes to rest after collision, while ball B moves with the same velocity as ball A was moving before.
    II. A bullet moving with speed v hits a suspended mass and it is imbedded in it and the combination moves with common velocity.
    III. Two cars moving in opposite direction become at rest after collision.

    Which of the following statements is correct about these collisions?
    A. II and III are inelastic collisions.
    B. I and II elastic collisions.
    C. I, II and III are inelastic collisions.
    D. I and III are elastic collisions.
19. An object of mass M is attached to a spring of spring constant K, as shown in the figure below.



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If the mass is pulled to a point P, which is at a distance A from equilibrium position O, and released, which one of the following statements is correct about energy of the system? (assume the surface is frictionless)

A. At point O, the system has both kinetic and potential energy.
B. At point P, the system has both kinetic energy and potential energy.
C. At point P, the mass attains its maximum velocity.
D. At point Q the system has both kinetic energy and potential energy.

20. A force $\vec{F} = (\hat{i} - 3\ \hat{k})$ N acts on a wooden bar and drags the bar through a displacement of $\vec{r} = 4\ \hat{i}$m. What is the torque due to the force in N .m?

A. $4\ \hat{i} + 12\ \hat{j}$
B. $-12\ \hat{j}$
C. $12\ \hat{j}$
D. $12\ \hat{k}$

21. Which one of the following statements does NOT describe coherent light or interference of coherent light?

A. In order for two waves to be coherent, they must be the same type of wave.
B. Coherent light waves have the same frequency and wavelength.
C. Two waves which maintain constant phase relationship can produce sustained interface.
D. Sustainable interference pattern can be formed by two waves with variable phase difference.

22. Kirchhoff's junction rule is a consequence of conservation of

A. Charge.
B. Electrical energy.
C. momentum.
D. mass.

23. Which one of the following statements describes the photoelectric effect interaction of radiation with matter?

A. In photoelectric effect, the photoelectron energy is directly proportional to the wavelength of incident radiation.
B. The intensity of radiation incident on a material can affect the energy of photoelectrons emitted from the metal surface.
C. When radiation of certain frequency f, interact with atoms of a material, electrons will be emitted with energies proportional to the radiation energy.
D. Any value of energy of photons (radiations) is capable of producing photoelectrons from a metal.

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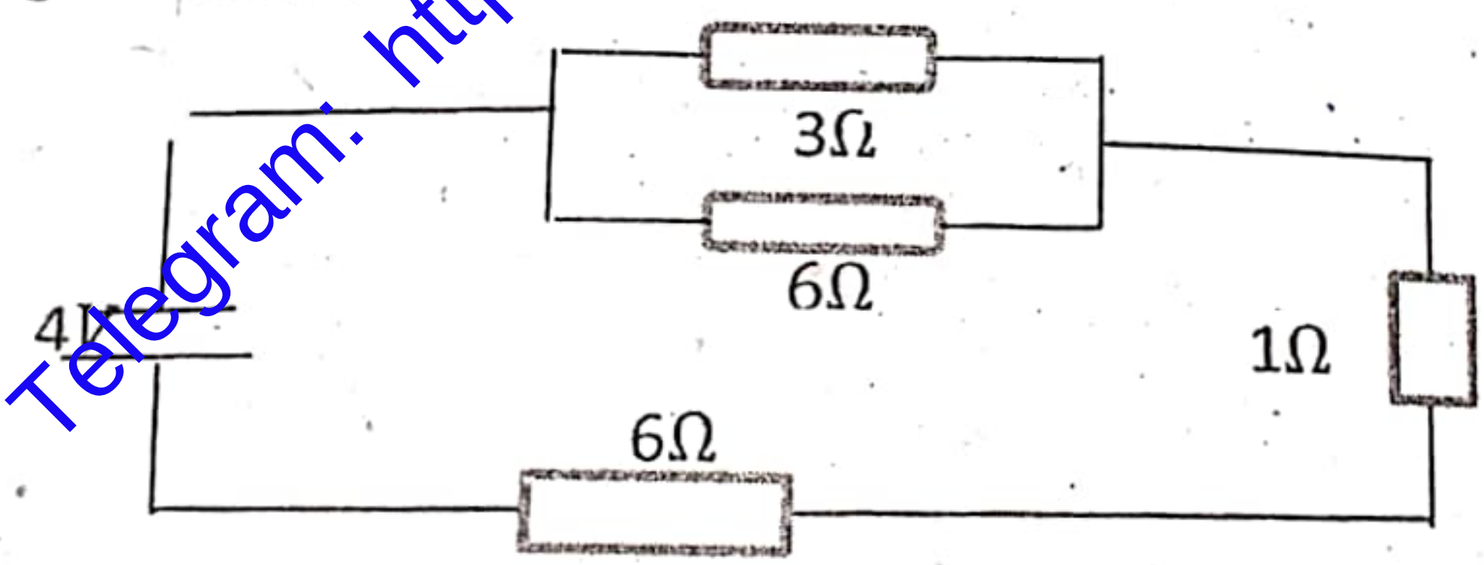
24. A positive point charge of 2μC is placed on the x-axis at the origin and a positive charge of 6μC and a negative point charge of -4μC are placed, respectively at x = -3cm and x = +3cm on the x-axis. What is the net electric force on the 2μC charge?
    A. 40N toward the positive x- axis
    B. 40N toward the negative x-axis
    C. 200 N toward the negative x-axis
    D. 200 N toward the positive x-axis
25. Consider a point charge Q in an electrostatic field and a point mass m in a gravitational field. Which one of the following statements is correct about the potential energy of the objects?
    A. The potential energies at distance r, from the field sources for both Q and m vary inversely with r^2.
    B. The gravitational force on mass m is in the same direction of the gravitational field while electric force on Q is in the same or opposite directions of electronic field depending the sign of the charge.
    C. The electric potential decreases as a positive charge moves opposite to the electric field while the gravitational potential of the mass increases when it moves along the gravitational field.
    D. The potential energy of the point charge and that of the point mass changes as they move over equipotential surfaces in their respective fields.
26. Four resistors are connected to a voltage source as shown in circuit diagram shown below



What are the current through the 3Ω resistor and the voltage drop across the 4 Ω resistor, respectively?
    A. $\frac{4}{3}$ A, and 8 V
    B. $\frac{3}{4}$ A and 8 V
    C. $\frac{2}{3}$ A and 4 V
    D. 2 A and $\frac{3}{4}$ V
27. An electron moving with a speed of 2.2×10^7 m/s east enters in the region of magnetic field strength of 5×10^{-4} T directed north. What is the magnetic force experienced by the electron?
    A. 3.64×10^{-30} N, towards east direction

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B. 1.76×10^{-15} N towards north direction
C. 1.76×10^{-15} N, perpendicular to north and east directions
D. 8×10^{-15} N perpendicular to north and east directions

28. A magnetic field intensity of 4×10^{-4} T crosses a coil having a face area of 100 cm^2 and 5×10^4 turns perpendicular to the face of the coil. If the magnetic flux in the coil is set to zero within 2ms. What is the magnitude of the induced emf in the coil?
A. 10 V
B. 2×10^{-5} V
C. 4×10^{-4} V
D. 100 V

29. In a simple a.c circuit, an inductor of 4μH and a capacitor of 9 pF are connected in series to each other and the combination is connected to an a.c source. What is angular frequency at resonance of the circuit?
A. 3.6×10^7 rad/s
B. 1.67×10^8 rad/s
C. 3.8×10^8 rad/s
D. 2.7×10^5 rad/s

30. Electrons in a Sodium atom emit radiation when they move from a higher energy level of -0.81eV to lower energy level of -2.21eV. What is the frequency of radiation emitted in the process?
A. 8.10×10^{15} Hz
B. 2.21×10^{14} Hz
C. 3.38×10^{14} Hz
D. 1.40×10^{15} Hz

31. Which one of the following units is NOT a possible unit of electric field strength?
A. Joule/Coulomb
B. Newton/ Coulomb
C. Joule/(Coulomb x meter)
D. Volt/meter

32. Which one the following statements does NOT describe magnetic nature of a material?
A. Magnetic dipoles of a material aligned parallel when the magnetic material is placed in external magnetic field.
B. There is no a single magnetic North or magnetic South Pole.
C. Magnetic molecules (dipoles) are oriented randomly when a material lose its magnetism.
D. Magnetic molecules in a magnet will align when they are heated or hammered.

33. Which one of the following statements is correct about conduction in semiconductors?
A. P-type semiconductors have holes as minority charge carriers in the crystal lattice of the material.
B. N-type semiconductor is made from half-filled shell atoms doped with accepter atoms.
C. When atoms with half-filled electrons are doped with atoms of five valance electrons, the majority charge carriers are holes.

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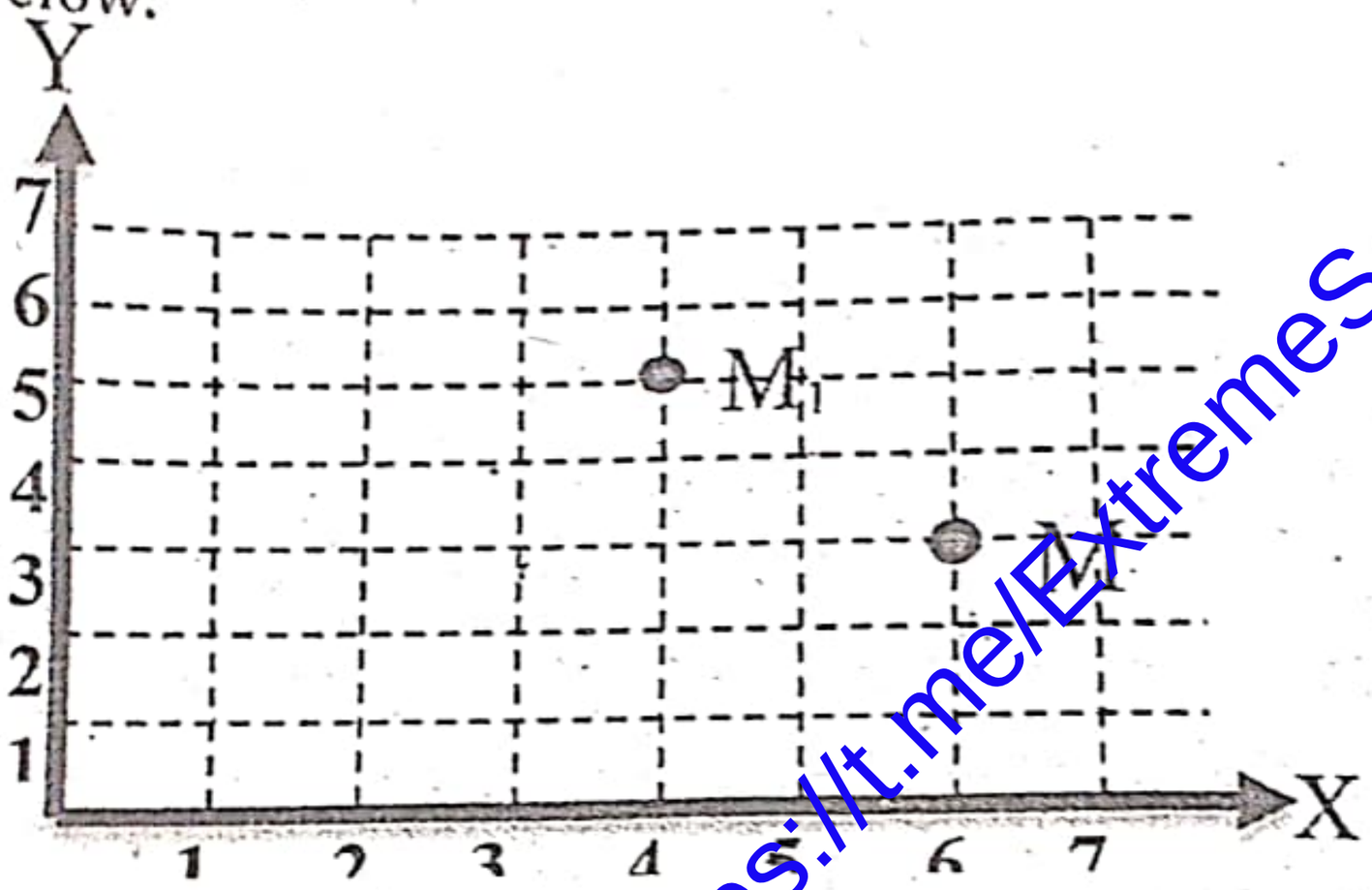
D. The majority charge carriers in N-type semiconductors are electrons due to doping.

34. A small source emits sound waves which are spherical. The intensity of the sound wave measured at a distance r = 3.0 m from the source is I= 5 W/m². At what distance from the source would the intensity be one-fourth as much as it is at r = 3.0 m?

A. 9.0 m
B. 12.0 m
C. 6.0 m
D. 4.5 m

35. Consider a system of two point masses M_1 and M_2, with $M_1 = 0.5\ M_2$. The two masses are located on the x-y plane as shown in the figure below.



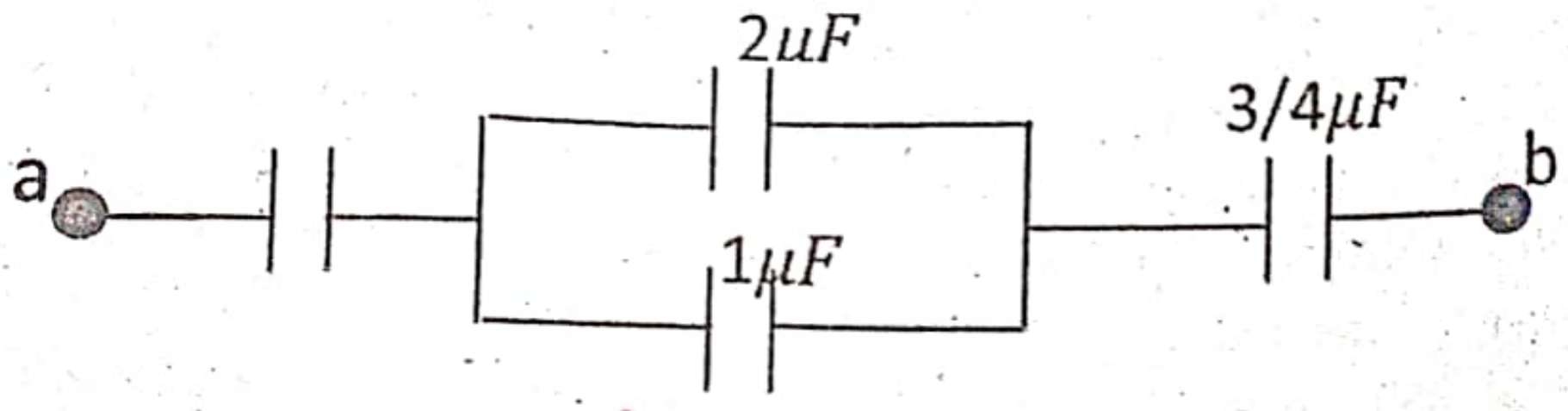
Which one of the following alternatives indicates the position of center of mass of the system?

A. $(\frac{5}{3}, \frac{7}{3})$ cm
B. (5 ,4) cm
C. $(\frac{11}{3}, \frac{16}{3})$ cm
D. $(\frac{16}{3}, \frac{11}{3})$ cm

36. What is the acceleration due to gravity at an altitude twice the radius of the Earth? (g_E is the gravitational acceleration on the surface of the Earth).

A. $\frac{g_E}{9}$
B. $\frac{g_E}{3}$
C. $\frac{g_E}{4}$
D. 3 g_E

37. Four capacitors are connected in a capacitor network as shown in the figure below.



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The equivalent capacitance between terminals a and b in μF is

A. 12
B. ¾
C. 1/4
D. 4

38. A 64 V battery is connected to a series combination of 3 kΩ bulb and 4 kΩ electric iron. If the total current in the circuit is 8mA, what is the internal resistance of the battery?

A. 7 kΩ B. 1 kΩ C. 15 kΩ D. 8 kΩ

39. A solenoid with 5000 turns and length 1m carries 15A of current. If no iron core is inserted in the solenoid, what is the magnetic field strength at the center of the solenoid?

A. $6\pi \times 10^{-2}$ T
B. $20\pi \times 10^{-2}$ T
C. $4\pi \times 10^{-2}$ T
D. $3\pi \times 10^{-2}$ T

40. Light ray is allowed to enter from air medium (n_{air} = 1) into water medium (index of refraction n_w = 4/3) at an angle of 30^0 with the normal to air-water interface (boundary). Which one of the following is correct about the ray of light?

A. The sine of the angle of refraction of the light ray in water is equal to 0.38.
B. The sine of the angle of refraction of the light ray in water is equal to 0.5.
C. The speed of light ray in water is 0.9 times that of its speed in air.
D. The speed of light ray in water is 1.33 times that of its speed in air.

41. Suppose you are standing on horizontal floor, in a room, in a room, with one of your feet and leaning the wall of the room with the other foot. The number of action reaction pair of forces you experience is

A. 4 B. 2 C. 3 D. 1

42. The amount of heat energy required to change a substance from a liquid to a gaseous state is known as

A. Latent heat of vaporization
B. Latent heat of fusion.
C. Quantity of heat.
D. Specific heat capacity.

43. Which one of the following is correct about mechanical waves?

A. The number of complete waves that passes a given point per time is called the period of the wave.
B. The distance between two identical points on adjacent waves is known as the wavelength of the wave.
C. The time taken for one complete wave to pass a given point is called the frequency of the wave.

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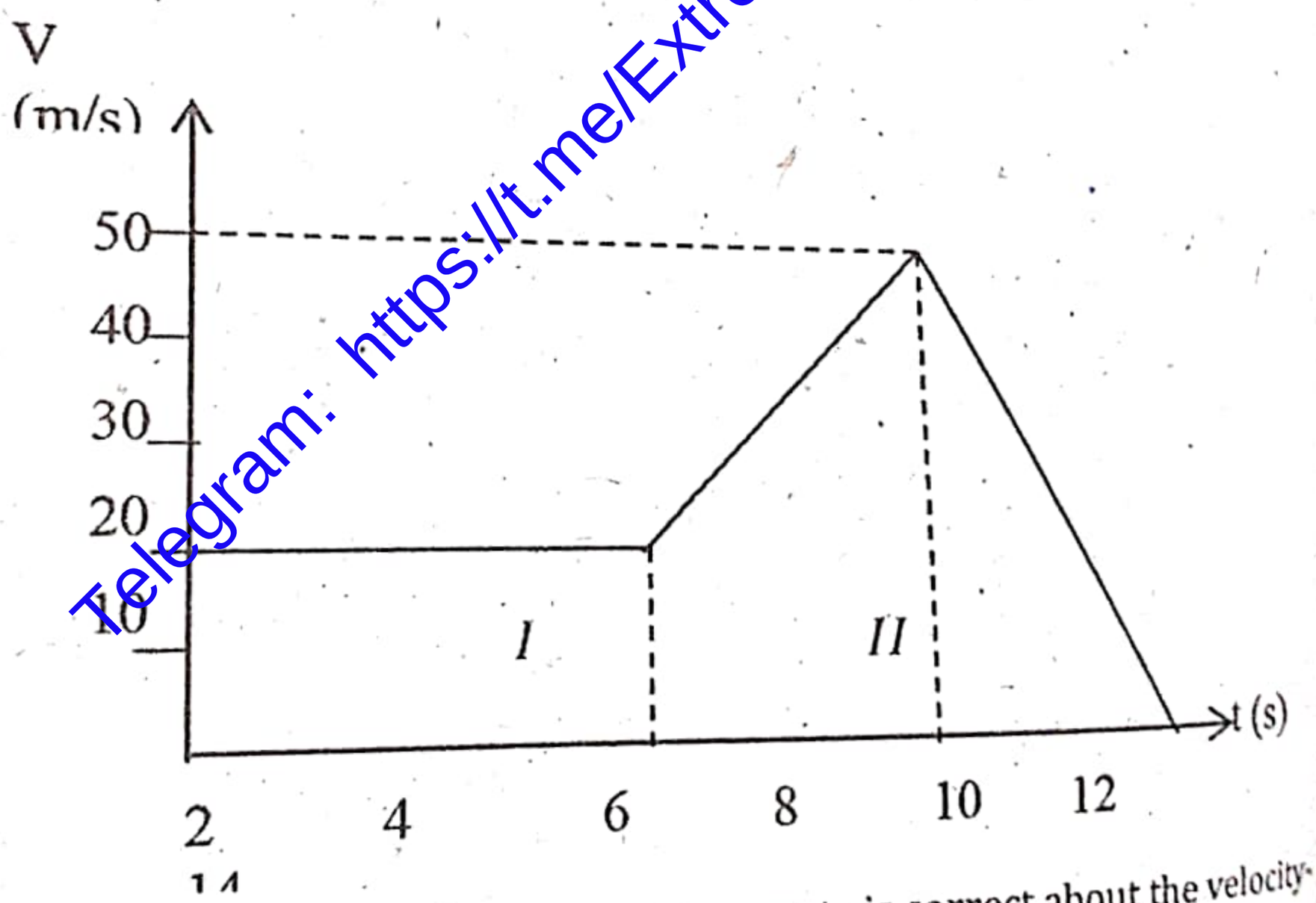
D. Specific heat capacity.

43. Which one of the following is correct about mechanical waves?
A. The number of complete waves that passes a given point per time is called the period of the wave.
B. The distance between two identical points on adjacent waves is known as the wavelength of the wave.
C. The time taken for one complete wave to pass a given point is called the frequency of the wave.
D. The maximum height in transverse wave is known as trough of the wave.

44. A boy in a journey covers his route by travelling 3.0 km east and 4.0 km north. What is the magnitude of his resultant displacement?
A. 2.7 km
B. 7.0 km
C. 1.0 km
D. 5.0 km

45. The graph given below shows the velocity of man travelling on a motorcycle as a function of time.



Which one of the following statements is correct about the velocity-time graph?
A. The acceleration in the first 6 s is equal to the gradient of the curve in the given interval and it is positive.
B. The acceleration during the time interval from t= 6 s to t = 10 s is equal to the gradient of the given interval and it is negative.
C. The displacement during the time intervals t = 0 to t = 6 s is equal to the area under the velocity-time graph in the given interval.

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D. 2.60×10^4 N

B. 2.64×10^4 N

47. Starting from rest, a block of mass 5.0 kg slides 2.5 m down a rough 37^0 incline. The coefficient of kinetic fiction between the block and the incline is $\mu_k = 0.44$. What is the work done by the friction force on block?

C. -18.92 J

A. -59.00 J

D. -75.00 J

B. -44.00 J

48. An inclined plane is 20m long and is inclined at an angle of 15^0 to the horizontal. If the effort required to push a block of weight 3,000 N up the slope is 900 N, what is the efficiency of the inclined plane?

C. 86.7 %

A. 30.0 %

D. 100 %

B. 25.9 %

49. A wooden block of volume 5.00×10^{-4} m^3 floats partially submerged in water, and small steel object of mass m is placed on top of the block. When m= 0.25 kg, the system is in equilibrium and the top of the wooden block is at the level of water. What is density of the wood?

A. 800 $kg.m^{-3}$

C. 2500 $kg.m^{-3}$

B. 500 $kg.m^{-3}$

D. 1000 $kg.m^{-3}$

50. A bimetallic strip made up of iron and brass bonded together is used in a thermostat. The coefficient of linear expansions for iron and brass are 1.1×10^{-5} /K and 1.9×10^{-5} /K, respectively. Which of one the following phenomena occurs when the temperature changes?

A. When the temperature decreases, the iron section of the bimetallic strip contracts faster than the brass and the strip bends towards the iron section.

B. When the temperature increases, the brass section of the bimetallic strip expands faster that the iron and the strip bends towards the iron section.

C. When the temperature increases, the iron section of the bimetallic strip expands faster than the brass and the strip bends towards the brass section.

D. When the temperature decreases, the brass section of the bimetallic strip contracts slowly than the iron and strip bends towards the brass section.

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1. In the absence of air resistance an object moving vertically downward loses PE, and gains an equal amount of KE.

   - An object moving vertically upward loses KE, and gains an equal amount of PE.
   - For a wooden block sliding over a rough horizontal surface mechanical energy is not conserved.
   - The mechanical energy of a wooden block sliding over a rough horizontal surface is NOT conserved due to dissipative force.

Answer; C

2. Given; $\omega_o = 20rev/s, \theta = 40rev$ and when the wheel is brought to rest final angular speed is $\omega = 0$. Required; $\alpha = ?$

Solution; the rotating wheel decelerates and the acceleration of a point on the rim of the wheel is obtained from

$$\alpha = \frac{\omega^2 - \omega^2_o}{2\theta} = \frac{0^2 - (20rev/s)^2}{2(40rev)} = -\frac{400}{80} rev/s^2 = -5rev/s^2$$

Answer; D

3. **Solution;** According to the statement of parallel axis theorem the moment of inertia of a body of mass M about an axis that passes through a distance 'd' from the center of mass of the body is given by

$$I_P = I_{cm} + Md^2.$$

Answer; A

4. **Solution;** Based on the information on the diagram

a) when the axis of rotation pass through point A, the net torque is,

$$\tau_A = \frac{F}{2}(d + \frac{L}{2}) + \frac{F}{2}(d + \frac{L}{2}) - F(d + \frac{l}{2}) = 2(\frac{F}{2})(d + \frac{L}{2}) - \frac{F}{2}(d + \frac{L}{2}) = 0$$

b) when the axis pass through point B, the net torque is,

$$\tau_B = \frac{F}{2}(\frac{L}{2}) + \frac{F}{2}(\frac{L}{2}) - F(\frac{l}{2}) = 2(\frac{F}{2})(\frac{L}{2}) - \frac{F}{2}(\frac{L}{2}) = 0$$

c) when the axis pass through O the arm of the forces is zero i,e, r = 0.

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Therefore; $\tau_o = 0$.

From the result we get, $\vec{\tau}_A = \vec{\tau}_B = \vec{\tau}_O = 0$.

Answer; C

5. Tensile stress is defined as perpendicular component of applied force divided by the cross – sectional area of the object. Thus, Tensile stress = $\frac{F}{A}$. Its unit is the same as the unit of pressure which is Pa.

Note; ratio of change in length to the original length describes the tensile strain.

Tensile strain = $\frac{\Delta L}{L_o}$. Tensile strain has no unit, it is a dimensionless quantity. Answer; A

6.Given; $\rho = 1080 kg/m^3$, $h = 2.1cm = 2.1x10^{-2} m$,

$d = 1.0mm = 10^{-3} m$, $\theta = 0^o$ Required; f =?

Solution; The surface tension of the fluid is obtained from

$$f = \frac{\rho g r h}{2\cos\theta} = \frac{1080kg/m^3 x10m/s^2 x5x10^{-4} m x2.1x10^{-2} m}{2(\cos 0^o)}$$

$= 0.0594 N/m$

Answer; No answer. According to choice B, f = 0.0567 N/m is closer to the answer.

7.Given; A = 0.1 m^2, L = 2 cm = 2 x 10^{-2} m, K = 0.1 J/m.s ^oC, $\Delta T = 100^o C$.

$\Delta t = 24h = 8.64x10^4 s$, Required; $\Delta Q = ?$

Solution; The amount of heat flowing through the slab is given by

$$\frac{\Delta Q}{\Delta t} = KA\frac{\Delta T}{\Delta L} \Rightarrow \Delta Q = KA(\frac{\Delta T}{\Delta L})\Delta t = (0.1 J/m.s^oC)\frac{100^o C}{2x10^{-2} m} x8.64x10^4 s$$

$\Delta Q = 4.32 \times 10^6 J$.

Answer; D

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8. Given; According to the sign convention when work is done on the system $\Delta W = 100J$,
When heat is extracted from the system $\Delta Q = -418.6J$, Required; $\Delta U = ?$

Solution; The change in internal energy of the system is given by;

$$\Delta U = \Delta Q + \Delta W = -418.6J + 100J = -318.6J.$$

Answer; D

9. - During a reversible process a) the entropy of a system remains constant and
there is no change in entropy. b) a system undergoing a reversible process is always close to thermodynamic equilibrium, where as a system undergoing irreversible process is not in thermal equilibrium and its entropy also increases. c) during a reversible process a system can move from one thermodynamic state to another and back to its initial state and change in internal energy is zero if the process is cyclic.

Answer; D

10. Resonance occurs when the frequency of a periodic driving force is equal to the natural frequency of the oscillator the amplitude of oscillation becomes maximum.

Answer; C

11. When object is placed in front of a plane mirror, the image formed is a) virtual and upright, b) laterally shifted, c) same size as the object an found as far as object is in front of the mirror.

Answer; A

12. Given; object distance = $d_o = 25cm$, focal length of the mirror = f = 10 cm.

Required; $M = ?$

Solution; The linear magnification of a mirror is given by;

$$\text{Magnification} = \frac{sizeofimage}{sizeofobject} = \frac{imagedistance}{objectdistance}.$$

$M = \frac{d_I}{d_o}$, the image distance is obtained from the mirror formula which is written as; $\frac{1}{f} = \frac{1}{d_I} + \frac{1}{d_o} \Rightarrow d_i = \frac{fxd_o}{d_o - f} = \frac{10cm(25cm)}{25cm - 10cm} = \frac{250}{15}cm.$

and using magnification formula you get

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Answer; A

14. Given; $\vec{A} = 3\vec{i} + 4\vec{j}$. Required; a unit vector in the direction of $\vec{A}$.

Solution; a unit vector in the direction of $\vec{A}$ is given by,

$$\vec{u} = \frac{\vec{A}}{|\vec{A}|} = \frac{3\vec{i} + 4\vec{j}}{\sqrt{3^2 + 4^2}} = \frac{3\vec{i} + 4\vec{j}}{5} = \frac{3}{5}\vec{i} + \frac{4}{5}\vec{j}.$$

Answer; C

15. Solution; Two coplanar vectors are collinear, if the vectors have opposite direction or the angle between the vectors is 180°.

$$\cos\theta = \frac{\vec{A} \bullet \vec{B}}{|\vec{A}||\vec{B}|} = -1, \text{ suppose } \vec{A} = 2\vec{i} + 3\vec{j} \text{ and } \vec{B} = -4\vec{i} - 6\vec{j}.$$

$$\vec{A} \bullet \vec{B} = A_x B_x + A_y B_y = 2(-4) + 3(-6) = -26 \qquad |\vec{A}| = \sqrt{2^2 + 3^2} = \sqrt{13},$$

$$|\vec{B}| = \sqrt{(-4)^2 + (-6)^2} = \sqrt{52}, \Rightarrow \cos\theta = \frac{-26}{\sqrt{13x52}} = -1 \Rightarrow \theta = \cos^{-1}(-1) = 180^o.$$

Answer; B

16. For an object moving in a straight line if the velocity increases by equal amount in equal time interval,a) the velocity time graph is a straight line, b) the slope under v - t graph is the acceleration and positive, c) the acceleration - time graph is a horizontal line parallel to the time axis.

Answer; A

17. Solution; For equal orbital radius, a satellite moving around the Earth, and for a satellite moving around the Moon their respective orbital speed are, respectively, given by;

$$v_E = \sqrt{\frac{GM_E}{R}}, \text{ and}$$

$$v_M = \sqrt{\frac{GM_M}{R}} \Rightarrow \frac{v_E}{v_M} = \sqrt{\frac{M_E}{M_M}} > 1 \Rightarrow v_E > v_M.$$

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17. **Solution;** For equal orbital radius, a satellite moving around the Earth, and for a satellite moving around the Moon their respective orbital speed are, respectively, given by;

$$v_E = \sqrt{\frac{GM_E}{R}}, \text{and}$$

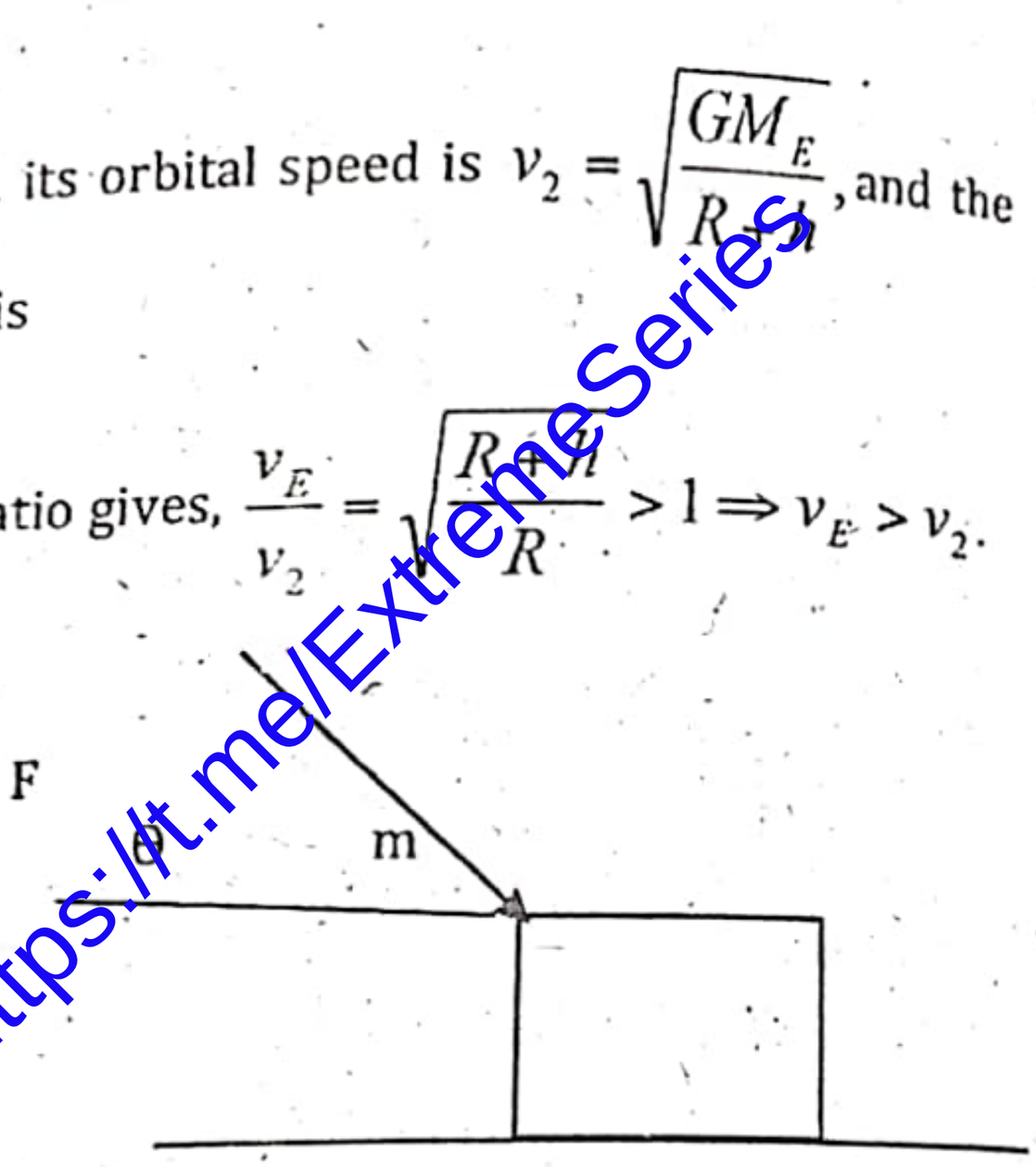
$$v_M = \sqrt{\frac{GM_M}{R}} \Rightarrow \frac{v_E}{v_M} = \sqrt{\frac{M_E}{M_M}} > 1 \Rightarrow v_E > v_M.$$

For a satellite far from Earth its orbital speed is $v_2 = \sqrt{\frac{GM_E}{R+h}}$, and the orbital speed closer to Earth is

$v_E = \sqrt{\frac{GM_E}{R}}$. Taking their ratio gives, $\frac{v_E}{v_2} = \sqrt{\frac{R+h}{R}} > 1 \Rightarrow v_E > v_2$.

Answer; B

18.



Solution; The components of the applied force are, respectively, given by;

$F_X = F\cos\theta, and F_Y = F\sin\theta,$ the normal reaction force is, $N = mg + F\sin\theta.$

The static friction force between the surfaces in contact is, $f_S = \mu(mg + F\sin\theta)$.

The net force is $F_{net} = F\cos\theta - (\mu mg + \mu F\sin\theta) = 0$

$$F\cos\theta = \mu mg + \mu F\sin\theta \Rightarrow F(\cos\theta - \mu\sin\theta) = \mu mg \Rightarrow F = \frac{\mu mg}{\cos\theta - \mu\sin\theta}$$

Answer; C

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19. Solution; the x - and y - components of the initial velocity before collision, are, respectively $u_{ix} = v\sin\theta$, and $u_{iy} = v\cos\theta$. After collision, the x - and y - components are, respectively $v_{fx} = -v\sin\theta$ and $v_{fy} = v\cos\theta$. The initial and final momentum of the ball are, respectively

$P_i = mu_{ix} + m_{iy} = mv\sin\theta + mv\cos\theta$, $P_f = mv_{fx} + mv_{fy} = -mv\sin\theta + mv\cos\theta$.

The change in momentum of the ball is,

$\Delta P = P_f - P_i = -mv\sin\theta + mv\cos\theta - mv\sin\theta - mv\cos\theta$.

$\Delta P = -2mv\sin\theta$. The average force exerted on the wall for a short time t is given by,

$$F_{ave} = \frac{|\Delta P|}{t} = \frac{|-2mv\sin\theta|}{t} = \frac{2mv\sin\theta}{t}.$$

Answer; B

20. Solution; According to the work - energy theorem, the work done by gravity for an object thrown upward and moving upward is negative, but the work done against gravity is positive.Work done by gravitational force on a satellite moving around the Earth is negative and is given by,

$$W_g = -\frac{GM_E m_s}{r}.$$

Work done by tension on an object tied to a string moving in a horizontal circle is zero because the force is perpendicular to the path. Work done by gravity on an object thrown downward is negative because the object loses potential energy and the object gains kinetic energy because its kinetic energy increases.

Answer; D

21. Given; Let $y_1 = A\sin(kx - \omega t)$, and $y_2 = A\sin(kx + \omega t)$. Required;

$y = y_1 + y_2$.

Solution; From trigonometric identity, we have the following

$y_1 = A\sin(kx - \omega t) = A\sin(kx)\cos(\omega t) - A\sin(\omega t)\cos(kx)$.

and

$y_2 = A\sin(kx + \omega t) = A\sin(kx)\cos(\omega t) + A\sin(\omega t)\cos(kx)$.

Adding the above two gives;

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$$y = y_1 + y_2 = 2A\sin(kx)\cos(\omega t).$$

Answer; D

22. **Given;** d = 0.06 mm = 6 x 10^{-5} m, D = 1.2 m, $\lambda = 563nm.$

Let the width of the bright fringe be, ω.

Solution; According to Young's double slit experiment to observe constructive interference pattern The slit width (d), distance of the screen from slits (D), the wavelength of light used (λ) and the width of the bright fringe formed are related according to the equation

$$\frac{\omega}{D} = \frac{\lambda}{d} \Rightarrow \omega = \frac{\lambda xD}{d} = \frac{5.63x10^{-7}mx1.2m}{6x10^{-5}m} = 11.26x10^{-3}m = 11.26mm.$$

The location of the fourth bright fringe from the central maximum is given by;

$$s = 4\omega = 4x11.26mm = 44.94mm = 4.5cm$$

Answer; B

23. Electric dipole is a system that consists of two point charges of equal magnitude and opposite sign separated by a small distance d and it is written as, $P = qd$. Dipole moment is a vector quantity and its direction is from the negative charge towards the positive charge.

Based on the figure we can see that

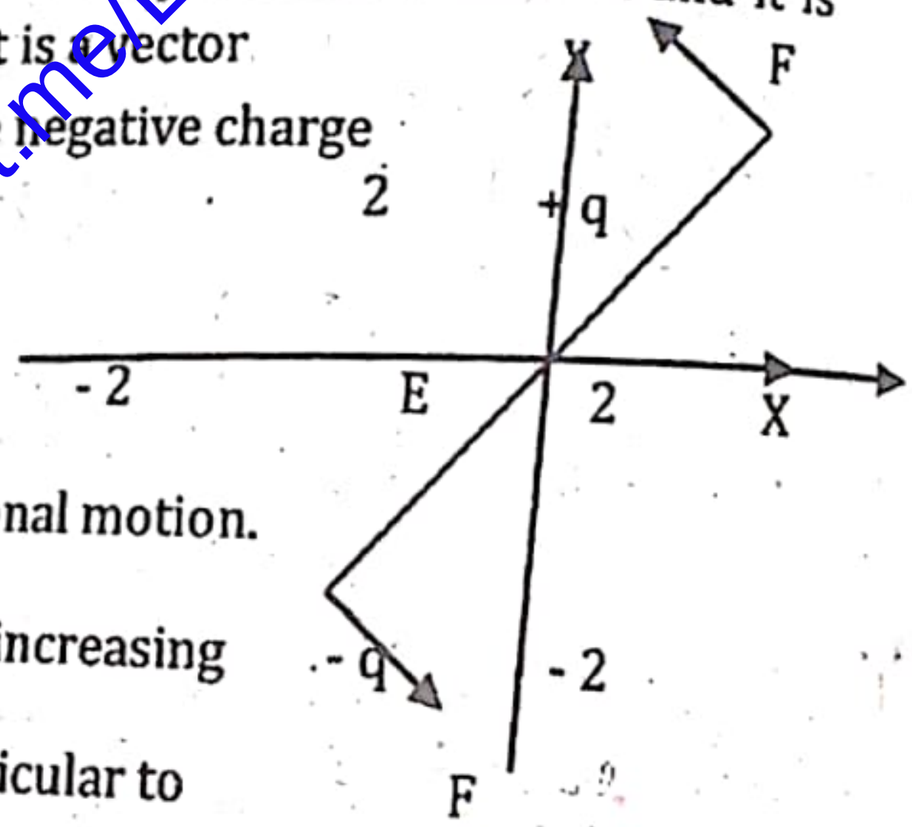
the forces can produce counter -

clock wise rotation without translational motion.

so the field rotates in the direction of increasing

the angle θ, until the dipole is perpendicular to

the field.



Answer; D

24. Given; r = 2 cm, q = - 1.6 x 10^{-9} C, K = 9 x 10^9 Nm^2/C^2, Required; V = ?

Solution; the electric potential at a distance r from a point charge q is given by;

$$V = K\frac{q}{r} = 9x10^9\,Nm^2/C^2\left(\frac{-1.6x10^{-9}C}{2x10^{-2}m}\right) = -720V.$$

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Answer; B

25. Given; $R_1 = 1\Omega$, $R_2 = 4\Omega$, $R_3 = 2\Omega$ and V = 14 V. Required; $V_1 = ?$

Solution; Let the combined resistance of the resistors connected in parallel be R_{12} and is given by;

$$R_{12} = \frac{R_1 x R_2}{R_1 + R_2} = \frac{1\Omega(4\Omega)}{5\Omega} = \frac{4}{5}\Omega = 0.8\Omega.$$

From the circuit shown $R_{12} = 0.8\Omega$ and $R_3 = 2\Omega$ are connected in series and the total resistance is

$R_T = R_{12} + R_3 = 0.8\Omega + 2\Omega = 2.8\Omega$. The total current in the circuit will be,

$I = \frac{V}{R_T} = \frac{14V}{2.8\Omega} = 5A.$ The voltage drop across the resistors in connected in parallel is the same.

Therefore, $V_{12} = V_2 = V_1 = IR_{12} = 5A(0.8\Omega = 4V.$

Answer; B

26. The magnetic field strength inside a current carrying solenoid becomes stronger by wounding more turns of wire, by increasing the current through the coil and by decreasing the length of the coil. Therefore, the magnetic field inside the solenoid is constant and is parallel to the solenoids axis, whereas the field outside is weak and negligible.

Answer; D

27. According to the law of electromagnetic induction, the magnitude of the emf induced in a circuit is directly proportional to the rate of change of magnetic flux crossing any surface bounded by the circuit. An emf is induced if there is a changing magnetic flux and magnetic flux lines must cross the bounded surface.

Answer; C

28. The average power dissipated in RLC series circuit is given by, $P = I^2_{rms}\, Z\cos\theta = I_{rms}V_{rms}\cos\theta$. For RLC series circuit the total impedance Z of the circuit is given by, $Z = \sqrt{R^2 + (X_L - X_C)^2}$.

Where the term $\cos\theta = \frac{V_R}{V_P} = \frac{I_{rms}R}{I_{rms}Z} = \frac{R}{Z} = \frac{R}{\sqrt{R^2 + (X_L - X_C)^2}}$.

$\cos\theta$ is the power factor.

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On average, only the resistance consumes power. That is, $P = P = I^2_{rms} R$.

Substituting for $\cos\theta$ in the power equation gives,

$$P = I_{rms} V_{rms} \frac{R}{\sqrt{R^2 + (X_L - X_C)^2}}.$$

Answer; A

29. Heisenberg's uncertainty principle states that; it is impossible to know both the exact position and exact velocity and therefore exact momentum of a particle at the same time. The uncertainty in position or momentum is therefore, $\Delta P \Delta x \geq \hbar$. If the position is more accurately determined, the momentum is less accurately determined. The opposite is also true. If the momentum is more accurately determined, then the position must be less accurately determined..

Answer; A

30. As nuclei get heavier than helium, their net binding energy per nucleon (BE/A) grows more and more slowly and it reaches its peak at iron (A=56). As nucleons are added, the total binding energy increases but so does the total disruptive energy of the electrostatic force and, once nuclei are heavier than that of iron (A=56), the increase in disruptive energy has more effect than the increase in binding energy.

Answer; A

31. Reflection of sound (echo) is the bouncing of a sound wave from a boundary surface. Reflection of sound occurs in a uniform medium. (mediums having the same density)

Answer; D

32. When a torque rotates an object of mass m and moment of inertia I through an angle θ ,the work done by the torque is given by, $W = \tau\theta$.

Answer; A

33. Given, $r = 4x10^{16} m$, $G = 6.67x10^{-11} Nm^2/kg^2$, $M_E = 6x10^{24} kg$.

Required; T =?

Solution; For a satellite that revolves around the Earth in a circular orbit of radius 'r' its orbital speed is

$v = \sqrt{\frac{GM_E}{r}} = \sqrt{\frac{6.67x10^{-11} x6x10^{24}}{4x10^{16}}} m/s = 0.1m/s$. The period of the satellite is given by,

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$$T = \frac{2\pi r}{v} = \frac{2(3.14)(6x10^{16})m}{0.1m/s} = 2.5x10^{18}s.$$

Answer; B

34. In the process of charging a neutral body by induction method, a) body contact is involved and b) the bodies carry similar charge. In the process of charging a neutral body by induction method, a) touching is not involved and b) after charging the bodies carry charges of opposite sign. When a positively charged object is momentarily brought into contact with a neutral metal sphere, the sphere acquires a net positive charge.

Answer; A

35. When a parallel plate capacitor is charged with a battery and when the battery is disconnected, and a dielectric material is inserted between the plates of the capacitor, a) the capacitance will increase, b) both electric field and energy stored in the capacitor decrease, c) potential difference deceases, and d) the charge on the capacitor will remain the same.

Answer; A

36. The resistance of a conductor with circular cross - section depends on the following factors. a) length of the conductor; as we increase the length of the conductor resistance increases. b) as we increase the cross - sectional area of the conductor resistance decreases. c) the nature of the material the wire is made from.For a conductor with circular cross - section resistance is given by; $R = \frac{\rho l}{A}$ .So the resistance of the conductor increases as its length increases and its diameter decreases.

Answer; D

37. In an electric circuit to investigate Ohm's law and to measure the current, and
the potential difference, using an ammeter (A) and a voltmeter (V) the ammeter should be connected in series and the voltmeter should be connected in parallel to the resistor (R). Therefore, the circuit shown i choice D is the right connection.

Answer; D

38. The charges q_1 and q_2 follow a circular path with the same speed and they have the same mass.The magnetic field is perpendicular to the speed of the charges and the magnetic force due to the charge is given by, $F_1 = q_1 vB$, and $F_2 = q_2 vB$. The magnetic forces will provide the centripetal force. Thus,

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$q_1 vB = \frac{mv^2}{r_1} \Rightarrow q_1 = \frac{mv}{r_1 B}$, in the same way, $q_2 = \frac{mv}{r_2 B}$. Taking their ratio gives,

$\frac{q_1}{q_2} = \frac{mv/r_1 B}{mv/r_2 B} = \frac{r_2}{r_1} > 1 \Rightarrow q_1 > q_2$. The path followed by q_1 is clock wise means its sign is positive and the sign of q_2 is negative because it is moving in counter clock wise direction.

Answer; A

39. Given; $N_p = 250,\ N_s = 1500,\ V_P = 220V$. Required; $V_S = ?,\ I_P / I_S = ?$

Solution; Using the transformer equation we have,

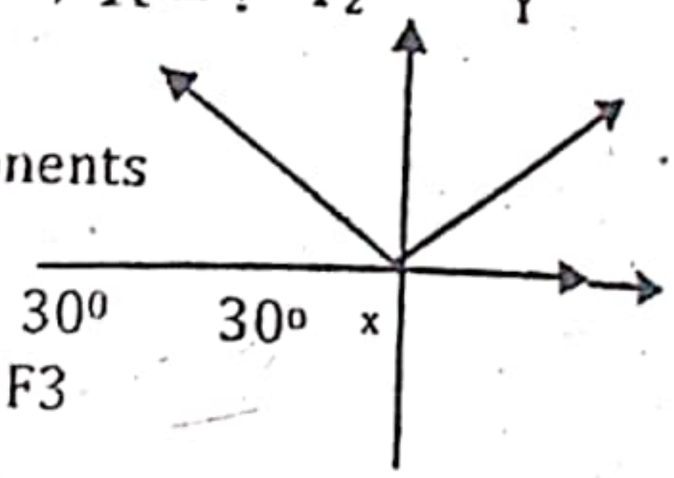
$$\frac{V_P}{V_S} = \frac{N_P}{N_S} = \frac{I_S}{I_P} \Rightarrow \frac{220V}{V_S} = \frac{250}{1500} = \frac{1}{6} \Rightarrow V_S = 6(220V) = 1320V.$$

$$\frac{I_P}{I_S} = \frac{N_S}{N_P} = \frac{1500}{250} = 6. \quad \textbf{Answer; C}$$

40. A diode is an electronic component that allows electric current to pass through one direction.

Answer; D

41. Given; $|\vec{F}_1| = |\vec{F}_2| = |\vec{F}_3| = 10N, \theta = 30^o$. Required; $\vec{R} = ?$



Solution; Resolving the forces into x - and y - components gives

$$\vec{F}_{1X} = F_1 \cos\theta(\vec{i}) = 10Nx\cos 30^o(\vec{i}) = 8.7N(\vec{i}),$$
$$\vec{F}_{1Y} = F\sin\theta(\vec{j}) = 10Nx\sin 30^o(\vec{j}) = 5N(\vec{j}).$$
$$\vec{F}_{2X} = -F\cos 30^o(\vec{i}) = -10Nx0.87(\vec{i}),$$
$$\vec{F}_{2Y} = F\sin 30^o(\vec{j}) = 10Nx0.5(\vec{j}) = 5N(\vec{j}).$$

$$\vec{F}_X = 10N(\vec{i}),\ \vec{F}_{3Y} = 0$$

$$\vec{R}_X = \vec{F}_{1X} + \vec{F}_{2x} = 8.7N9\vec{i}) + 8.7N(-\vec{i}) + 10N(\vec{i}) = 10N(\vec{i}),$$
$$\vec{R}_Y = \vec{F}_{1Y} + \vec{F}_{2Y} = 5N(\vec{i}) + 5N(\vec{j}) = 10N(\vec{j})$$

The magnitude of the resultant is calculated using Pythagoras theorem. Thus,

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30⁰ 30⁰ x
F3

$$\vec{F}_{1x} = F_1 \cos\theta(\vec{i}) = 10Nx\cos 30^\circ(\vec{i}) = 8.7N(\vec{i}),$$
$$\vec{F}_{1r} = F\sin\theta(\vec{j}) = 10Nx\sin 30^\circ(\vec{j}) = 5N(\vec{j}).$$
$$\vec{F}_{2x} = -F\cos 30^\circ(\vec{i}) = -10Nx0.87(\vec{i}),$$
$$\vec{F}_{2r} = F\sin 30^\circ(\vec{j}) = 10Nx0.5(\vec{j}) = 5N(\vec{j}).$$
$$\vec{F}_{3x} = 10N(\vec{i}),\ \vec{F}_{3r} = 0$$

$$\vec{R}_x = \vec{F}_{1x} + \vec{F}_{2x} = 8.7N9\vec{i}) + 8.7N(-\vec{i}) + 10N(\vec{i}) = 10N(\vec{i}),,$$
$$\vec{R}_y = \vec{F}_{1r} + \vec{F}_{2r} = 5N(\vec{i}) + 5N(\vec{j}) = 10N(\vec{j})$$

The magnitude of the resultant is calculated using Pythagoras theorem.

Thus,

$$|\vec{R}| = \sqrt{R^2x + R^2r} = \sqrt{(10N)^2 + (10N)^2} = \sqrt{200N^2} = 10(1.41)N = 14.1N.$$

The direction of the resultant is obtained from tangent relation. Thus,

$$\tan\theta = \frac{R_r}{R_x} = \frac{10N}{10N} = 1 \Rightarrow \theta = \tan^{-1}(1) = 45^\circ. \quad \text{So,} \quad \vec{R} = 14.1N, 45^\circ$$

above the positive x - axis. **Answer; B**

42. Given; u = 20 m/s, v = 0, t = 3 s, Required; h = ? and v_f = ?

Solution; The maximum height reached by the ball is given by,

$$h_{max} = \frac{v^2 - u^2}{-2g} = \frac{0 - (20m/s)^2}{-20m/s^2} = \frac{400}{20}m = 20m$$

$$v_f = u - gt = 20m/s - 10m/s^2(3s) = -10m/s$$

The negative sign tell us that the ball is moving downward **Answer; B**

43. Static friction is the maximum friction force between two surfaces in contact. For an object at rest on a horizontal surface, the static friction becomes equal to the limiting friction, when a constant force is applied on the object.

Answer; A

44. Given; m = 60 kg, g = 10 m/s², real weight = W = mg = 600 N, Required; W_{app} =?

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46. When an object of mass 'm' is raised through a height 'h' over a frictional inclined plane of length 'l' the ideal MA, and the actual MA are, respectively, given by,

$$IMA = \frac{S_E}{S_L} = \frac{l}{h} = \frac{l}{l\sin\theta} = \frac{1}{\sin\theta}, \text{ and}$$

$$AMA = \frac{F_L}{F_E} = \frac{mg}{mg(\sin\theta + \mu\cos\theta)}.$$

When the inclined plane is frictionless, $IMA = AMA = \frac{1}{\sin\theta}$.

Answer; C

47. According to the principle of floatation, a body submerged in a fluid floats under the following circumstances. a) when the density of the submerged body is less than the density of the fluid and b) when the weight of the displaced fluid is equal to the entire weight of the submerged body.

Answer; B

48. The unusual behavior of water is its abnormal expansion. As temperature of water increases from 0° C to 4° C, its density increases.

Answer; B

49. Given; m_w = 40 g, T_{hw} = 50° C, m_c = 30 g, T_c = 20° C, c_w = 4200 J/kg k, c_c = 420 J/kg k.

Solution; According to the law of heat exchange, when thermal equilibrium is attained

Heat lost by hot water = Heat gained by cold calorimeter.

$$m_w c_w \Delta T_w = m_c c_c \Delta T_c \Rightarrow m_w c_w (T_w - T_m) = m_c c_c (T_m - T_c)$$

$$40gx4200J/kgk(50^\circ C - T_m) = 30gx420J/kgk(T_m - 20^\circ C)$$

$$40(50^\circ C) - 40T_m = 3T_m - 60^\circ C \Rightarrow 43T_m = 2060^\circ C$$

$$T_m = \frac{2060^\circ C}{43} = 47.9^\circ C.$$

Answer; D

50. Given; $f = 1.5x10^8/s$, $\lambda = 2.00m$, Required; v =?

Solution; The speed of the FM radio signal is calculated from

$$v = \lambda f = 2.00m(1.5x10^8/s) = 3.0x10^8 m/s.$$

Answer; A

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provide a principle which will be helpful in the study of a real
more complicated which is necessary to study properties of thermal
[illegible] energy

answer:- B

Specific latent heat of vaporization is the quantity of heat required to convert the unit of a substance from liquid to vapor state without any change of temperature. From this statement we can clearly understand that measuring time for latent heat of vaporization is not necessary

Answer:- C

The boiling point increases with increased pressure up to the critical point. Whereas the critical temperature for a pure substance is the temperature above which the gas cannot become liquid regardless of the applied pressure

Answer:- C

The moment of inertia of a body about an axis parallel to the body passing through its center is equal to the sum of the moment of the body about the axis passing through the center and the product of the mass of the body times the square of the distance between the two axes

Mathematically $I = Ic + Mh^2$

Given: $Ic = 45kgm^2$, $m = 30kg$, $h = 2m$

Solution

$I = Ic + Mh^2$

$= 45km^2 + 30kg \times (2m)^2 = 165kgm^2$

Answer:- B

3. The moment of force on the beam is

$t_f = F_1 d_1 + F_2 d_2$

$= (30N \times 0.4m) + (30N \times 0.4m = 12Nm + 12Nm = 24Nm$

Answer:- D

4. Equation continu is $P_1A_1V_1 = P_2A_2V_2$

Since an ideal fluid is incompressible and there is no variation in the density of the fluid the above equation becomes:-

$A_1V_1 = A_2V_2$

$\frac{A_1}{A_2} = \frac{V_2}{V_1}$ $\quad A_1 = 5cm^2$ $\quad A_2 = 25cm^2$

$\frac{V_2}{V_1} = \frac{A_1}{A_2} = \frac{5cm^2}{25cm^2} = \frac{1}{5} = 0.2$

Answer:- C

7. The relation b/n volume and Temperature of an ideal gas undergoing a diabatic process

$T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1} = \text{Constant}$

Where γ is the ratio of specific heat capacity

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is the projection of uniform circular motion along the diameter of the circle on which the circular motion is taking place.

Answer; - C

9. For tubes open at one end and closed at the other, there is a pressure antinode at the closed end of the tube and a pressure node at the open end. The equation used for such pipe are: -

$$\lambda n = \frac{4L}{n}, \qquad f_n = n f_1, \qquad f_1 = \frac{V}{4L}$$

n must be odd

Therefore, the correct statement about the fundamental resonance frequency is $V/4L$

Answer: - A

10. In Constructive interference for a double slit is the path length difference must be an integral multiple of the wave length. $P = \lambda n$ where $n = 0,1,2$ -----

- Destructive interference for a double slit is the path length difference must be a half-integral multiple of the wavelength.

$P = (n + \frac{1}{2})\lambda$ Where $n = 0,1,2,3$ ----

Answer:- B

11. Power of Lens $= \frac{1}{Focal length}$

The power of the lens is calculated in Diopters (D) if the focal length is given in meters. Based on the above equation the statement shown on choose B is not correct.

Answer:- B

12. Examples of scalar quantities are: -

- Length, area, volume, mass, density, energy, entropy, work, power

Examples of vector physical quantities are; -

- Displacement, velocity, acceleration, momentum, force, weight, thrust,

Answer:- D

13. Significant figures

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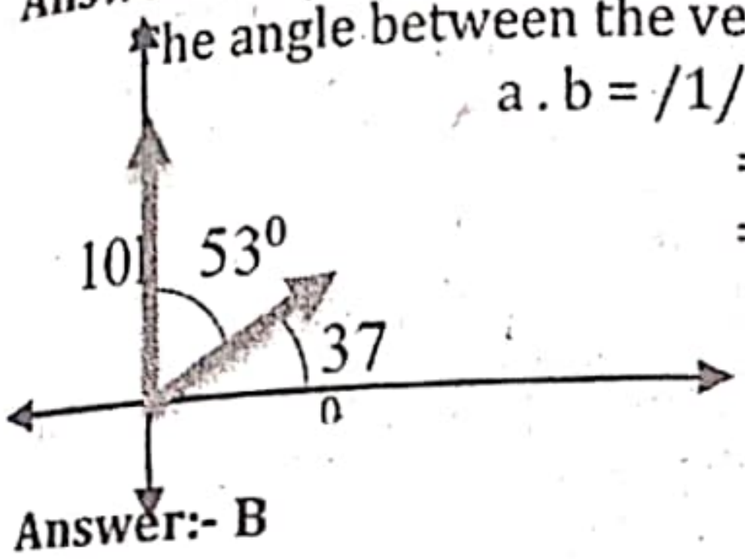
- All non-zero numbers ARE significant
  Eg;- 33.2 has three significant figures
- Zeros between two non-zero digits ARE significant
  Eg:- 3042 has four significant figure
- Leading zeros are NOT significant
  Eg:- 0.65has only two significant figure
- Trailing zeros to the right of the decimal ARE significant
  Eg:- 86.00 has four significant figure
- Trailing zeros in a whole number with the decimal shown ARE significant figure
  Eg:- 650 has three significant figure

Answer:- A

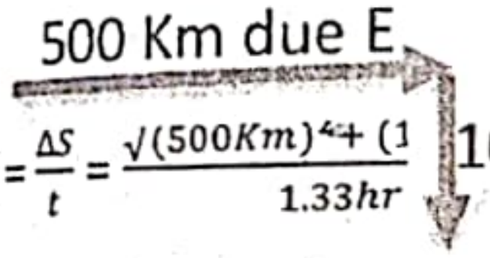
14. The angle between the vectors is 53^0 as shown below



$$a \cdot b = |1|\,|b|\ \cos\theta$$
$$= 10 \times 6 \times \cos 53^0$$
$$= 60 \times 0.6$$
$$= 36\ u^2$$

Answer:- B

15. 500 Km due E



$$V_{av} = \frac{\Delta S}{t} = \frac{\sqrt{(500Km)^2 + (1}}{1.33hr}$$

100 Km due $\quad 5\sqrt{26}\ Km/hr$

Answer:- D

16. For a bullet fired with an initial speed u at an angle θ with the horizontal.

A. Total time of flight = 2T where $T = \frac{usin\theta}{g}$

B. Maximum range is achieved when

$$R = u2 \sin\frac{2\theta}{g}$$

at $\theta = 45^0$ or $\sin 2\theta = 1$

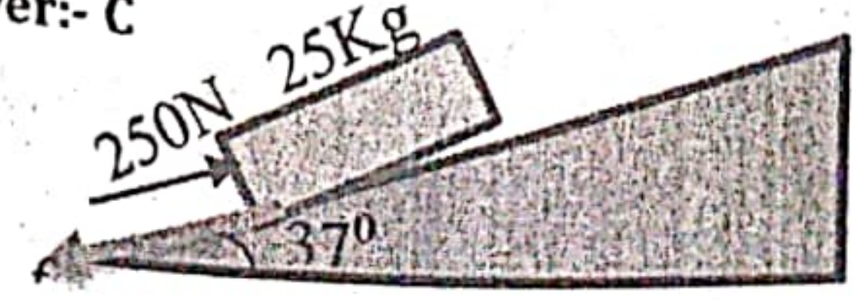
C. Maximum height achieved when

$$h\ max = \frac{u^2 sin2_\theta}{29}$$

h max is directly proportional to initial speed u.

Answer:- C

17.



To find the frictional force exerts on the block and in the inclined plane

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From figure above we can extract a mathematical equation

$ma + mg\sin\theta + f = F$

but a = 0 as the object moves with constant speed

$mg\sin\theta + f = F$

$f = F - mg\sin\theta$

$= 250N - 25Kg \times 10\frac{m}{s^2} \times \sin 37^0$

$= 250N - 150N$

$= 100N$ downward the plane

Answer:- C

18. An elastic or 'bouncy' collision is one where kinetic energy is conserved, or the same before and after a collision. In an inelastic, or 'sticky' collision, there is no kinetic energy conserved it's absorbed by its surroundings. As a result, both objects stick together and move forward as one. From the given three collision examples, *I* is elastic collision but *II* and *III* are showing inelastic collision.

Answer;- A

19. When the spring-mass system attains the extreme position 'P' it will stop and its K.E is transformed into elastic potential energy.

When the spring-mass system attains at 'O' (at the equilibrium position) its elastic potential energy is transformed into K.E and the velocity of the object becomes maximum.

When the spring-mass system attains at Q has both K.E and elastic potential energy.

Answer;- D

20. Given $\vec{F} = (\hat{i} - 3\hat{k})N$ required $\tau = ?$

$\hat{r} = 4\hat{i}m$

Solution;-

Torque is a vector quantity which is the cross product of $\hat{F}$ and $\hat{r}$

$\tau = \hat{F} \times \hat{r}$

$= (\hat{i} - 3\hat{k}) \times 4\hat{i}$

$= (\hat{i} - 4i) - (3\hat{k} - 4\hat{i})$

$= 0 - 12\hat{j} = -12\hat{j}$

Answer;- B

21. Coherent sources of light are those sources that emit a light wave having the same frequency, wavelength and in the same phase, or they have a constant phase difference. A coherent source forms sustained interference patterns when the superimposition of waves occurs, and the positions of maxima and minima are fixed. Two independent sources are never coherent, or they cannot be considered coherent sources.

Answer;- D

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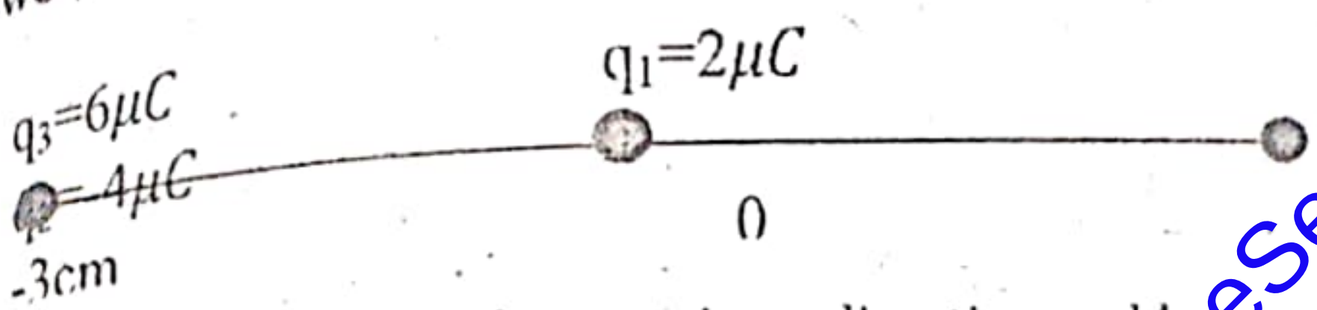
23. Because charges are always conserved, the amount of charge flowing into a junction must come out, i.e., there should not be any charges stored at the junction. Hence Kirchhoff's junction law is a consequence of the law of conservation of electric charge.

Answer;- A

23. The photoelectric effect is the emission of electrons when electromagnetic radiation, such as light hits a material. The photons of a light beam have a characteristic energy called photon energy, which is proportional to the frequency of the light.

Answer;- C

24.



The force on q_1 due to q_2 is in the negative x direction and has a magnitude of

$$F_{21} = K\frac{q_2q_1}{r_{21}^2} = (9 \times 10^9 \frac{Nm^2}{C^2})(\frac{(-4X10^{-6}C)(2X\ 10^{-6}C)}{(0{,}03m)^2} = -80N$$

The force on q_1 due to q_3 is in the positive x direction. Its magnitude is

$$F_{31} = K\frac{q_3q_1}{r_{31}^2} = (9 \times 10^9 \frac{Nm^2}{C^2})(\frac{(6X10^{-6}C)(2X\ 10^{-6}C)}{(-0.03m)^2} = 120N$$

The resultant force on q_1 due to q_2 and q_3 is

$F = F_{21} + F_{31} = -80N + 120N = 40N$ in the positive x direction.

Answer;- A

25. - The gravitational potential energy of mass m at a point which is at a distance r from a source mass is given by $P.E = -G\frac{M_m}{r}$

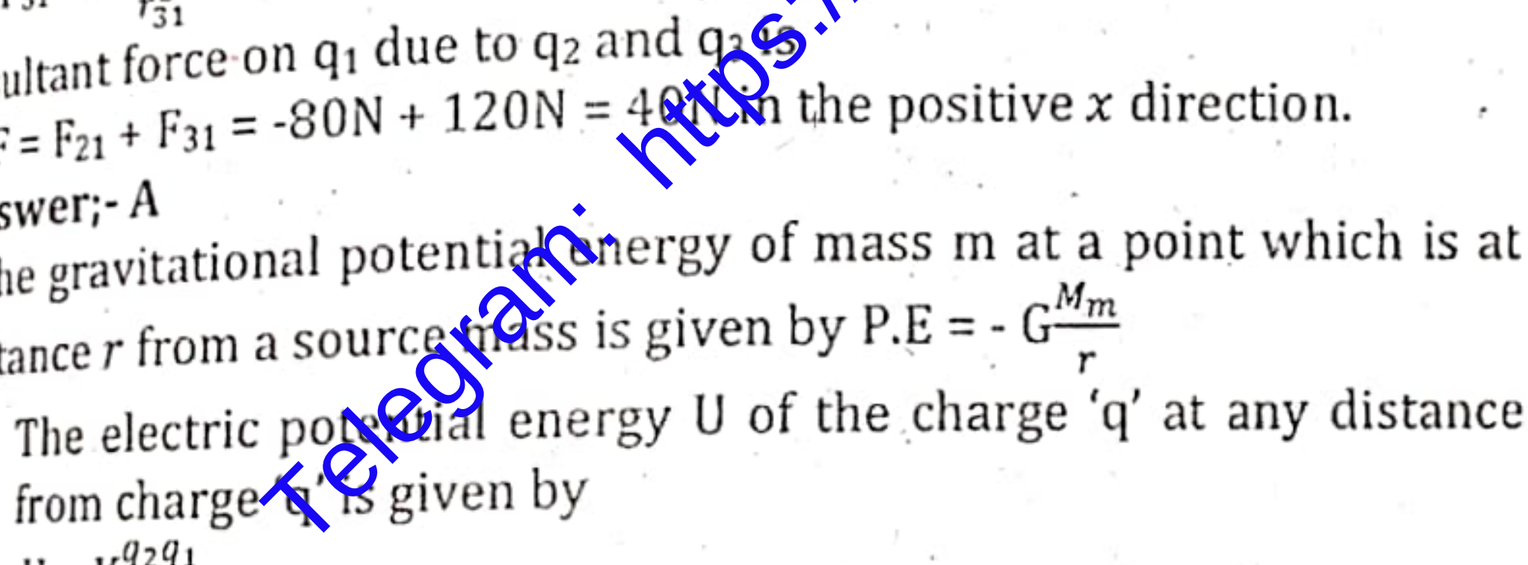
- The electric potential energy U of the charge 'q' at any distance 'r' from charge 'q' is given by

  $U = K\frac{q_2q_1}{r}$

- From the given alternatives only choose B is correct.

Answer;- B

26.



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R_1 and R_2 are connected in parallel

$$\frac{1}{R_{12}} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{3\Omega} + \frac{1}{6\Omega} = \frac{2+1}{6} = \frac{3}{6\Omega}$$

$R_{12} = 2\Omega$

R_3 and R_4 are connected in series

$R_T = R_{12} + R_3 + R_4$

$= 2\Omega + 1\Omega + 4\Omega$

$= 7\Omega$

$\therefore I_T = I_{12} = I_3 = I_4$

$I_T = \frac{V_T}{R_T} = \frac{14V}{7\Omega} = 2A$

$I_3 = I_4 = I_{12} = 2A$

$V_4 = I_4 R_4 = 2Ax4\Omega = 8V$

$V_3 = I_3 R_3 = 2A \times 1\Omega = 2V$

$V_T = V_{12} + V_3 + V_4$

$14V = V_{12} + 2V + 8V$

$V_{12} = 14V - 10v = 4V$

$V_{12} = V_1 = V_2 = 4V$ since R_1 & R_2 parallel circuit

$\therefore I_1 = \frac{V_1}{R_1} = \frac{4V}{3\Omega} = \frac{4}{3} A$

Current through 3Ω resistor is $4/3A$

The voltage drops across 4Ω resistor is 8V

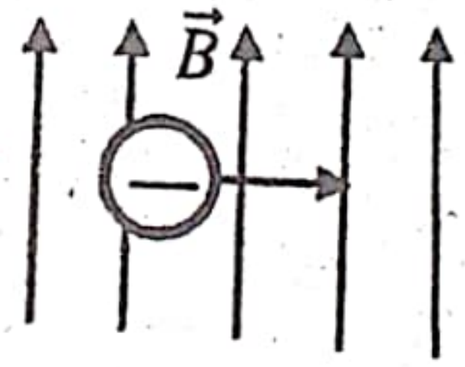
Answer;- A

27.

Given;- $V = 2.2 \times 10^7 \, m/s \, \hat{\imath}$

$\hat{B} = 5 \times 10^{-4} T\hat{\jmath}$

$q = e = 1.6 \times 10^{-19} C$



Solution;- The magnetic force experienced by the electron is given by

$F_m = -e\hat{V} \times \hat{B}$

$= -1.6 \times 10^{-19} \, m/s \, \hat{\imath} \times 2.2 \times 10^7 \, m/s \, \hat{\imath} \times 5 \times 10^{-4} T\hat{\jmath}$

$= (-1.76 \times 10^{-15}) N\hat{R}$

it is perpendicular to north and east direction.

Answer;- C

28. Applying Faraday's law, we can write that the magnitude of ε, the magnitude of the emf induced, equals the number of turns in the coil times $\Delta\phi$ sub Bovert $\Delta\phi$ sub, we've seen, can be written as ΔB times A.

Given;- $\hat{B} = 4 \times 10^{-4} T$

$A = 100 \, cm^2 = 10^{-2} m^2$

$N = 5 \times 10^4$

$\Delta t = 2ms$

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Solution;- $\varepsilon = N\frac{\Delta B}{\Delta t}A = \frac{5x10^{4}x\,4x10^{-4}T\,x\,10^{-2}m^{2}}{2\,x\,10^{-3}s} = 100V$

Answer;- D

29. The resonant frequency, f ,of a circuit with inductance L and capacitance C is given by

$$w = 2\pi f = \sqrt{\frac{1}{LC}}$$

Given ;- L $4\mu H$ and C = 9PF

Solution;- $w = 2\pi f = \sqrt{\frac{1}{4\,x\,10^{-6}H\,x\,9\,x\,10^{-12}F}} = \sqrt{\frac{1}{36\,x\,10^{-18}}}$ rad/sec

$= 1/6 \times 10^{9}$ rad/sec $= 1.67 \times 10^{8}$ rad/sec

Answer;- B

30. $\Delta E = -0.81eV - (-2.21eV) = 1.4eV$

The difference in enery between the two energy levels are 1.4

To find- The frequency of radiation emitted when the atom makes a transition between the energy levels

The energy between the energy levels is

$E = 1.4eV = 1.4 \times 1.6 \times 10^{-19}J$

The frequency of the radiation emitted during transition is

$E = h\mathcal{V}$

$\mathcal{V} = \frac{E}{h} = \frac{1.4\,x\,1.6\,x\,10^{-19}}{6.626\,x\,10^{34}} = 0.338 \times 10^{15}Hz = 3.38 \times 10^{14}Hz$

Answer;. C

31. The SI unit of the electric field is volts per meter (V/m). This unit is equivalent to Newton per coulomb (N/C). As Joule is equal to Newton meter, Joule per meter coulomb (J/m.C) is also unit of energy.

Answer;- A

32. Magnetic nature of a magnetic material can lose or magnetic molecules are oriented randomly when heated and/or hammered.

Answer;- D

33. A P-type semiconductor is a type of extrinsic semiconductor which is obtained by adding a trivalent impurity in an intrinsic semiconductor. Therefore, the majority charge carriers in the P-type semiconductor holes.

An N-type semiconductor is also extrinsic semiconductor, which is obtained by adding a pentavalent impurity in an intrinsic semiconductor. In case N-type semiconductor, the majority charge carriers are electrons.

Answer;- D

34. The Intensity of sound can be calculated using $I = \frac{P}{4\pi R^{2}}$

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Let us calculate power first

$P = 14\pi R^2 = 5\,{}^{W}/_{m^2}\, x\,(3m^2) x\, 4\pi = 180\pi W$

The intensity of sound $5/4 W/m^2$ at

$$R = \left(\sqrt{\frac{180\pi}{4\pi x \frac{5}{4}}}\right) = \sqrt{36} m = 6m$$

Answer;- C

35. The center of mass between two-point masses can be calculated

$x_{cm} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$

Given - position of m_1 = (4, 5) cm

- Position of m_2 = (6, 3) cm
- $m_1 = 0.5 m_2$ $m_2 = 2m_1$

Solution;-

$$x_{cm} = \frac{m_1 x\, 4cm + 2m_1 x\, 6cm}{m_1 + m_2} = \frac{16m_1}{m_1 + 2m_1} = \frac{16}{3} cm$$

$$y_{cm} = \frac{m_1 x\, 5cm + 2m_1 x\, 3cm}{m_1 + 2m_1} = \frac{11m_1}{3m_1} = \frac{11}{3} cm$$

∴ The position of center of mass of the system is

$$(x_{com}, y_{com}) = \left(\frac{16}{3} cm, \frac{11}{3} cm\right)$$

Answer;- D

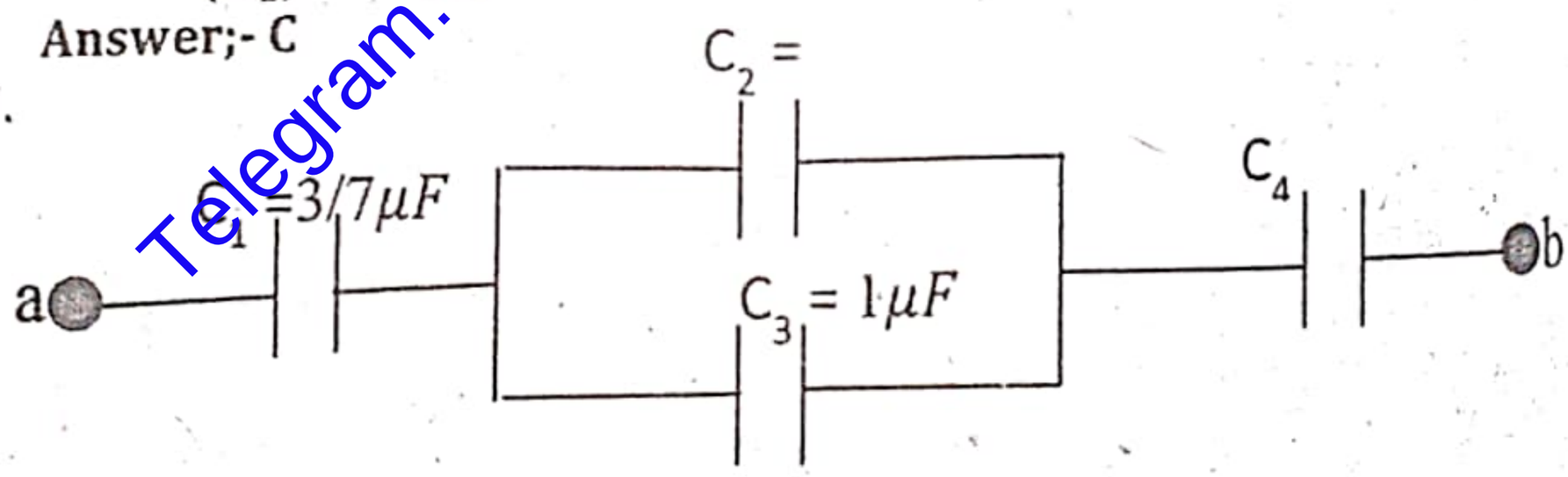
36. Gravitational acceleration on the surface of the Earth is $g_E = G\frac{M_E}{(r_E)^2}$

Therefore, g at $2r_E$ is

$$g_{2rE} = G\frac{M_E}{(2r_E)^2} = G\frac{M_E}{4r_E^2} = \frac{1}{4} g_E = g_{E/4}$$

Answer;- C

37.



C_2& C_3 are connected in parallel

$C_{23} = C_2 + C_3 = 2\mu F + 1\mu F = 3\mu F$

C_1, C_{23} & C_3 are connected in series

$$\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_{23}} + \frac{1}{C_3} = \frac{1}{\frac{3}{7}\mu F} + \frac{1}{3\mu F} + \frac{1}{\frac{3}{4}\mu F} = \frac{12}{3}\mu F$$

$$C_T = \frac{3}{12}\mu F = 1/4\mu F$$

Answer;- C

38. Total Resistance of the circuit is

$$R_T = \frac{V_T}{I_T} = \frac{64V}{8mA} = 8K\Omega$$

The external Resistance is

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$R_T = R_1 + R_2 = 3K\Omega + 4K\Omega = 7K\Omega$

Internal resistance of the battery is

$R_I = R_T - R_e = 8K\Omega - 7K\Omega = 1K\Omega$

Answer; – B

39. Magnetic field strength, B, inside the center of solenoid is found using the equation; $\vec{B} = \frac{\mu 0 NI}{L}$

Given ; –

$N = 5000$

$L = 1m$

$I = 15A$

$\mu o = 4\pi \times 10^{-7}\, T.m/A$

Answer;- D

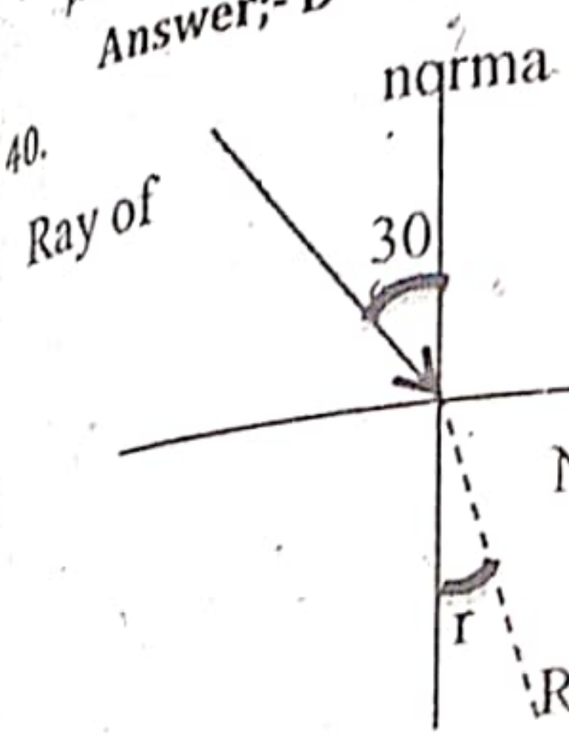
Solution; -

$$\vec{B} = \frac{4\pi \times 10^{-7} T.m/A \times 5000 \times 15A}{1m}$$

$= 300000\pi \times 10^{-7}T$

$= 3.0\pi \times 10^{-2}T$

40.



$n_1 \sin\theta_1 = n_2 . \sin\theta_2$

$\frac{n_2}{n_1} = \frac{\sin\theta_i}{\sin\theta_r}$

$n_{air} = \sin\theta_r = \frac{n_1}{n_2}\sin\theta_i$

$= \frac{1}{4/3}\sin 30^0 = \frac{3}{4x0.5} = 0.375$

$N_{water} = \frac{n_1}{n_2} = \frac{V_w}{V_a}$

$\frac{V_w}{V_a} = \frac{1}{4/3} = \frac{3}{4}$

$V_w = 3/4V_1 = 0.75Va$

Answer;- A

41. The number of action reaction pair of forces are two in number. These are 1) between the floor and the foot and
2) between the foot and the wall

Answer;- B

42. The amount of heat energy required to change a substance from a liquid to a gaseous state is known as latent heat of vaporization.

Answer;- A

43. A mechanical wave is a wave that is an oscillation of matter and is responsible for the transfer of energy through a medium.

- ***Frequency*** is the number of completed wave cycles in a given period
- ***Wavelength*** is the distance traveled by the wav during one complete oscillation
- ***Period;*** - the time taken by the wave to complete one oscillation
- ***Amplitude;*** - the maximum displacement of the wave from its mean or equilibrium position

Answer;- B

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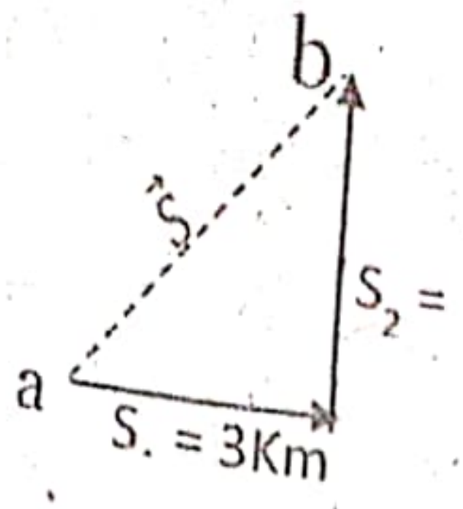


44. The displacement between a and b

$\vec{S} = \sqrt{(3Km)^2 + (4Km)^2} = 5.0Km$

Answer;- D

45. Based on the given graph, during the first 6s the velocity is constant and has zero acceleration. The motion during 6s to 10s velocity is changing and has positive acceleration, The motion during 10s to 14s velocity is changing and has negative acceleration. The displacement during the time interval t=0 to t=6s is equal to the area under the velocity-time graph in the given interval.

Answer;- C

46. Given;- $m = 1.5 \times 10^3$ Kg $V_i = -15.0$m/s $V_f = 2.6$ m/s $\Delta t = 0.15s$

Solution;-

- Force is the rate of change of momentum and is mathematically described as

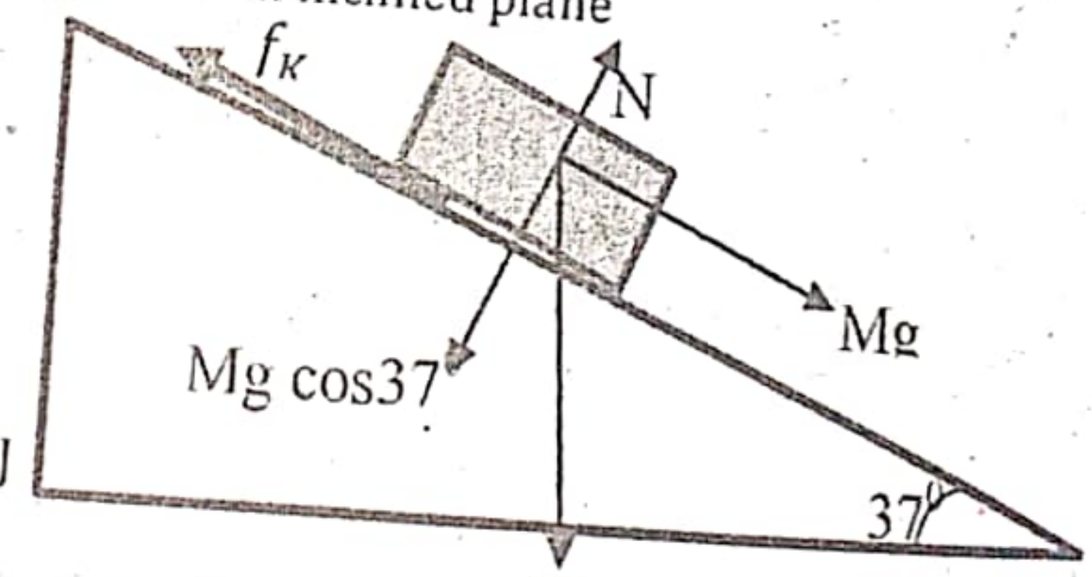
$$F = \frac{P_f - P_i}{\Delta t} = \frac{mV - mU}{\Delta t} = \frac{1.5x10^3 Kg \times 2.6^{m}/_{s} - 1.5x10^3 x(-1.5.0)^{m}/_{s}}{0.15s}$$

$= 1.76 \times 10^5$N

Answer;- A

47. Given;- $U = 0$ $m = 5.0$Kg $\mu_k = 0.44$ $S = 2.5$m $\theta = 37^o$

Solution; - Work done by friction on an inclined plane



N = mgcos37⁰

$f_k = \mu_k N$

$= 0.44 \times 5 \times 10 \times \cos 37^0$

$= 17.56$ N

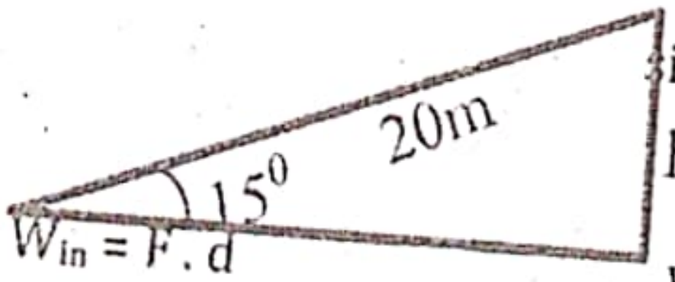
Work done by friction

$W_f = \vec{f_k} \cdot \vec{s} = f_k s \cos 180^0$

$= 17.569$N x 2.5m x -1 = -44J

Answer;- B

48. Given; - $\ell = 20m$ $\theta = 15^0$ $E = 900N$ $w = 3{,}000N$



$\sin 15^0 = \frac{h}{20m}$ h = 5.176m

$W_{in} = F \cdot d$

$W_{out} = mgh = w \times h$

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$= 900N \times 20m$

$= 18000J$

$= 3000N \times 5{,}176m$

$= 15529J$

$$\eta = \frac{W_{out}}{W_{in}} \times 100\% = \frac{15529J}{18000J} \times 100\% = 86.27\%$$

Answer;- C

Given;- $V = 5.00 \times 10^{4}\ m^3$ $\quad m = 0.25Kg$

Solution;-

Since the wood is totally submerged, the volume of displaced water is the volume of the wood. The systems in equilibrium that is the net force is zero. Downward force is due to the gravitational force on the steel and the wood. Upward force is the buoyancy.

Let $\rho_{wat} = 1000\ {}^{Kg}/_{m^3}$

$$\rho_{wat} Vg = \rho_{wood} vg + mg$$

$$\rho_{wat} = \rho_{wood} + \frac{m}{v}$$

$$\rho_{wood} = \rho_{wat} - \frac{m}{v} = 1000\text{Kg} - \frac{0.25Kg}{5x10^{-4}m^3} = 500\text{Kg/m}^3$$

Answer;- B

When the bimetallic strip is heated, brass expands more than iron due to higher thermal coefficient than steal. Higher thermal expansion coefficient means that substance will expand more as compared to any other substance per degree rise in temperature, and hence due to this phenomenon brass side shows higher expansion especially linear expansion than iron and hence forcing the whole bimetallic strip to bend.

Answer;- B

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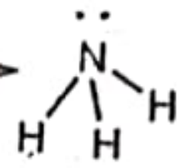
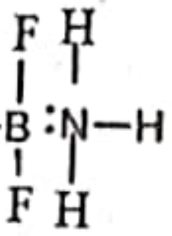
1. Which of the following postulates of Dalton's atomic theory was modified by the modern atomic theory?
   A. Atoms are the smallest particles of all elements.
   B. Atoms are indivisible and indestructible.
   C. Atoms are neither created nor destroyed in chemical reactions.
   D. Compounds are formed when atoms of more than one element combine.
2. Which of the following includes all of the known atomic models?
   A. Bohr Model, Quantum Mechanical Model, Dalton's Atomic Model, Valance Bond Model
   B. Bohr Model, the Quantum Mechanical Model, Thomson Model, Rutherford Model
   C. Bohr Model, the Quantum Mechanical Model, Rutherford Model, Thomson Model, Dalton's Atomic Model.
   D. Bohr Model, Thomson Model, Rutherford Model, Dalton's Atomic Model, de Broglie Model.
3. The electron configuration of Cu(Z=29) is
   A. $[Ar]4s^23p^8$ B. $[Ar]4s^23d^9$ C. $[Kr]4s^23d^9$ D. $[Ar]4s^13d^{10}$
4. What does Mendeleev's Periodic Law state? Properties of elements are periodic function of their
   A. atomic number
   B. atomic mass
   C. electron configuration
   D. atomic radius
5. Among the following reactions which one shows a coordinate covalent bond formation?

   A. $:\dot{N}\cdot + 3H\cdot \longrightarrow$ 

   C. $H\cdot + \cdot\ddot{Cl}: \longrightarrow H:\ddot{Cl}:$

   B. $BF_3 + NH_3 \longrightarrow$ 

   D. $H\cdot + \cdot HH \longrightarrow H:H$
6. Which statement describes the formation of covalent bond? In covalent bond electrons are
   A. shared between metallic atoms
   B. transferred from metallic atoms to non-metallic atoms
   C. shared between non-metallic atoms
   D. transferred from non-metallic atoms to metallic atoms
7. How do elements of group IIA form ions?
   A. By gaining one electron
   B. By losing one electron
   C. By losing 2 electrons
   D. By gaining 2 electrons

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8. How is hydrogen bondinding formed? It is formed when a hydrogen atom is bonded to
   A. highly electronegative elements, fluorine, chlorine and oxygen.
   B. highly electronegative elements, fluorine, oxygen and phosphorus.
   C. highly electronegative elements, chlorine, oxygen and nitrogen.
   D. highly electronegative elements, fluorine, oxygen and nitrogen.
9. Which of the following forms of energy change occurs from the combustion of carbon in oxygen is
   A. light energy B. electrical energy C. heat energy D. solar energy
10. What is the molar volume of a gas?
   A. The volume of a gas at STP (standard temperature 0°C and pressure, 1atm.), which is equal to 22.4L/mol.
   B. The volume of a gas at STP (standard temperature 25°C and pressure, 1 atm.), which is equal to 24L/mol.
   C. The volume of a gas at STP (standard temperature 0°C and pressure, 1kPa), which is equal to 22.4L/mol.
   D. The volume of a gas at STP (standard temperature 25°C and pressure, 1 kPa), which is equal to 22.4L/mol?
11. What is the mole ratio of oxygen gas to carbon dioxide gas in the following reaction? $C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(l)$
   A. 10 mole oxygen gas to 6 mole carbon dioxide
   B. 5 mole oxygen gas to 3 mole carbon dioxide
   C. 2 mole oxygen gas to 3 mole carbon dioxide
   D. 7 mole oxygen gas to 5 mole carbon dioxide
12. Consider the following chemical reaction for combustion of ethane (C_2H_6) in the presence of oxygen to produce carbon dioxide and water. $C_2H_6(g) + 7O_2(g) \rightarrow 4CO_2(g) + 6H_2O(l)$ In this reaction:
   A. the rate of consumption of ethane is 7 times faster than the rate of consumption of oxygen.
   B. the rate of formation of CO_2 equals the rate of formation of water?
   C. water is formed at a rate equal to 2/3 the rate of formation of CO_2
   D. CO_2 is formed twice as fast as ethane is consumed.
13. Which of the following factors **DOES NOT** affect rate of a reaction?
   A. Concentration B. Surface area C. Temperature D. Enthalpy
14. How does a catalyst increase the rate of a reaction?
   A. By forming intermediate complex
   B. By increasing the activation energy
   C. By lowering the activation energy
   D. By changing the equilibrium constant
15. Which of the following statements correctly describes Boyele's Law?
   A. The volume of a gas at a given temperature is directly proportional to the applied pressure.

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B. The volume of a gas at a given temperature is inversely proportion to the applied pressure.
C. The volume of a gas at a given temperature is proportion to the square root of the applied pressure.
D. The volume of a gas at a given temperature is inversely proportional to the square root of the applied pressure.

16. Which of the following assumption of the kinetic molecular theory of gases explains the high compressibility of gases?
A. The actual volume of gas molecules is negligible
B. There is no force of attraction between gas molecules
C. Particles are always in random motions
D. Different particles have different speeds

17. Consider a bottle containg water, water vapor and air in equilibrium. If the temperature decreases slightly, the amount of water vapors;
A. and vapor pressure will increase.
B. will increase and vapor pressure will increase.
C. will decrease and vapor pressure will increase.
D. and vapor pressure will decrease.

18. Why do water droplets collect on outside of a glass of water on a hot day?
A. there is heat exchange between water in the glass and the surrounding.
B. Water from the glass starts to vaporize.
C. There will be a condensation of water vapor on the glass.
D. There will be a sublimation of water vapor on the outside of a glass.

19. What are hydrocarbons? Compounds that contain
A. hydrogen and carbon with no benzene ring.
B. hydrogen and carbon only.
C. benzene rings only.
D. benzene and its derivatives only.

20. Which of the following statement is INCORRECT regarding the boiling points of alkanes?
A. Boiling points increases with stronger Van der Waals forces.
B. Surface area is the only factor that determines the boiling points of alkanes.
C. Boiling points of straight chain alkanes is greater than that of branched chain alkanes.
D. The boiling point of cycloalkanes is higher than that of linear alkanes.

21. Which of the following statements is correct regarding alkenes and alkynes?
A. Alkenes alkynes are soluble in polar solvents.
B. Alkenes are soluble in polar solvents.
C. Alkynes have longer bond length less surface area than alkenes.

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D. The molecules of alkenes and alkynes are held together by weak inter-molecular force.

22. What is the general formula of alkenes?
A. $C_nH_{2n-2}(n \geq 2)$ B. $C_nH_{2n-2}(n > 2)$ C. C_nH_{2n} D. C_nH_{2n+2}

23. Which functional group containing compound is named using the suffix "-ol"?
A. Carboxyl B. Aryl C. Hdroxyl D. Phenyl

24. Which of the following represent the general formula of monohydric alcohols?
A. $C_nH_{2n-2}OH$ B. $C_nH_{2n-2}(OH)_2$ C. $C_nH_{2n}OH$ D. $C_nH_{2n+1}OH$

25. What is Lewis acid? It is
A. a proton donor C. an electron pair donor
B. an electron pair acceptor D. a proton acceptor

26. Which of the following is a polyprotic acid?
A. HCl B. H_3PO_4 C. HF D. HNO_3

27. Which of the following does respect a general property of a base?
A. Solutions of strong bases are good conductors of electricity.
B. Aqueous solutions of bases turn blue litmus paper to red.
C. Soluble bases release hydronium ion in aqueous solution.
D. Aqueous solution of bases turns methyl orange in to red.

28. What are salts? salts are
A. Ionic compounds formed by complete replacement of ionizable hydrogen by anions of the aid.
B. ionic compounds that result the reaction between meta hydroxide and water.
C. ionic compounds that derive their cations from a base and their anions from an acid.
D. ionic compounds formed by partial replacement of ionizable hydroxide by metal cations.

29. The process in which a compound absorbs electrical energy and decomposes in to its element is
A. electrolysis B. solvolysis C. hydrolysis D. electroplating

30. Which of the following is true about galvanic cells used in commercial lead storage batteries? In lead storage batteries,
A. galvanic cells are connected in series.
B. galvanic cells are connected in parallel.
C. Half of galvanic cells are connected in series.
D. Half of galvanic cells are connected in parallel.

31. What is the purpose synthetic cryolite in the extraction of aluminum by the Hall's process?
A. Cryolite increases the purity of the extracted aluminum?
B. Cryolite reduces the melting point of alumina.

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C. Cryolite increases the melting point of alumina.
D. Cryolite is used as a catalyst in the extraction of aluminum.

32. Which of the following is the chemical property of aluminum?
A. Aluminum liberates hydrogen gas from dilute acids like HCl and H_2SO_4
B. Aluminum is a grey metal in color.
C. Aluminum is a good conductor of heat and electricity
D. Aluminum has a lower density than any other commercial metals.

33. How is tanning carried out? In tanning the hides or skins are
A. soaked in water for several days.
B. soaked in lime and water for several days.
C. treated with sodium salts.
D. treated with dilute hydrochloric acid to remove the lime.

34. How can a project establistioning an industry that consumes caol affect the environment?
A. Increament in concentration of nitric oxide.
B. Rise in global temperature.
C. Production of high level of lead in the atmosphere.
D. By changing pH of water.

35. Which of the following is a fundamental quantity SI unit?
A. speedB. Force C. Pressure D. Length

36. For the conversion of 3.5 yards in to meter using the factor label method, which of the following is the correct step?
1 yard=3.5 feet; 1 foot=12 inch; 1inch=2.54cm; 1m=10cm
A.(3.5 yards/1)(3.5 feet/1 yard)(12 in/1 foot)(2.54 cm/1 in)(1m/100cm)
B. (3.5 yards/1)(1 yard/3.5 feet)(12 in/1 foot)(2.54 cm/1 in)(1m/100cm)
C. (3.5 yards/1)(3.5 feet/ 1 yard)(12 in/ 1 foot) (1in/2.54 m)(1m/100cm)
D. (3.5 yards/1)(3.5 feet/1 yard) (1 foot/1 12in)(2.54cm/1in) (1m/100cm)

37. In the scientific method, a predictive statement that can be tested experimentally is
A. truth B. evidence C. proof D. hypothesis

38. Which of the following is NOT true about cathode rays? Cathode rays
A. contain particles, which are negatively charged.
B. travel in straight lines from cathode to anode.
C. contain particles, which are positively charged.
D. are common constituents of all matter and their m/e is constant for all gases.

39. Which of the following statements is correct about the energy of a photon?
A. It is directly proportional to the observed frequency of the light.
B. It is inversely proportional to the observed frequency of the light.
C. It is directly related to the intensity of the light.

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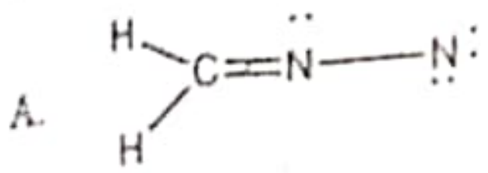
D. It is directly proportional to the wave length of the light.

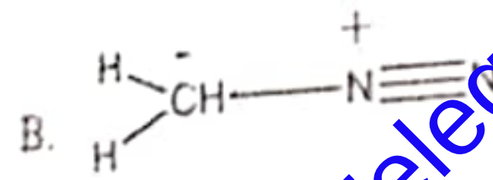
40. Which of the following explains the emission spectra of atoms consisting of a series of fine lines?
A. An electron can have specific energy values in an atom.
B. An electron under goes a transition by energy changes corresponding to photon of specific wave length.
C. The energy of an atoms is continuous with discreet frequencies.
D. An electron that revolves around the nucleus emits radiation.

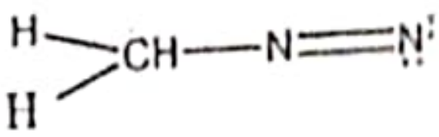
41. What is the significance of electron probability distribution? It tells as about
A. the uncertainty to assign fixed path for electrons.
B. finding an electron in its orbit at particular time.
C. finding the electron very close to the nucleus at all times.
D. the probability of finding an electron is very far away from the nucleus.

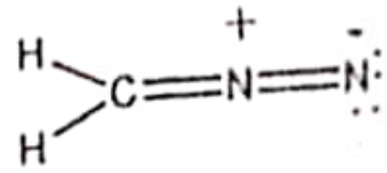
42. Which group in the periodic table contains elements with valance electron configuration of ns^2np^1?
A. I B. II C. III D. IV

43. Which of the following is a correct Lewis structure for diazomethane (CH_2N_2)?

A. 

B. 

C. 

D. 

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44. Which of the following is the property of covalent compounds? Covalent compounds
A. is non volatile.
B. has high melting and boiling points
C. is generally insoluble in water.
D. is crystalline solids at room temperature.

45. Which of the following is a covalent molecule?
A. PCl_5 B. NaCl C. $[NH_4]^+$ D. CO_3^{2-}

46. What are the intermolecular forces? They are forces due to the attraction between
A. molecules.
B. cations and anions
C. nuclei and electron pair.
D. cations and delocalized electrons.

47. Which of the following belongs to chemical bonding theories?
A. Kinetic-molecular theory and valance shell electron pair repulsion theory

B. Valance bond theory and molecular bond theory.
C. Molecular orbital theory and kinetic-molecular theory.
D. Valence bond theory and valance shell electron pair repulsion theory.

48. In which of the following molecules is the central atom sp^2 hybridized?
A. CO_2 B. N_2O C. SO_2 D. CO

49. For the reaction: $^1/_2 A \rightarrow 2B$ the rate disappearance of 'A' is related to the appearance of 'B' by the expression:
A. $-\frac{\Delta[A]}{\Delta t} = \frac{\Delta[B]}{\Delta t}$ B. $-\frac{\Delta[A]}{\Delta t} = \frac{4\Delta[B]}{\Delta t}$ C. $-\frac{\Delta[A]}{\Delta t} = {}^1/_2\frac{\Delta[B]}{\Delta t}$ D. $-\frac{\Delta[A]}{\Delta t} = {}^1/_4\frac{\Delta[B]}{\Delta t}$

50. The half-life of a reaction is the time taken for the
A. reactant concentration to decrease by 50% of its initial value.
B. rate constant to increase by 50% of its initial value.
C. rate constant to decrease by 50% of its initial value.
D. complete conversion of a reactant into a product.

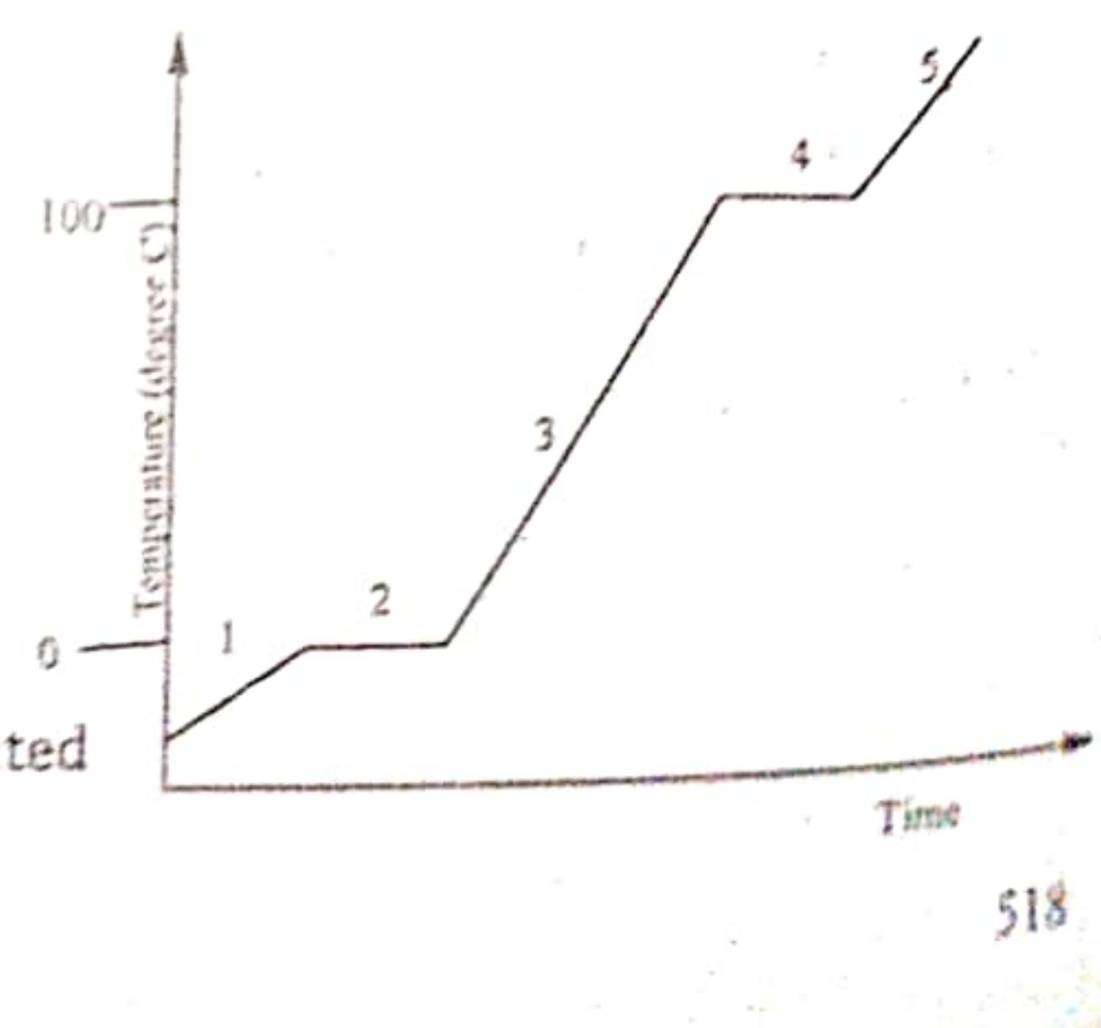
51. In the first order decomposition of dinitrogen pentaoxide at 335K $N_2O_5(g) \rightarrow 2NO_2(g) + \frac{1}{2} O_2(g)$: A reaction was carried out with 2.5 g of N_2O_5 sample at 335K and after 109 seconds, 1.50g of N_2O_5 remained. What is the value of the rate constant?
A. $2.25\times10^{-5}s^{-1}$ B. $4.69\times10^{-3}s^{-1}$ C. $6.25\times10^{-2}s^{-1}$ D. $7.63\times10^{-2}s^{-1}$

52. What is the difference between reversible and irreversible reactions?
A. In reversible reactions, there is equilibrium between reactants and products.
B. In irreversible reactions, there is equilibrium between reactants and products.
C. In irreversible reactions, the reaction goes to completion.
D. In irreversible reactions, there is a mixture of products and reactants.

53. Consider the reactions all given bellow are in gaseous state. For which of the following reactions is K_p less than K_c?
A. $2HI \rightleftharpoons H_2+I_2$ B. $N_2+O_2 \rightleftharpoons 2NO$ C. $N_2O_4 \rightleftharpoons 2NO_2$ D. $2SO_2+O_2 \rightleftharpoons 2SO_3$

54. The heating curve shown below shows a substance in the solid, liquid and gas phase. Which segment represents only the liquid phase?

A. 1 B. 2 C. 3 D. 5



55. The general formula for saturated monocarboxylic acids is

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A. RCHO B. ROR C. RCOOH D. RCOR

56. Which of the following carboxylic acids is a solid substance at room temperature?
A. Heptanoic acid B. Octanoic acid C. Nonanoic acid D. Decanoic acid

57. What is the IUPAC name of an ester having the molecular formula $C_6H_{12}O_2$?
A. Ethyl butanoate
B.Methyl butyrate
C. Ethyl butyrate
D. Pentyl butyrate

58. What is the formula of pentanoic acid?
A. $CH_3(CH_2)_4COOH$
B. $CH_3(CH_2)_3COOH$
C. $CH_3(CH_2)COOH$
D. $CH_3(CH2)_5COOH$

59. What is the rate of dissolution? It is
A. the rate at which a solid turns to liquid.
B. the speed at which a solid substrate goes into solution.
C. the rate at which a solid dissociate.
D. the speed at which a solid sublimes.

60. Which one of the following is correctly stated regarding solubility as an equilibrium process?
A. Unsaturated solution is a solution that is in equilibrium with respect to a given dissolved substance.
B. Saturated solution is a solution in which more of a solute can be dissolved.
C. Super saturated solution is a solution that contains more than the equilibrium amount of dissolved solute.
D. Supersaturated solution is more stable than saturated solution.

61. What is the concentration of a solution? It is
A. the quantity of solute in a given quantity of solution.
B. the amount of solvent needed to dissolve a solute.
C. the amount of solute needed to make a concentrated solution.
D. number of moles of solute and number of moles of solvent.

62. How many moles of $C_{12}H_{22}O_{11}$ (Sucrose molecular weight=342.91g/mol) is needed there to prepare 2.0L of 0.3M solution?
A. 0.15 B. 0.43 C. 0.6 D. 6.6

63. At 75°C , the vopor pressure of pure water is 39 kPa. Which one of the vapor pressure of a solution obtained by dissolving 34.2 g sucrose in 180 mL of water?(Assume that density of water is 1.000g/mL)
A. 31. 2kPa B. 38.6 kPa C. 41.1 kPa D. 44. 5 kPa

64. Which of the following statement is NOT true about colligative properties electrolysis and non-electrolytes?
A. the effect of electrolytes on colligative properties is very high compared to non-electrolytes.

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B. The colligative properties of non-electrolytes are not considerably changed.
C. The effect of non-electrolytes on colligative properties is very low compared to electrolytes.
D. The effect of non-electrolytes on colligative properties is the same as those of electrolytes.

65. For the reaction shown below, which of the following is a conjugate acid-base pair? $HC_2O_4^-(aq) + H_2O(l) \rightleftharpoons H_3O^+ + C_2O_4^{2-}$
A. $HC_2O_4^-$ and H_2O
B. H_2O and $C_2O_4^{2-}$
C. H_3O^+ and $HC_2O_4^-$
D. $HC_2O_4^-$ and $C_2O_4^{2-}$

66. The self-ionization of water can be expressed by the equation:
$H_2O(l) + H_2O(l) \rightleftharpoons H_3O^+(aq) + H_3O^-(aq)$
Which of the following is the correct expression of the ionic product of water?
A. $K_w=K_c(H_2O)^2$
B. $K_w=K_c(H_2O)$
C. $K_w=[H_3O^+][OH^-]$
D. $K_w=[H_3O^+][OH^-]/[H_2O]$

67. What is the effect of a common ion in an electrolytic solution?
A. It suppresses the imitation of a weak acid and raises the pH.
B. It suppresses the imitation of a weak base and lowers the pOH.
C. It decreases the solubility of a solute.
D. It has no effect on the pH of a solution.

68. Sodium carbonate is a basic salt because it is a salt of
A. a strong base and a weak acid and the anion of the weak acid can be hydrolyzed.
B. a strong base and a strong acid and the anion of the strong acid can be hydrolyzed.
C. a weak acid and a weak base and the cation of the weak base can be hydrolyzed.
D. a strong acid and a weak base and the anion of the strong acid can be hydrolyzed.

69. Which of the following statements is true for an acid-base titration?
A. Indicators catalyze an acid-base titration to show the inflection point of the titration curve.
B. Acid-base titration cannot be carried out in the absence of indicators.
C. An acid-base titration is a method of quantitative analysis for determining of the concentration of an acid or base.
D. An acid-base titration is a method of qualitative analysis for determing of the concentration of an acid or base.

70. What is the change in internal energy (ΔE) for a reaction, in which a system absorbs 650J of heat and 125J of work is done on the system?
A. 775J B. 535J C. -775J D. -535J

71. What is enthalpy change? It is

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A. the amount of heat given out in a reaction.
B. the amount of heat given out absorbed in a reaction at constant pressure.
C. the amount of heat given out or absorbed in a reaction at constant pressure and volume.
D. the amount of heat absorbed in a reaction.

72. What is oxidation? Oxidation is
A. the addition of hydrogen to a substance and removal oxygen from a substance.
B. the loss of electrons by a species, accompanied by an increase in oxidation number.
C. the addition of electron(s) to the oxidizing agent.
D. the reduction of the oxidation state of oxidizing agent.

73. What is meant by preferential discharge, if a number of ions are present in a solution? It means that the one requires
A. least energy will be liberated first at a given electrode.
B. least energy will be liberated last at a given electrode.
C. highest energy will be liberated first at a given electrode.
D. highest energy will be liberated last at a given electrode.

74. Faraday's second law states:
A. passing the same quantity of electricity through solution of different electrolytes produces masses of substances that are directly proportional to their molecular masses.
B. passing the same quantity of electricity through solution of different electrolytes produces masses of substances that are directly proportional to their equivalent masses.
C. the amount of substance consumed or produced at an electrode is directly proportional to the quantity of the electricity passing through the solution.
D. the amount of substance consumed or produced at an electrode is inversely proportional to the quantity of electricity passing through the solution.

75. Which of the following is true about Zn-Cu voltaic cell?
A. The anode half-cell consist of Cu electrode dipped in to a Cu^{2+} solution.
B. After the cell runs for a several hours, the Zn electrode weight less.
C. The electrons flow from right to left through the external circuit.
D. As the cell operates, electrons are continuously generated at the cathode and consumed at the anode.

76. Which of the following statements is NOT correct about minerals and ores?
A. All minerals are ores.

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B. All ores are minerals.
C. Minerals are the natural forms in which various materials exist.
D. Ores are usually used to extract metals economically.

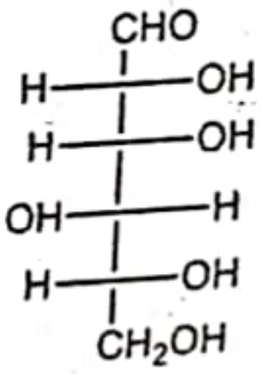
77. Why is $CaCl_2$ added to NaCl in in a 1:1½ ratio in in the Down's cell for the manufacturing of sodium metal?
A. $CaCl_2$ is used as a catalyst.
B. $CaCl_2$ is used to lower melting temperature of NaCl.
C. $CaCl_2$ is used to lower density of the mixture.
D. $CaCl_2$ is used as an electrolyte.

78. Which of the following is NOT a major step in metallurgical processes?
A. Mining the ore
B. Concentrating the ore
C. Pretreatment
D. Extracting the metal and refining

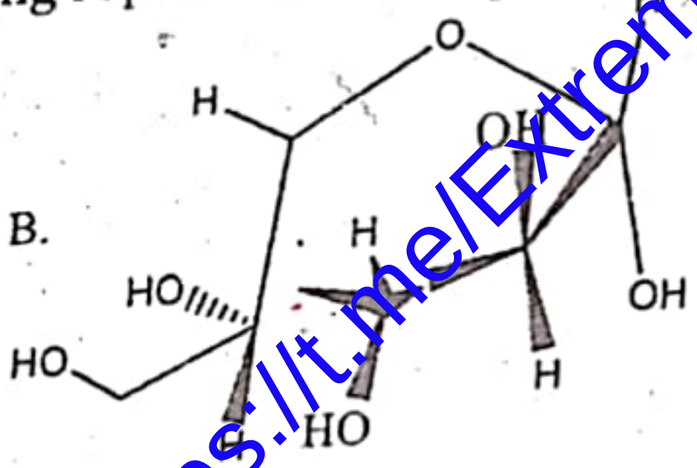
79. Which monomers are used for the synthesis of Teflon?
A. Isoprene B. Propylene C. Styrene and phenol D. Tetrafluoroethylene

80. Which of the following represents the structure of a disaccharide?

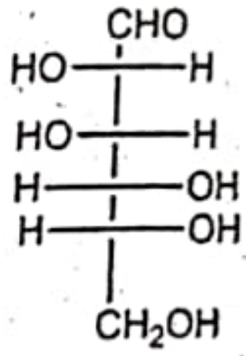
A.



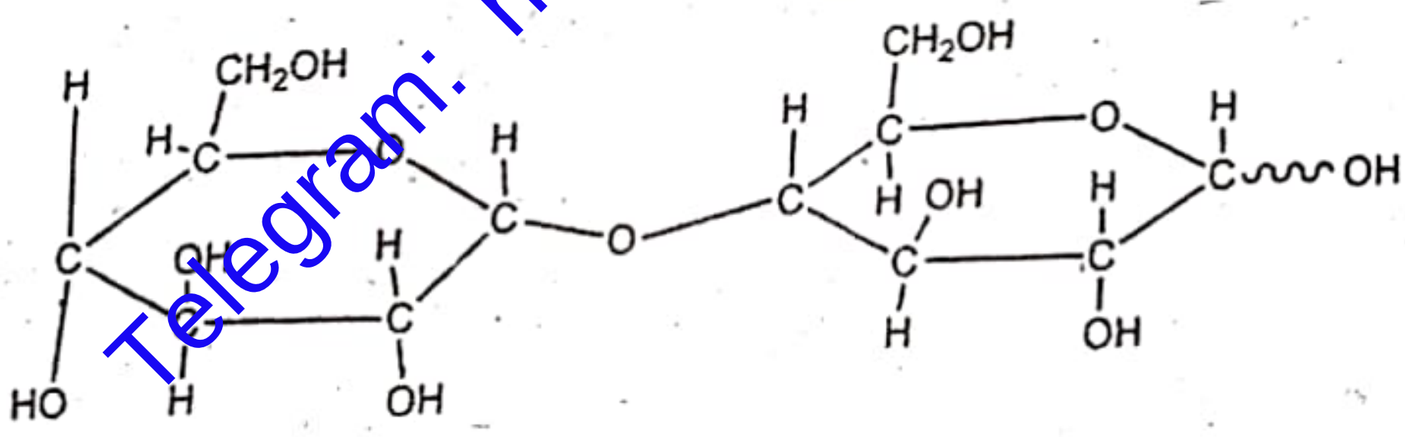
B.



C.



D.



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1. Which one of the following is the main reason for the large difference in dipole-moment between the triatomic molecules water (H_2O) and carbon dioxide (CO_2)?
   A. The presence of multiple bonds in carbon dioxide (CO_2) while single bonds in water.
   B. The bonds in carbon dioxide (CO_2) are directed 180 and bond polarity reinforces each other.
   C. The presence of lone pairs of electrons in water (H_2O) partially reinforces bond polarities.
   D. The higher electronegativity difference among atoms in water (H_2O) than in carbondioxide (CO_2).
2. How many non-bonding pairs of electrons are present in the valance shell of the central atom in ClF_3?
   A. 3
   B. 2
   C. 1
   D. 0
3. Which one would be observed in a laboratory activity that is applied on naphthalene?
   A. It shows relatively high melting point.
   B. It is soluble in water.
   C. It is soluble in benzene.
   D. It conducts electricity in molten state.
4. Which one of the following is correct?
   A. In Lewis model, electrons repel one another and a pair of bonding electros lies far apart as possible.
   B. Valance bond model explain molecular shape from the interaction of atomic orbitals.
   C. Valance bond model simply shows the linkage between atoms and the presence of lone pairs.
   D. Lewis model, predicts the shapes of molecules based on the electron-pair repulsions around a central atom.
5. The energy required to completely separate one mole of a solid ionic compound into gaseous ions is a/an
   A. Lattice energy
   B. Bond energy
   C. Ionization energy
   D. Activation energy
6. Which one of the following **IS TRUE** about isotopes?

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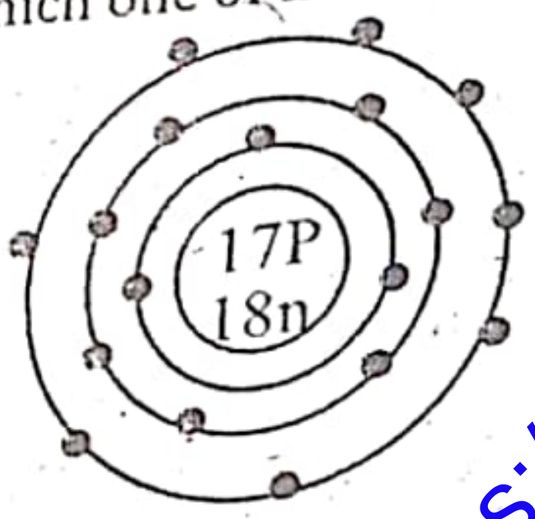
A. Isotopes of the same elements contain the same number of neutrons.
B. Isotopes of elements have the same chemical properties.
C. Isotopes of elements are atoms that have different number of electrons and protons.
D. Isotopes of an element are identical in mass number.

7. Which one of the following describes the quantum mechanical model?
A. An atom is a sphere with a positively charged matter, in which electrons are embedded.
B. Electrons moved around the nucleus of an atom in circular orbits.
C. The location of electrons in an atom are described interms of probability.
D. An atom is represented by a solid indestructible sphere.

8. Which one of the following is diagrammatic representation of $^{35}_{17}Cl$?

A.



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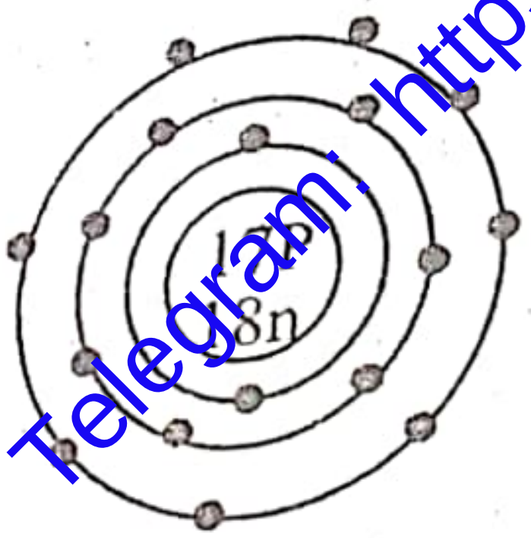

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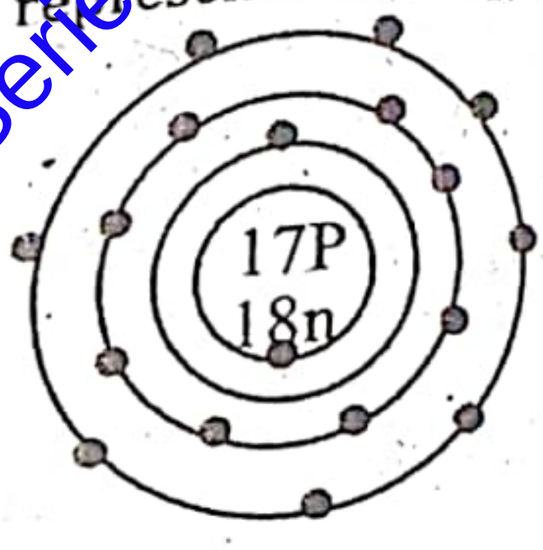

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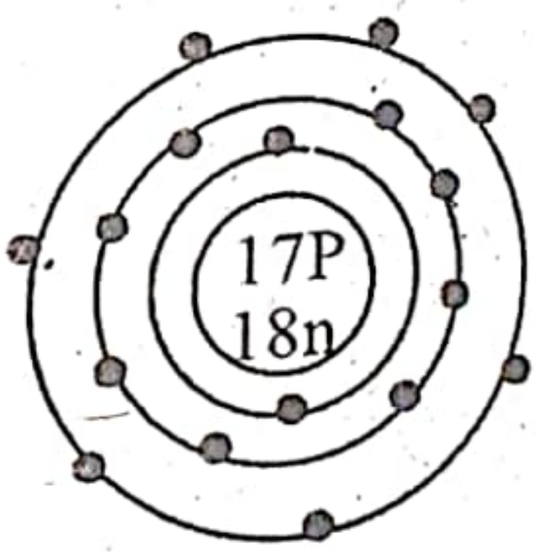

9. Which one of the following is ionic compound?
A. PCl_3
B. H_2O_2
C. $CaCl_2$
D. CO_2

10. An element is lqcated in Group VIIA and 3rd period of the periodic table. On the basis of this the element;
A. forms an acidic oxide
B. has extremely low electro affinity.
C. generally forms an ion with-2 charges
D. has very low ionization energy.

11. Which one of the following is a colorless. flammable volatile liquid and miscible with non-polar solvents?

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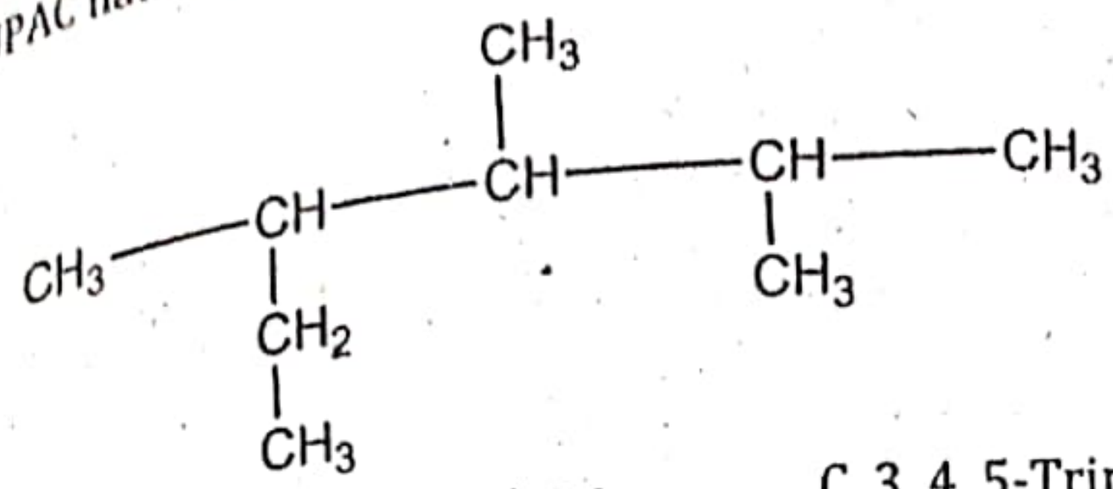
B. Methane C. Benzene D.

A. Methanol
Ethanol

12. The IUPAC name of the compound with the following structure is:



A. 4-Ethyl-2, 3-dimethyl pentane
B. 2,3,4-Trimethylhexane
dimethylpentane
C. 3, 4, 5-Trimethylhexane
D. 2-Ethyl-3, 4-

13. Which one of the following petroleum products is used as a solvent and dry cleaning?
A. Paraffin wax
B. Diesel oil
C. kerosene
D. Petroleum ether

14. Which one of the following is **NOT TRUE** about laboratory preparation of ethylene by dehydration of ethanol?
A. Combustion of ethylene produces a non-luminous flame.
B. The collected ethylene gas in the experiment decolorizes alkaline $KMnO_4$
C. The collected ethylene gas in the experiment decolorizes Br_2 in CCl_4.
D. Concentrated sulphuric acid is used as dehydrating agent.

15. Which one of the following is the essential nutrient of plants?
A. Sodium
B. Potassium
C. Aluminum
D. Lead

16. The heat change at constant pressure that results when 1 mole of a compound is formed from its elements in their standard state is
A. standard reduction potential reaction
B. standard enthalpy of formation
C. standard entropy of
D. standard enthalpy of

17. What is a redox reaction? It is a reaction in which
A. atoms or ions of two compound exchange bonding partners.
B. an acid and a base neutralize each other.
C. Two insoluble ions in compounds form an insoluble products.
D. Electrons are transferred between species.

18. Which of the following explains how electroplating take place?

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A. The cations in the electrolyte are supplied either by the added salt or from oxidation of the anode.
B. Metal is deposited on the anode by reduction of ions in the electrolyte solution.
C. The electrolyte is an aqueous solution of salts of the plated metal.
D. The object to be electroplated either metal or graphite-coated plastic constitute the anode.

19. How does the equivalent mass of a base is obtained?
A. By multiplying its number of equivalent by the number of OH^- ions furnished by one formula unit of a base.
B. By dividing its number of equivalent by the number of OH^- ions furnished by one formua unit of a base.
C. By multiplying its molar mass by the number of OH^- ions furnished by one formula unit of a base.
D. By dividing its molar mass by the number of OH^- ions furnished by one formula unit of a base.

20. Which one of the following reactions would occur spontaneously in an aqueous solution at 25 °C ? Assume that the initial concentrations of the dissolved species are all 1.0 M.

Half reaction	E^0(V)
$Sn^{2+}(aq) + 2e^- \rightarrow Sn(s)$	-0.14
$Br_2(l) + 2e^- \rightarrow 2Br^-(aq)$	+0.107
$Ag^+(aq) + e^- \rightarrow Ag(s)$	+0.80
$Ni^{2+}(aq) + 2e^- \rightarrow Ni(s)$	-0.25
$Ca^{2+}(aq) + 2e^- \rightarrow Ca(s)$	-2.87
$Cd^{2+}(aq) + e^- \rightarrow Cd(s)$	-0.40
$Cu^{2+}(aq) + e^- \rightarrow Cu^+(aq)$	+0.15
$Fe^{3+}(aq) \rightarrow Fe^+(aq)$	+0.77

A. $Cu^{2+}(aq) + Fe^{2+}(aq) \rightarrow Cu^+(aq) + Fe^{3+}(aq)$
B. $Ca(s) + Cd^{2+}(aq) \rightarrow Ca^{2+} + Cd(s)$
C. $2Ag(s) + Ni^{2+}(aq) \rightarrow 2Ag^+(aq) + Ni(s)$
D. $2Br^-(aq) + Sn^{2+}(aq) \rightarrow Br_2(l) + Sn(s)$

21. Which of the following compounds react with basic oxide to form salt and water?
A. Base oxide B. Neutral oxide C. Acid D. Acidic

22. Which one of the following reaction represents formation of an acid from its salt?
A. $NaNO_3(s) + H_2SO_4(l) \rightarrow NaHSO_4(s) + HNO_3(l)$
B. $AgNO3(aq) + NaCl(aq) \rightarrow NaNO3(aq) + AgCl(s)$
C. $CH_3COOH(aq) + CH_3OH(aq) \rightarrow CH_3CO_2CH_3(aq) + H_2O(l)$
D. $P_2O_5(s) + 6NaOH(aq) \rightarrow 2Na_3PO_4(s) + 3H_2O(l)$

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23. The negative logarithm of the molar concentration of hydrogen ion to the base ten is known as _____
A. K_a B. pOH C. pH D. K

24. Which one of the following galvanic cell is an example of secondary cell?
A. Leclanche dry cell
B. Zinc-carbon dry cell C. Daniel's cell D. Lead storage cell

25. Which one of the following is responsible for electrolytic conductivity?
A. Electros held in a fixed position.
B. Movement of ions.
C. Ions strongly held in a fixed position.
D. Free electrons.

26. Which one of the following is the reducing agent in the following reaction?

$$8NH_3(g) + 6NO_2(g) \rightarrow 7N_2(g) + 12H_2O\ (l)$$

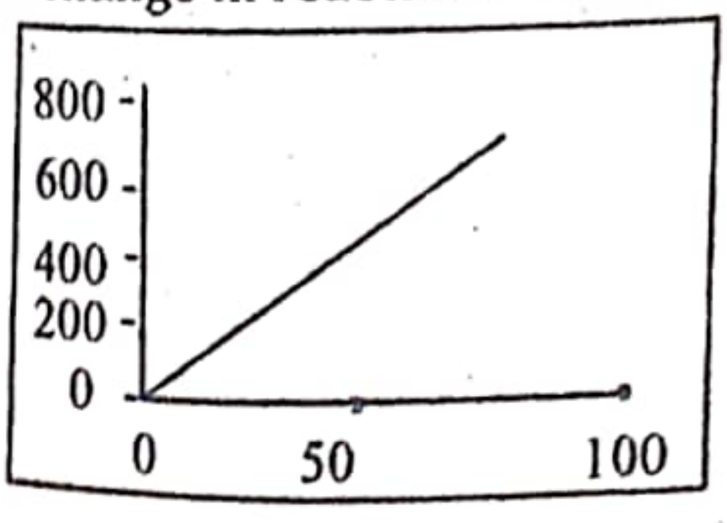
A. H_2O B. N_2 C. NO_2 D. NH_3

27. Carbon forms two stable compounds with oxygen, namely, carbon monoxide and carbon dioxide. For a fixed mass of carbon, the ratio of oxygen in the carbon monoxide to oxygen in carbon dioxide is 1:2. This result is consistent with the law of
A. constant composition C. definite proportion
B. multiple proportion D. conservation of mass

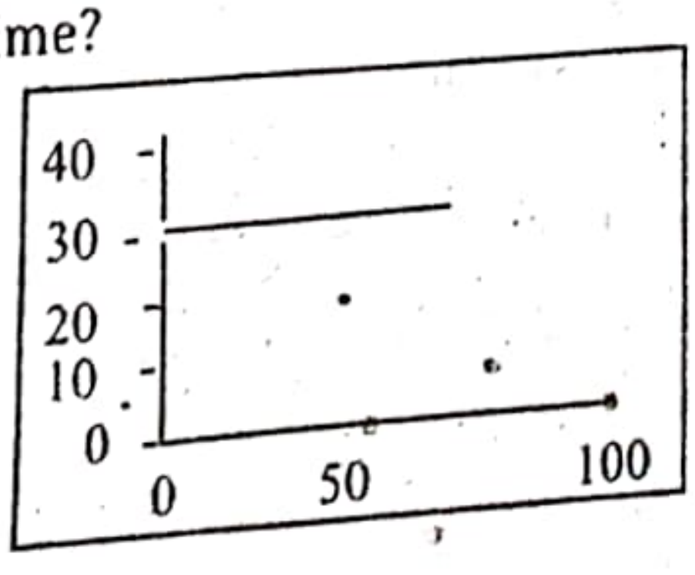
28. When a system at equilibrium is disturbed by a change of variable, the system shifts the equilibrium position in a way that tends to counter act this change of variable. This is a statement for
A. Heisenberg uncertainty principle C. Pauli exclusion principle
B. Le Chatelier's Principle D. Aufbau principle

29. Which one of the following graphs illustrates the rate change interms of change in reactant concentration with time?

A.



C.



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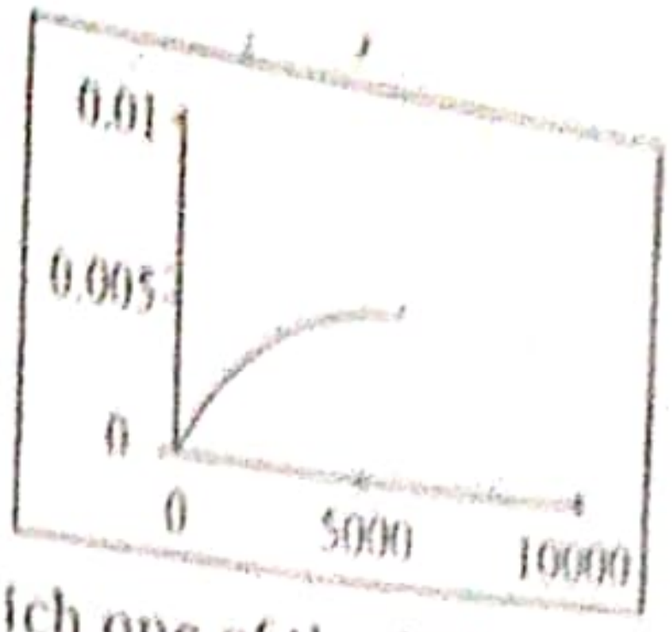
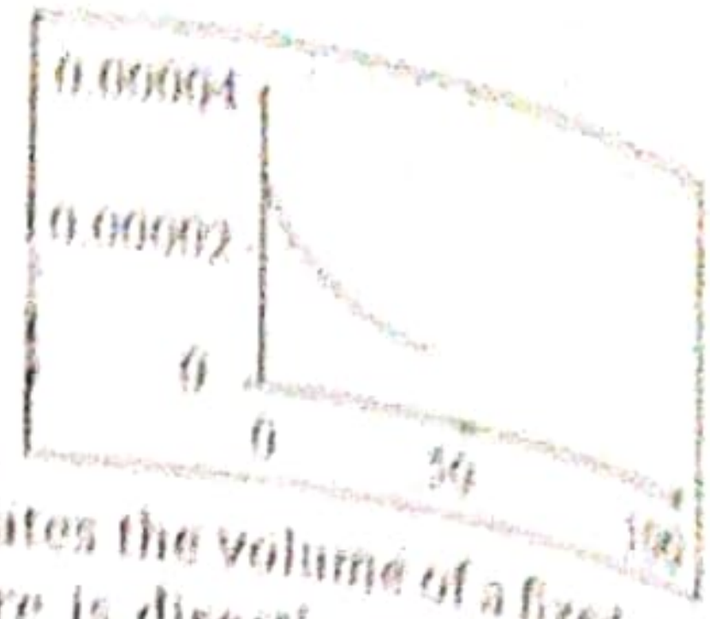
B.



D.



30. Which one of the following gas laws states the volume of a fixed amount of gas maintained at constant pressure is directly proportional to the absolute temperature of the gas?
    A. Avogadro's law B. Combined gas law C. Charles's law D. Boyle's law
31. Which of the following would occur during the formation of ions?
    A. The electronic configuration of an atom forming the ions remains unaffected.
    B. Valance electrons are transferred from one atom to another.
    C. Inner most shell electrons are involved.
    D. Valance electrons are shared between atoms.
32. In order to illustrate the structure and properties of a solid, a student put hundred marble balls in to a box that representing metal cations and the empty space between and around the marble representing the electron sea. Which one of the following bonding is represented by this model?
    A. Metallic bonding
    B. Dative bonding C. Covalent bonding
    D. Ionic
33. Which one of the following explains dipole-dipole force?
    A. An attractive force resulting from temporarily dipoles induced in molecules.
    B. An electrostatic attraction between metal ions and valance electrons.
    C. A type of intermolecular force that exists between non-polar molecules.
    D. The intermolecular attraction between oppositely charged poles of nearby polar molecules.
34. Which one of the following reactions creates energy in our bodies during respiration?
    A. $C_6H_{12}O_6(s) \rightarrow 2C_2H_5OH(l) + 2CO_2(g)$
    B. $N_2H_4CO(g) + H_2O(g) \rightarrow 2NH_3(l) + CO_2(g)$
    C. $C_6H_{12}O_6(s) + 6O_2(g) \rightarrow 6CO_2(g) + 6H_2O(l)$
    D. $6CO_2(g) + 6H_2O(l) \rightarrow C6H_{12}O_6(s) + 6O_2(g)$

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35. "A reaction that involves the exchange of positive and negative ions of each reactant". This is a definition of a
A. Double displacement reaction C. Decomposition reaction
B. Single displacement reaction D. Combination reaction

36. In a kinetic experiment, a student placed crystals of iodine in a closed reaction vessel, introduced a given quantity of hydrogen gas, and collected data to calculate the rate of formation of hydrogen iodide. What would be the effect obtained on the rate of the reaction, if the student takes more amount of iodine from that used in the first experiment.
A. The formation of hydrogen iodide will be decreased.
B. The rate of the reaction will be constant.
C. It speeds up the rate of the reaction.
D. It lowers the rate of the reaction.

37. Which of the following **CORRECTLY** relates the vapor pressure of a liquid with its boiling point? Boiling point is
A. the temperature at which molecules present in the vapor phase go back to the liquid state.
B. the temperature at which the kinetic energy of molecules is much less than the intermolecular force.
C. the temperature at which all the three physical states co-exist.
D. the temperature at which the vapor pressure becomes equal to the external pressure.

38. Consider the equilibrium system: $CO(g) + Fe_3O_4(s) \rightleftharpoons CO_2(g) + 3FeO(s)$ Which of the following disturbances shift the equilibrium position to the right?
A. Removing $FeO(s)$ C. Removing $CO_2(g)$
B. Adding $Fe_3O_4(s)$ D. Removing $CO(g)$

39. Which one of the following resulted from interaction of the atomic orbitals of bonding atoms to produce new electron distribution associated with the entire molecule?
A. Electron orbit B. Bohr orbit C. Hybrid orbital D. Molecular orbital

40. Which type of organic substances would be produced by heating a mixture of a carboxylic acid and an alcohol in the presence of H_2SO_4?
A. Esters B. Ethers C. Ketones D. Aldehydes

41. Which of the properties of solvents **WOULD NOT** be affected by the formation of a solution with non-volatile solute?

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A. Chemical identity
B. Freezing point
potential

42. The partial pressure of ethylene, C_2H_4 in a container is 10 atm at 25°C. What is the solubility in molarity of C_2H_4, assume that it obeys Henry's law? (K_H= 1.04 mol/L atm)
A. 104 B. 10.4 C. 9.62
0.104

43. What is the normality of a solution that contains 19.6 g of H_2SO_4 in 2L mL of the solution?
A. 20 B. 10 C. 0.02

44. Which one of the following illustrates an extensive property?
A. Doubling the volume of a sample also doubles the mass; so the ratio of mass to volume remain the same.
B. If water in two beakers at the same temperature are mixed to make a single quantity of water and the temperature of the combined quantity of water is the same as it was in the two separate beakers.
C. The electrode potential at standard condition for the half reaction $2Cu^{2+} + 4e^- \rightarrow 2Cu$ is the same as $Cu^{2+} + 2e^- \rightarrow 2Cu$
D. If a system is divided into parts and it is found that the property of the complete system has a value that is the sum of the values of the property of all the parts.

45. The point in a titration at which the number of moles of OH- ions added to a solution is equal to the number of moles of H+ ions originally present is
A. Triple point B. Equivalence point C. Zero point D. End point

46. The rate law for: $(CH_3)_3\text{-}C\text{-}Br(aq) + H_2O\ (l) \rightarrow (CH_3)_3\text{-}C\text{-}OH(aq) + H^+ + Br^-(aq)$ is rate = $k[(CH_3)_3\text{-}C\text{-}Br]$. The accepted mechanism for the reaction is
I. $(CH_3)_3\text{-}C\text{-}Br(aq) \rightarrow (CH_3)_3\text{-}C^+(aq) + Br^-(aq)$ [slow]
II. $(CH_3)_3\text{-}C^+(aq) + H_2O(l) \rightarrow (CH_3)_3\text{-}C\text{-}OH_2^+(aq)$ [fast]
III. $(CH_3)_3\text{-}C\text{-}OH_2^+(aq) \rightarrow H^+ + (CH_3)_3\text{-}C\text{-}OH(aq)$ [fast]
Why DOES NOT H_2O appear in the rate law? Because
A. water exists in the liquid state in the reaction .
B. water is missing in the rate determining step.
C. water is a catalyst in the reaction.
D. water is an intermediate species.

47. Which type of substances has the following general formula? (R_1, R_2 and R_3 hydrocarbon groups)

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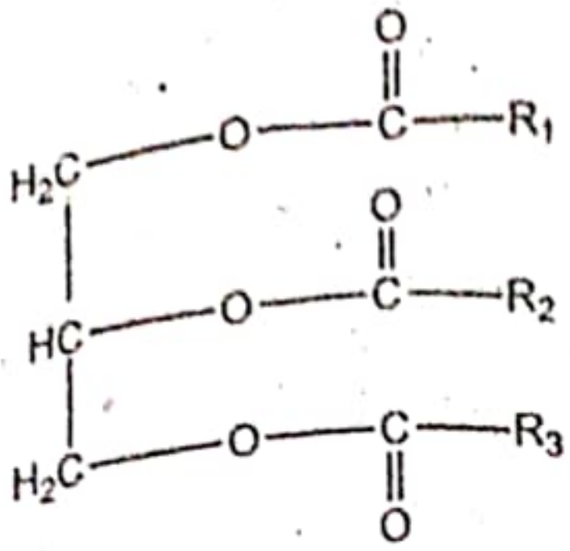

A. Alcohols B. Fatty acids C. Carboxylic acids D. Fats

48. The rate law for the reaction $CO(g) + NO_2(g) \rightarrow CO_2(g) + NO(g)$ is $rate = K[NO_2]^2$. Which one of the following could be a plausible mechanism for the reaction?

A.	$NO_3(g) + NO(g) \rightarrow NO_2(g) + NO_2(g)$	[slow]
	$NO_2(g) + CO_2(g) \rightarrow NO_3(g) + CO(g)$	[fast]
B.	$NO_2(g) + NO_2(g) \rightarrow NO_3(g) + NO(g)$	[fast]
	$NO_3(g) + CO(g) \rightarrow NO_2(g) + CO_2(g)$	[slow]
C.	$NO_2(g) + NO_2(g) \rightarrow NO_3(g) + NO(g)$	[slow]
	$NO_3(g) + CO(g) \rightarrow NO_2(g) + CO_2(g)$	[fast]
D.	$NO_3(g) + NO(g) \rightarrow NO_2(g) + NO_2(g)$	[slow]
	$NO_3(g) + CO(g) \rightarrow NO_2(g) + CO_2(g)$	[fast]

49. The speed with which a solute goes into solution is
A. rate of dissolution C. degree of solvation
B. extent of dissolution D. heat of solution

50. Which one of the following explains how 2 L of 1 M NaOH aqueous stock solution is converted into a 0.25 M NaOH solution?
A. Evaporating 0.25 L of water from the stock solution
B. Evaporating 1L of water from the stock solution
C. Adding 6 L water to the stock solution
D. Adding 8 L water to the stock solution

51. The pH of a 0.1 M solution of formic acid (HCOOH) is 3. What is the K_a of the acid?
A. 9.99×10^{-4} B. 9.99×10^{-5} C. 1.01×10^{-5} D. 91.01×10^{-7}

52. A system does 15 J of work and no other changes take place. Which of the following function has fallen by 15 J?
A. Enthalpy B. Entropy C. Internal energy D. Specific he

53. Which of the following is **CORRECTLY** stated
A. For an endothermic reaction with positive reaction entropy, the reaction is non-spontaneous at low temperature.
B. For an exothermic reaction with negative reaction entropy, the reaction is spontaneous at high temperature.

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C. For an endothermic reaction with negative reaction entropy, the reaction is spontaneous at all temperature.

D. For an exothermic reaction with positive reaction entropy, the reaction is non-spontaneous at all temperature.

54. In thermodynamics: solution in a beaker, a gas in a cylinder, a reaction vessel, a biological cell are examples of a

A. path function variables
B. thermodynamic system
C. thermodynamic
D. state function

55. Given the following:

	Bond energies (kJ/mol)
H-H	436
N-N	193
N=N	409
N≡N	942

and the enthalpy for the reaction $\frac{1}{2}N_2(g) + \frac{3}{2}H_2(g) \rightarrow NH_3(g)$; $\Delta H = -45$ kJ/mol. What is the bond energy for NH_3?

A. 960 B. 1080 C. 1125 D. 1170

56. Which one of the following statement is **TRUE** about gases?

A. When a gas sample is heated at constant pressure, its volume decreases.
B. The volume of a gas is equal to the volume of its container.
C. The particles of gases are closely packed together.
D. The volume of a gas is kept constant as pressure increases.

57. Which one of the following mathematical expressions is derived from the combination of Boyele's law, Charles's law and Avogadro's law?

A. $V \alpha \frac{nP}{T}$ B. $V \alpha \frac{P}{nT}$ C. $V \alpha \frac{nT}{P}$ D. $V \alpha \frac{T}{nP}$

58. 180 g of calcium carbonate ($CaCO_3$) is allowed to react with 156 gram of hydrochloric acid(HCl), according to the following reaction. Which of the following substances is the limiting reagent? (Atomic masses; Ca=40, Cl=35.5, O=16, C=12, H=1): $CaCO_3 + 2HCl \rightarrow CaCl_2 + CO_2 + H_2O$

A. H_2O B. CO_2 C. HCl D. $CaCO_3$

59. Which one of the following isomers differs in the arrangement of carbon atoms in the longest continuous carbon chain?

A. Cis-tans B. Structural C. Geometric D. Position

60. Which one of the following methods can be used in the laboratory for the preparation of alkynes?

A. Dehdrohalogenation of adjacent dihalides with sodium hydroxide.

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B. Hydrogenation of alkenes with a metal catalyst.
C. Dehydration of alcohols with concentrated sulphuric acid.
D. Heating of sodium salt of an organic acid with soda lime.

61. Which one of the following is TRUE about the electrolysis of a molten hypothetical electrolyte, MX?
A. Electrons are not transferred from one species to the other.
B. The overall reaction is redox reaction.
C. X is produced at the negative electrode.
D. M is produced at anode.

62. During the extraction of an unknown metal, the metallic ore was fused with NaOH in air and then dissolved in water to obtain the soluble unknown metal containing species. Reduction with carbon is followed after the solution is treated with HCl. Finally the obtained oxide is reduced to its metallic form by aluminum. The most probable unknown metal would be
A. Cr B. Pb C. Ca D. Zn

63. Why tin does NOT react with cold water but calcium reacts with cold water?
A. Due to its physical state.
B. Due to the pH of water.
C. Due to its less reactive nature.
D. Due to the formation of a thin oxide layer.

64. Why is vulcanization performed in natural rubber?
A. To decrease the number of monomer unit in the natural rubber.
B. To increase the number of monomer unit in the natural rubber.
C. To convert the monomer unit in natural rubber in to polymer.
D. To establish cross-links between polymer chain in the natural rubber.

65. Which one of the following explains the determination of the cell potential of n-Cu cell using voltmeter?
A. Each U-tube served as a salt bridge that can be filled with saturated KCl.
B. After the experiment the mass of zinc is increased.
C. In the Cu half cell, Cu metal will be dipped in $ZnSO_4$ solution.
D. In the Zn half cell, Zn metal will be dipped in $CuSO_4$ solution

66. Which of the following statements describes chemical industries? It is
A. a process in which dissolved solute comes out of solution and forms crystals.
B. A firm that involves usage of raw material to produce desired products by using chemical reactions and refining methods.
C. an area of chemistry concerned with the speeds, or rates at which chemical reaction occur.

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D. an expression for the chemical composition of a compound interms of the symbols of the atoms of the elements involved.

67. What will be the product if phosphorus reacts with excess amount of oxygen?
A. P_4O_{10} B. P_4O_6 C. H_3PO_3 D. H_3PO_4

68. One of the basic differences between metallic conduction and electrolytic conduction is:
A. The charge carriers in metals are localized electrons while strongly held ions are in electrolytic conduction.
B. The conduction of electricity is due to the movement of ions in the metallic conduction while the flow of mobile electrons in electrolytic conduction.
C. The conduction of electricity is due to the movement of ions in an electrolytic conduction while the flow of mobile electrons in metallic conduction.
D. The charge carriers in metals are cations and anions while mobile electrons are in electrolytic conduction.

69. Which one of the following gases is released from combustion of a fuel in an industrial machinery causing global warming
A. NO B. SO_2 C. CO D. CO_2

70. The number of digits to the right of the decimal point denotes:
A. Conversion factor C. Significant figures
B. Scientific notation D. Decimal places

71. Which one of the following is an example of molecular crystal?
A. Solid carbon dioxide B. Magnesium C. Diamond D. Sodium chloride

72. Which one of the following COULD NOT be considered as factor influencing the rate of chemical reaction?
A. The rate constant of the reaction C. Presence of catalyst
B. Temperature of reactants D. Surface area of reactants

73. Which of the following shows the hybridization scheme of the central atom in H_2O?

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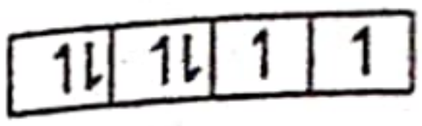
Ground state

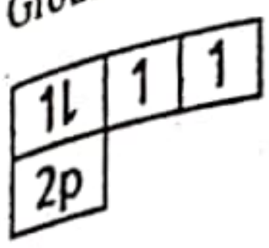
mixing →



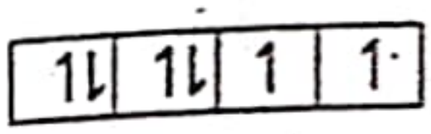
SP^3

A.

Ground state

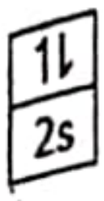


mixing →

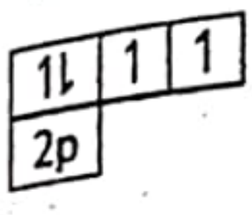


SP^3

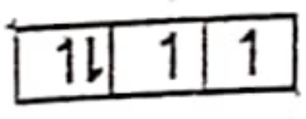
B.



Ground state

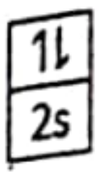


mixing →

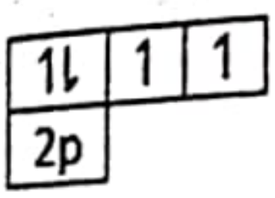


SP^3

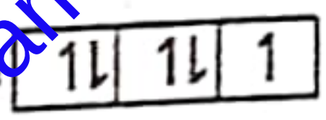
C.



Ground state



mixing →



SP^3

D.

74. In which of the following order of a reaction, a plot of $\ln[A]_t$ versus time is a straight line whose slope is $-k$?

A. Third B. Second C. First D. Zero

75. A forward and reverse reaction continues to take place at equal rate in microscopic level without change in composition at macroscopic level. This statement the definition of

A. degree of freedom C. Chemical kinetics

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B. the law of mass action
equilibrium
D. Dynamic

76. What are four quantum numbers for an electron in a 2s orbital?
A. $2, 0, 0, +\frac{1}{2}$ B. $2, 0, 1, +\frac{1}{2}$ C. $2, 1, 0, +\frac{1}{2}$ D. $2, 1, 1, +\frac{1}{2}$

77. A student uses digital balance (±0.02), a thermometer (±0.1), a measuring cylinder (±0.05), and a beaker (±0.5) during an experiment. Which of the above is more precise?
A. The beaker
C. The thermometer
B. The measuring cylinder
D. The digital balance

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78. Which one of the following explains that light has BOTH wave and particle nature?
A. Microscopic system, such as atoms and molecules can passes wave properties.
B. Light consists of electromagnetic waves which have an oscillating electric and magnetic disturbance that spreads electromagnetic radiation.
C. A beam of light is a stream of particles of electromagnetic energy, with energy proportional to the observed frequency of the light.
D. An electron that is bound to the nucleus behaves like a standing wave and its position is given interms of probability.

79. A nuclear breakdown in which particles or electromagnetic radiation is emitted is _____.
A. Radio wave
C. Radioactive isotope
B. Radioactive decay
D. Radioactivity

80. Which statement describes valance shell electron pair repulsion (VSEPR) theory?
A. Atomic orbitals are capable of combining to produce new electron distribution.
B. A covalent bond is formed when orbitals of two atoms overlap and the overlap region is occupied by a pair of electrons.
C. Each degenerate orbitals are occupied by a single electron before the second electrons of opposite spin enters the orbitals.
D. Electron pairs will be as far apart from each other in the three dimensional spaces as possible.

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1. B According to Dalton's atomic theory atoms are indivisible. However, according to modern atomic theory, atoms are divisible to sub-atomic particles (electron, proton and neutron).

2. C There are five atomic models

i. **Dalton's Model**

Is the first atomic model.

It is also called billiard ball model

It considered atom as a solid indestructible sphere.

ii. **Thomson's model**

It considered atom as a sphere with a uniform positive charge in which negatively charged electrons were embedded.

It is better atomic model than Dalton's model because it included existence of electron and proton.

It is also called plum-pudding model.

It is considered as chocolate chip ice-cream model.

iii. **Rutherford's model**

It explains atoms are mostly empty space with a dense positive nucleus

It is better atomic model than Thomson's atomic model as it shows how electros and protons are arranged in an atom.

It is also called nuclear model.

iv. **Bohr's atomic model**

This is the better model that introduced the concept of orbit and principal quantum number (n).

According to this model, electrons follow circular orbits around the nucleus.

v. **Quantum mechanical model**

This is the most modern atomic model.

It modified Bohr's atomic model by replacing orbit with orbital.

It introduced sub energy levels (s, p, d, f....)

3. D Copper atom has 29 electrons and the normal electron configuration take the form: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^9$ or $[Ar]4s^2 3d^9$, but the experimentally confirmed electron configuration is $[Ar]4s^1 3d^{10}$. This is

possible because 4s and 3d sub-shells are comparable in energy so that electros can easily jump from 4s to 3d and the full filled d sub-shell (d^{10}) is more stable.

4. B According to Mendeleev's periodic law, elements are periodic function of their **atomic masses.** But according to the modern periodic low, elements are periodic function of their **atomic number.**

5. B In coordinate covalent bond, the shared electron pair is contributed

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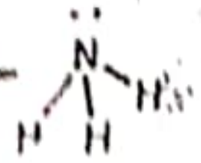
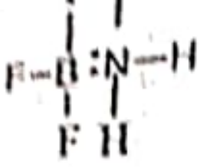
only by one of the atoms.

A. $:\dot{N}\cdot + 3H\cdot \longrightarrow$ 

In this reaction, the shared electron are contributed both by N and H atoms.

B. $BF_3 + NH_3 \longrightarrow$ 

In this reaction, the electrons shared between B and N atoms are contributed by nitrogen atom.

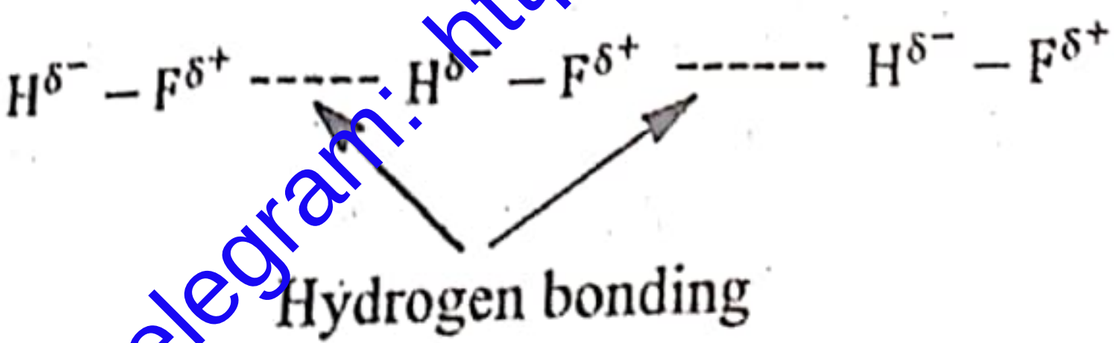
C. $H\cdot + \cdot\ddot{Cl}: \longrightarrow H:\ddot{Cl}:$

In this reaction, the shared electrons are contributed both by H and Cl atoms.

D. $H\cdot + \cdot H \longrightarrow H:H$

In this reaction, the shared electrons are contributed by the two hydrogen atoms.

6. **C** Covalent bond is formed when electrons are shared between atoms of non-metallic elements.
7. **C** Group IIA elements will have electron configurations of their nearest noble gas by losing two electrons and become more stable.
8. **D** Hydrogen bond is intermolecular bond that is formed when hydrogen is bonded to one of the three most electronegative elements (F, O or N).



9. **C** During combustion reaction heat energy is released.
10. **A** At STP (T=0°C and P=1atm) volume of 1 mole of any gas is 22.4L and this is called molar volume of a gas.
11. **B** From the balanced chemical equation coefficient of oxygen 5 and that of carbon dioxide is 3, thus the ratio is 5:3.
12. **D** From the balanced chemical equation the rate expression is:

$$\text{Rate} = -\frac{1}{2}\frac{\Delta[C_2H_6]}{\Delta t} = -\frac{1}{7}\frac{\Delta[O_2]}{7\Delta t} = \frac{1}{4}\frac{\Delta[CO_2]}{\Delta t} = \frac{1}{6}\frac{\Delta[H_2O]}{\Delta t}$$ From this we can read

Rate of consumption of C_2H_6 is 2/7 times rate of consumption of O_2
Rate of formation of CO_2 is 4/6=2/3 times rate of formation of H_2O.
Rate of formation of H_2O is 6/4=3/2 times rate of formation of CO_2.
Rate of formation of CO_2 is 4/2=2 times rate of consumption of C_2H_6

13. **D** Factors that affect rate of chemical reaction include temperature, concentration, surface area, nature of reactant, catalyst, pressure (for

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gases). Enthalpy is heat of .

14. C Positive catalysts increase rate of reaction by lowering activation energy, while negative catalysts decrease rate of reaction by increasing activation energy.

15. B According to Boyle's Law, pressure and volume of a gas are inversely proportional at constant temperature and this is mathematically represented as: $P_1V_1=P_2V_2$

16. A Gases are highly compressible because volume of particles are negligible compared to total volume the gas.

17. D As temperature decrease the number of liquid water molecule that inters gas phase (vapor) decreases and as a result the vapor pressure decreases.

18. C Droplets of water can be formed when there is condensation of water vapor.

19. B Compounds of carbon and hydrogen such as alkanes, alkenes, alkynes and aromatics are called hydrocarboons. Examples

20. D The boiling point of cycloalkanes is lower than that of linear alkanes. This is because surface area of cycloalkanes is lower and their intermolecular force is lower.

21. D Alkenes and alkynes are non-polar and held together by weak inter molecular forces and soluble in non-polar solvents such as benzene, carbon tetrachloride and toluene. Alkenes have longer bond length than alkynes and alkynes have higher bond energy than alkenes.

22. A The genera formula of alkynes is C_nH_{2n-2}, n=2, 3, 4, ... (The simplest alkyne contain two carbon atoms).

23. C Alkohols are characterized by having hydroxyl functional group (-OH) and named using the suffix "-ol".

24. D -Monohydric alcohols contain only one –OH group and their general formula is $C_nH_{2n+1}OH$.
- Carboxyl contain functional group of carboxylic acid (-COOH).
- Aryl contain aromatic functional group (C_6H_5-).
- Phenyl is side chain of $C_6H_5CH_2$- type.

25. B Acids can be defined in three different ways.
Are substances that release H+ in water (according toArhenius).
Are substances that can donate proton (according to Bronsted-Lowry).
Are substances that can accept pair of electron (according to Lewis)

26. B Acids can be classified into mono protic, diprotic, triprotic and poly protic acids based on their number of ignitable hydrogen atoms.
mono protic acides contain one ionizable hydrogen atom. eg. HCl
diprotic acids contain two ionizable hydrogen atoms. eg. H_2SO_4
triprotic acids contain three ionizable hydrogen atoms. eg. H_3PO_4

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diprotic and triprotic acids are called polyprotic acids.

27. **A** Soluble bases release hydroxide ion (OH^-) in water. Their Aqueous solutions can conduct electricity. Bases change color of red litmus paper to blue and red methyl orange to yellow.

28 **C** Salts are ionic compounds that contain positive ions (cations) derived from bases and negative ions (anions) derived from acids.

29. **A** **Electrolysis:** Process in which electrical energy is used to bring about chemical change.

Solvolysis: Is reaction between solute and solvent.

Hydrolysis: Is reaction of anion or captain or both with water.

Electroplating: Covering surface of a metal with another thin sheet of metal using electricity.

30. **A** Lead storage battery or car battery contains six identical cells joined together in series.

31. **B** The purpose of adding Cryolite is to reduce melting point of the ore (from 2045°C to 1000°C).

32. **A, C and D** Al is soft silvery white light metal that liberate hydrogen from dilute acids and it is good conductor of heat and electricity.

Physical properties of Al

- Soft silvery white metal having density 2.7g/cm³ and melting point 660°C.
- It is good conductor of heat and electricity
- It is malleable and ductile.

Chemical properties of Al

- It react with oxygen forming aluminum oxide (Al_2O_3).
- It can react with dilute HCl and H_2SO_4 and liberate hydrogen gas.
- Does not react with dilute or concentrated HNO_3.
- Can react with chlorine gas and sodium hydroxide solutions.

33. **C** Tanning is the process of converting raw animal hides and skin to leather. Is a process that converts the protein of raw hide or skin to stable material. There are two types of tanning: Vegetable or natural tanning and mineral tanning.

Vegetable (Natural) tanning:

- Solution of tannin which occur naturally in barks and leaves of many plants are used.
- Primary barks used in modern times include chestnut, oak, tanoak, hemlock, querabcho, mangrove, wattle(acacia)and myrobalan.
- Hides tanned naturally are flexible and used for making shoes, luggage and furniture.

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Mineral tanning
Solutions such as chromium sulphate and other salts of chromium are used.
Faster than natural tanning which require only 24 hours.
Greenish-blue leather which can be used to make garments and hand bags is produced.

34. B Coal releases huge amount of carbon dioxide which accumulated in atmosphere and cause of global warming due to greenhouse effect.

35. D The seven basic physical quantities are length, time, temperature, mass, current, luminous intensity and amount of substance. Other units such as volume, area, pressure, speed, velocity, force, density and molarity are derived units.

36. A In dimensional analysis, conversion factors are used and thus all the units except the intended unit are canceled out. The correctly labeled Conversion factor is

$(3.5\ \text{yards}/1)(3.5\ \text{feet}/1\ \text{yard})(12\ \text{in}/1\ \text{foot})(2.54\ \text{cm}/1\ \text{in})(1\text{m}/100\text{cm})$

and the unit remain is m.

37. D The main steps in scientific method are observation, hypothesis, experiment and theory.
Observation: Way of gathering information about phenomenon.
Hypothesis: Tentative explanation for observation.
Experiment: Testing the hypothesis.
Theory: Well tested hypothesis.
Hypothesis is tentative explanation for observation that can be tested experimentally.

38. C Cathode rays are negatively charged particles travel in straight line from cathode to anode possess mass and energy and exist in all elements.

39. A The higher the frequency, the higher the energy and the smaller the wave length ($\Delta E = h\nu = \frac{hc}{\lambda}$).

40. B The emission spectra of an atom is due to electron transition between energy levels.

41. C Probability of finding an electron is maximum near the nucleus.

42. C Number of valance electron is equal group of the element. Since there are 3 valance electrons, the group of the element is III.

43. A Though diazomethane has different resonance structures, one can obtain the structure given in "A" following steps of writing Lewis structure.

44. C Covalent compounds are volatile, generally have low melting and

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boiling points, insoluble in polar solvents such as water, soluble in non-
polar solvents such as benzene, non-electrolyte (do not conduct electricity).

45. A NaCl is ionic crystal. $[NH_4]^+$ and CO_3^{2-} are ions.

46. A Inter molecular forces are forces between molecules such as Van der Waals forces, dipole- diploe forces and hydrogen bonding.

47. B The two modern atomic theories are Valance Bond Theory (VBT) and
Molecular Orbital Theory (MOT).
Valance Bond Theory: Explains that a convent bond is formed by overlapping of atomic orbitals and the region of overlap is occupied by
pair of electron.
Molecular Orbital Theory: Explain that a covalent bond is formed by
combination of atomic orbitals and thus bonding and antibonding molecular orbitals are formed.

48. C In SO_2 there are three sets of electrons and has sp^2 hybridization the rest molecules have sp hybridization.

49. D From the given chemical reaction the rate expression is:
$\text{Rate}=-\frac{1}{1/2}\frac{\Delta[A]}{\Delta t}=-\frac{2[A]}{\Delta t}=\frac{\Delta[B]}{\Delta t}$ and thus $-\frac{\Delta[A]}{\Delta t}=1/4\frac{\Delta[B]}{\Delta t}$

50. A The time required for a reaction to be its initial concentration decrease by half is called half- life of the reaction.

51. B Given: m_o=2.5g, m_t=1.5g and t=109s Required: k=?
$\ln(\frac{m_0}{m_t})=\ln(\frac{2.5}{1.5})=k\times109s$, $0.511=k\times109s$,
$$K=\frac{0.511}{109s}=0.004688s^{-1}=4.688\times10^{-3}s^{-1}$$

52. D Reversible reaction is two way reaction in which reactant is converted to product, while the product is converted to reactant and thus
mixture of reactant and product exists. It do not go to completion. However irreversible reaction is one way reaction and it goes to completion.
Equilibrium can be reached between reactant and product

53. D Kp and Kc are related by the equation: $Kp=Kc(RT)^{\Delta n}$ and they will be equal when $\Delta n=0$, Kp>Kc when Δn>0 and Kp<Kc when Δn<0.
A. Δn=2-2=0 and Kp=Kc C. Δn=2-1=1 and Kp>Kc
B. Δn=2-2=0 and Kp=Kc D. Δn=2-3=_1 and Kp<Kc

54. C The substance exists in liquid phase only when its temperature is

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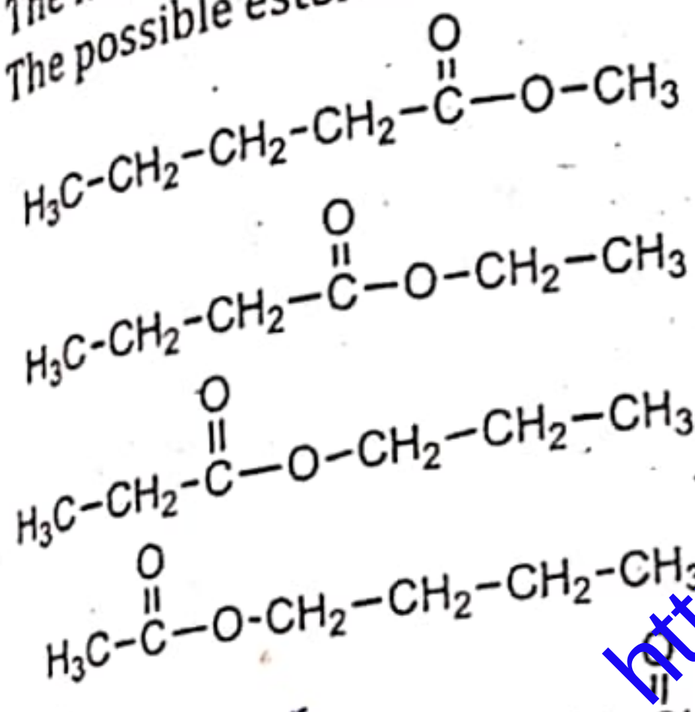
above melting point and below boiling point. In region 1 only pure solid exists. In region 2mixture of solid and liquid exists. In region 3, only pure liquid exists. In region 4 mixture of liquid and gas exists. In region 5, only gas exists.

55. C -COOH is the functional group for carboxylic acid. RCHO is general formula for aldehyde. ROR is general formula for ethers and RCOR is general formula for ketones.

56. D The first nine carboxylic acids are liquids and the others are solids.

57. A The possible esters that can be formed from $C_6H_{12}O_2$ are



methyl pentanoate

ethyl butanoate

propyl propanoate

butyl ethanoate

pentyl methanoate

58. B Pentanoic acid contains 5 carbon atoms.

59. B The speed at which solute goes into solution is said to be rate of dissolution.

60. C Supper saturated solution contain more solute than its saturated Solution and formed after equilibrium is reached. The solution we have before saturated solution is formed is unsaturated solution.

61. A Concentration is amount of solute present in a given quantity of solvent or solution.

62. C Given: V=2L, M=0.3M Required: n=?

$M=\frac{n}{V}$ and $n=MV=2L\times0.3mol/L=0.6mol$

63. B From density of water mass of water= 180g and number of mole of water=180g/18g/mol=10mol

number of moles of sucrose=34.2g/342.91g/mol=0.1mol

mole fraction of water=$\frac{10}{0.1+10}=\frac{10}{10.1}=0.99$

$P=X_{water}P^0=0.99\times39kPa=38.61kPa$

64. D Colligative properties of electrolyte and non electrolyte are not the same. Colligative properties of

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Non-electrolytes	Electrolytes	
$\Delta T_b = k_b m$	$\Delta T_b = ik_b m$	(boiling point elevation)
$\Delta T_f = k_f m$	$\Delta T_f = ik_f m$	(freezing point depression)
$\Delta P = x_2 P_1^0$	$\Delta P = ix_2 P_1^0$	(vopor pressure lowering)
$\pi = MRT$	$\pi = iMRT$	(osmotic pressure)

i is Van't Hoff factor and greater than 1.

65. **D** From the given reaction the conjugate acid –base pairs are either $HC_2O_4^-$ and $C_2O_4^{2-}$ or H_2O and H_3O^+.

$$\underset{\text{Acid}}{HC_2O_4^-(aq)} + \underset{\text{Base}}{H_2O(l)} \rightleftharpoons \underset{\text{Conjgate acid}}{H_3O^+} + \underset{\text{Conjugate base}}{C_2O_4^-}$$

66. **C** According ti the law of mass action, liquid and solid species don't involved in equilibrium expression and thus $K_w=[OH^-][H^+]$.

67. **A** Presence of common ion suppresses ionization of weak base and weak acid and decreases the concentration of OH^- and H_3O^+ and rises the pH of the acid and pOH of the base.

68. **A** carbonate ion is moderately strong and hydrolyzed with water molecule making the solution basic: $CO_3^{2-} + H_2O \rightarrow HCO_3^- + OH^-$

69. **C** Titration is technique in which a solution of known concentration is used to determine concentration of an unknown solution. In the process mostly indicators are used but not always.

70. **A** Given: q=650J, w=120J Required: ΔE=? ΔE=q+w=650J + 120J=775J.

71. **B** Heat gained or lost at constant pressure ($\Delta H = (q)_p$) is said to be enthalpy change.

72. **B** Oxidation is loss of electron or increase in oxidation number and reduction is gain of electron or decrease in oxidation number.

73. **A** Selection and discharge for one kind of ion during electrolysis based on energy required is called preferential discharge. It can be affected by concentration, position of the ion electrochemical series and the type of electrode.

74. **B** **Faraday's second law** of electrolysis states that when the same quantity of electricity is passed through different electrolytes, the masses of different substances deposited at respective electrodes will be proportional to their equivalent masses. But **Faradaday's first law** states that amount of substance consumed or produced at an electrode is directly

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proportional to the quantity of electricity passed through the cell.

75. B In Daniel cell the anode is Zn metal immersed in $ZnSO_4$ solution and the cathode is Cu metal immersed in $CuSO_4$ solution. Electrons flow from anode to cathode in external circuit. Electrons are generated at anode and consumed at cathode.

76. A All ores are minerals but all minerals are not ores. Ores are minerals from which elements can be extracted easily and economically.

77. B When $CaCl_2$ is added melting point of NaCl is lowered from 800°C to 600°C

78. A Mining is not considered as major metallurgical process. The major metalargical processes are concentrating ore, pretreatment and production of metal (extracting and refining).

79. D Teflon (Polytetraflouroethylene)(PTFE) is homopolymer whose monomer is Tetrafluoroethylene($F_2C=CF_2$).

80. D Disaccharides are made of two monosaccharaides joined through glycosidic linkage.

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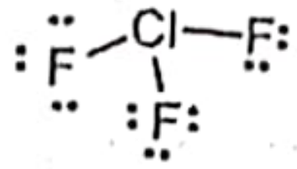
SOLUTION FOR CHEMISTRY HAMILE 2015 E.C (JULY 2023G.C)

1. **C** The main reason is due to presence of lone pair electrons on central atom in water molecule which is the cause for existence of net dipolmoment.
2. **B** From the Lewis structure of the molecule, we can see that the central atom chlorine contains two lone pairs of electron.



3. **C** Naphthalene is a non-polar organic molecule and by the rule of "Like dissolves like", it is soluble in non-polar solvents such as benzene.
4. **B** According to valance bond theory, bonds are formed from overlapping of atomic orbitals which can also determine shape of a molecule. The Leis model simply shows linkage of atoms an presence lone pair of electron.
5. **A** Energy required to convert ionic compounds to their ions is lattice energy.
6. **B** Isotopes are atoms of the same elements with different number of neutron or mass number but the same number of proton and electron. Their chemical properties are also the same.
7. **C** According to quantum mechanical model, the exact position of electrons cannot be known and only described interms of probability.
8. **D** The atom has 17 electros. Two of them are in the first shell, 8 of them are in the second shell and seven of the are in outer shell.
9. **C** Ionic compounds are usually formed between metals and non-metals.
10. **A** The element in group VIIA and period 3 is chlorine. It can react with oxygen to form oxides. The well-known oxide of chlorine is Cl_2O_7 which react with water to form very strong acid, $HClO_4$ (perchloric acid). It forms an ion with -1 charge. It has high electron affinity and ionization energy.
11. **C** From the given molecules methanol and ethanol are polar solvents and do not missile with non-polar solvents. Methane is non-polar molecule, but it is not liquid. So the liquid molecule that miscible with non-plar molecule is benzene.
12. **B** We follow the rules used to give IUPAC names.

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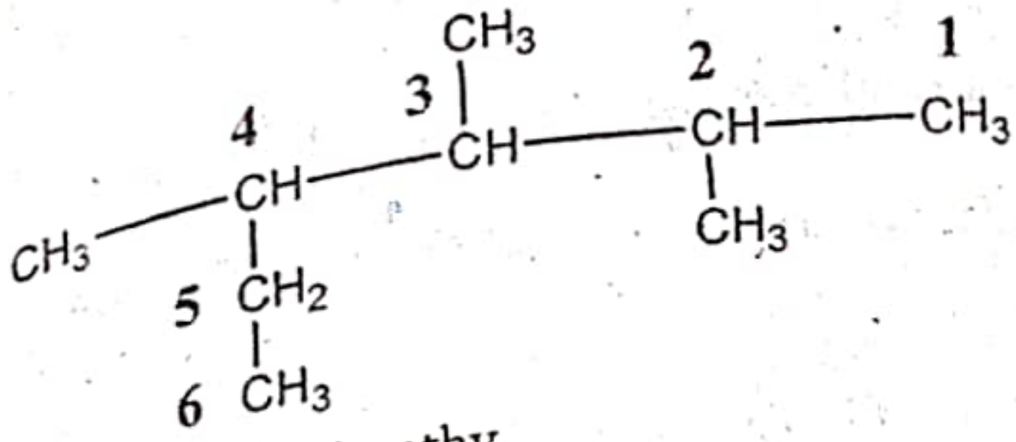

Side chains: 2, 3, 4-trimethy
Parent name: hexane
IUPAC name: 2, 3, 4-Trimehy hexane.

13. D Paraffin wax: is mainly used for production of candle and polishes.
Diesel oil: Is mainly used as furnace fuel and diesel engine fuel.
Kerosene: Is used for jet engine fuel and household fuel.
Petroleum ether: is used for dry cleaning and as a solvent.
14. A Alkenes generally burn with luminous flame.
15. B Essential plant nutrients include; C, H, O, S, N, P, K, Mg, Ca, Fe, Mn, B, Cu, Zn, Mo, Co and Cl.
16. B The standard enthalpy of formation, $\Delta H°_f$, is the change in enthalpy in the reaction when one mole of a substance is formed from the elements in their standard states.
17. D Redox reaction is oxidation reduction reaction in which one substance lose electron, while the other gain electron or simply it is a reaction in which electros transfer.
18. A In electroplating: Metal to be plated is used as cathode, the plating metal is used as anode and the electrolyte is salt of plating metal. In the process metal is deposited on material used as cathode.
19. D Dividing molar mass by number of OH^- relased by the base gives equivalent mass of the base.
20. B A cell reaction is spontaneous when its cell potential (E^0_{cell}) value is Positive.
A. $E^0_{cell} = E^0_{red}(cathode) - E^0_{red}(anode) = 0.15 - 0.77 = -0.62V$
B. $E^0_{cell} = -0.4 - (-2.87) = -0.4 + 2.87) = +2.47$
C. $E^0_{cell} = -0.25 - 0.8 = -1.05V$
D. $E^0_{cell} = -0.14 - 1.07 = -1.21V$
21. A Basic oxide react with water to form salt and water. But basic oxide react with acidic oxide and form salt only.
22. A Nitric acid (HNO_3) is formed from its sodium nitrate ($NaNO_3$) salt.
23. C $pH = -\log[H^+]$
24. D Secondary cells are those which can be recharged such as lead storage (car) battery, mobile battery and laptop battery.
25. B The charge carriers in electrolytic conductivity are ions (cations and anions).
26. D Oxidation number of nitrogen in NH_3 is changed from -3 to 0

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(increased) and changed from +4 to 0 in NO_2 (decreased). Thus NH_3 is

oxidized (it is reducing agent) and NO_2 is reduced (it is oxidizing agent).

$$\overset{-3}{8}NH_3(g) + 6\overset{+4}{N}O_2(g) \rightarrow 7\overset{0}{N}_2(g) + 12H_2O(l)$$

27. B The law of multiple proportion states that if two or more different compounds are formed from the same elements, the masses of one of the elements combined with fixed mass of the other element is in small whole number ratio.

28. B The effect of different factors (temperature pressure and concentration) on an equilibrium is explained by Le Chatlier's principle.

29. D Generally rate of the reaction is high at the beginning (when the concentration of the reactant is high) and the rate is low at end (when concentration of reactant is less).

30. C Charles's law states that at constant pressure, volume of gas is directly proportional to temperature.

31. B Ions are formed by transfer of valance electros of atoms.

32. A Electron sea model explains metallic bonding as an attraction force between metal cations and the surrounding sea of electrons.

33. D Dipole-dipole force is intermolecular force (force between molecules) which is the attraction force between polar molecules

34. C Respiration is cellular reaction in which energy is produced from combustion of simple sugar with oxygen.

35. A Double displacement reaction is a reaction of two different compounds to form two new compounds by exchanging their positive and negative ions.

36. D Decreasing concentration of a reactant, decreases rate of a reaction.

37. D Boiling point is the temperature at which vapor pressure of a liquid is equal to external (atmospheric) pressure.

38. A, B, C According to Le Chatelier's principle, removing the product and adding the reactant shifts the equilibrium to the product side (to the right).

39. D According to molecular orbital theory, atomic orbitals interact (overlap) to produce new electron distribution in new orbitals called molecular orbitals.

40. A The reaction between carboxylic acids and alcohols is called

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esterification in which esters are produced.

41. A Formation of a solution is physical process in which the solute and solvent retain their chemical identities.

42. B Given: P_g=10 atm, K_h=1.04 mol/L atm

Required: C_g=?

Solution: $C_g=K_hP_g$=1.04mol/L atm×10atm=10.4mol/L

43. A Given: m=19.6g MW=98g/mol V=20mL=0.02L Z=2 eq/mol

Required: N=?

Solution:

$$N=\frac{mZ}{MW\times V}=\frac{19.6g\times 2eq/mol}{98g/mol\times 0.02L}=20\ eq/L=20\ N$$

44. D Extensive properties are those which vary with the amount of the substance and properties which do not vary with the amount of the substance are extensive properties.

45. B Equivalence point is the point at a base and an acid neutralizes each other.

46. B The rate-determining step in the mechanism is the slowest step.

47. D The given structure is general formula of fats and oils (triesters).

48. C The mechanism given in alternative "C" satisfies the two requirements:The sum of the elementary reactions give the overall reaction and the rate law obtained from the slowest step agrees with given rate law.

First requirement:

$$NO_2(g)+\cancel{NO_2(g)} \rightarrow \cancel{NO_3(g)}+NO(g)$$
$$\cancel{NO_3(g)}+CO(g) \rightarrow \cancel{NO_2(g)}+CO_2(g)$$

Over all rxn: $NO_2(g)+CO(g) \rightarrow NO(g)+CO_2(g)$

Second requirement:

Rate law from the slow step is rate=$k[NO_2][NO_2]=k[NO_2]^2$

49. A The speed a solute goes into solution is called rate of dissolution.

50. C A less concentrated solution is prepared from more concentrated solution by dilution (adding more water). The amount of water added

can be obtained as follows: First calculate volume of the new solution.

Given: V_i=2L C_i=1M C_f=0.25M

Required: V_f=?

$C_fV_f=C_iV_i$ and $V_f=\frac{C_iV_i}{C_f}=\frac{1M\times 2L}{0.25M}=8L$

Therefore the amount of water added is $=V_f-V_i$=8L−2L=6L

51. C The acid is weak acid and dissociates partially.

$$HCOOH(aq)+H_2O(l) \rightleftharpoons HCOO^-(aq)+H^+(aq)$$

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	[HCOOH]	[HCOO⁻]	[H⁺]
Initial	0.1M	0	0
Change	-x	+x	+x
Equilibrium	0.1-x	x	x

$Ka = \frac{[HCOO^-][H^+]}{[HCOOH]} = \frac{x^2}{0.1-x}$ The value of x=[H⁺] and obtained from pH value.

$pH = -\log[H^+] = 3$ and $\log[H^+] = -3$. Thus $x=[H^+]=10^{-3}$

Therefore the value of Ka is

$Ka = \frac{x^2}{0.1-x} = \frac{(10^{-3})^2}{0.1-0.001} = \frac{10^{-6}}{0.099} = 1.01\times10^{-5}$

52. **C** Work is done by the system and thus the internal energy of the system decrease by 15J.

53. **A** In the expression $\Delta G = \Delta H - \Delta ST$; ΔG can be positive (non-spontaneous) towards low temperature when the reaction is endothermic ($\Delta H = +ve$) and $\Delta S = -ve$.

54. **B** Any thing under consideration is called thermodynamic system.

55. **D** $\Delta H = \sum\begin{pmatrix}\text{Bond energies of}\\ \text{bond broken}\end{pmatrix} - \sum\begin{pmatrix}\text{Bond energies of}\\ \text{bond formed}\end{pmatrix}$

$\frac{1}{2}N_2(g) + \frac{3}{2}H_2(g) \rightarrow NH_3(g)$

$-\Delta H = \frac{1}{2}(N \equiv N) + \frac{3}{2}(H-H) -$ bond energy of ammonia

$-45\ kJ = \frac{1}{2}(942kJ) + \frac{3}{2}(436kJ) -$ bond energy of ammonia

$-45kJ = 471kJ + 654kJ -$ bond energy of ammonia

$-45kJ = 1125kJ -$ bond energy of ammonia

Bond energy of ammonia $= 45kJ + 1125kJ = 1170kJ$

56. **B** Volume and temperature of a gas are directly proportional at constant pressure. Gases do not have definite volume and they take volume of their containers. The particles of gases are far apart. Volume of gases decrease when pressure decrease.

57. **C** The ideal gas equation (PV=nRT) is derived from Boyle's law, Charles's law and Avogadro's law. From the equation, one can show that $V = \frac{nRT}{P}$ which means volume is directly proportional to temperature and number of moles of the gas while it is inversely proportional to the pressure of the gas.

58. **D** Calculate mass of $CaCl_2$ formed using both reactants

$$\underset{180g}{CaCO_3} + \underset{156g}{2HCl} \rightarrow \underset{X}{CaCl_2} + CO_2 + H_2O$$

Using calcium carbonate

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$\frac{x}{111g} = \frac{100g}{100g} \Rightarrow x = \frac{180g \times 111g}{100g} = 199.8g$ of $CaCl_2$

Using hydrochloric acid

$\frac{x}{111g} = \frac{156g}{73g} \Rightarrow x = \frac{156g \times 111g}{73g} = 237.2g$ of $CaCl_2$

Limiting reactant is the which gives smaller amount of the product. Therefore calcium carbonate is the limiting reactant.

59. B Structural or chain isomerism is due to the difference in the arrangement of the carbon atoms in the longest continuous carbon chain.

60. A Alkynes can be prepared by dehdrohalogenation of vicinal(adjacent) dihalides with bases such as NaOH, and alkylation of sodium acetaylide with primary alkyl halide.

61. B Electrolysis is oxidation reduction reaction (redox reaction) in which a metal M is produced at cathode (negative electrode) by reduction and a no=metal element X is produced at anode (positive electrode) by oxidation.

62. A The metal extracted by the method described is chromium (Cr).

63. C Tin is less reactive and cannot react with cold water.

64. A Rubber is usually volcanized to increase its strength by establishing cross-links between polymer chain.

65. A Daniel's cell (Zn-Cu cell) is constructed by dipping Zn metal in $ZnSO_4$ solution in one compartment and Cu metal in $CuSO_4$ solution in another compartment and the compartments are joined by saltbirdge (U-shaped tube). After the experiment mass of Zn metal decreased while that of Cu is increased.

66. B Chemical industry is an institution which uses chemical reaction to produce finished and semi-finished products.

67. A Phosphorus react with limited oxygen to produce tetrahexoxide (P_4O_6) and react with excess oxygen to produce tetraphosphorus decoxide (P_4O_{10}).

68. C In metallic conduction
- ✓ The charge carriers are mobile (delocalized) electrons.
- ✓ Conduction of electricity is due to movement of electrons.
- ✓ The process is physical process.

In electrolytic conduction
- ✓ The charge carriers are mobile ions (cations and anions).
- ✓ Conduction of electricity is due to movement of ions.
- ✓ The process is chemical process.

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69. **D** Carbondioxide (CO_2) is a gas released during combustion of fuel in industries and automobile engine which is main cause of global warming.

70. **D** In reporting values of measurements, decimal points are used to indicate significant figures (precision of measurements).

71. **A** Carbon dioxide is molecular crystal, magnesium is metallic crystal, diamond is covalent network crystal and sodium chloride ionic crystal.

72. **A** Rate constant of a reaction is constant.

73. **B** H_2O is AX_2E_2 type molecule and its hybridization is sp^3 like that of AX_4 type molecules.

74. **C** Time integrated rate equation for first order is $\ln[A]_t = -kt + \ln[A]_0$ and thus its slope is $-k$.

75. **D** A chemical equilibrium is dynamic equilibrium in which the forward and backward reactions continue at equal rate.

76. **A** 2s subshell is given. Therefore $n=2$ and $l=0$. The only possible value of m_l is 0.

77. **D** The smaller the uncertainity, the higher the precision.

78. **A** According to De Broglie, matter such as atom and molecule have both particle nature (posses mass) wave nature (posses wave).

79. **B** Nuclear breakdown of an atom in which particles such as alpha and beta particles or radiations such as gamma radiations are emitted is called *radioactive decay*. The spontaneous emission of radiation from unstable nuclei of an atom is called *radioactivity*.

80. **D** Valance shell Electron Pair Repulsion Theory states that pair of valance shell electrons around the central atom of a molecule stays as far apart as from each other as possible to minimize repulsion between them.

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EDUCATIONAL ASSESSMENT AND EXAMINATIONS SERVICE ETHIOPIAN SECONDARY SCHOOL LEAVING CERTIFICATE BIOLOGY EXAMINATION 2014/15 (2022)

Booklet code: 154

1. Bacterial cell wall is unique due to its
   A. Chemical composition
   B. Freely permeability
   C. Selective permeability
   D. Mechanical support

2. Which genus of bacteria oxidizes hydrogen sulphide to sulphate and make sulphate ion available to be taken by plant roots from the soil?
   A. Klebsiella
   B. Desulphovibrio
   C. Azotobacter
   D. Thiobacillus

3. Which one of the following diseases is caused by food poisoning bacteria?
   A. Candidiasis
   B. Salmonellosis
   C. Tuberculosis
   D. Pneumonia

4. Which one of the following is cored concerning the reproductive cycles of viruses?
   A. Both lytic and lysogenic life cycles release new viruses without the genetic alteration of the host cell
   B. In lysogenic life cycle the DNA of the virus integrates with the host cell's DNA
   C. Both lytic and lysogenic life cycles undergo the same life cycle but ending with different results.
   D. In lytic life cycle, viruses are released without killing the host cell.

5. Unlike free living cells, viruses are completely
   A. Pathogenic to a living cell
   B. Mutual to a living cell
   C. Parasitic to a living cell
   D. Harmful to a living cell

6. Why are vaccines important? Because they help to
   A. Trigger the immune system
   B. Kill pathogens
   C. Prevent functional diseases
   D. Heal wounds

7. Assume you wanted to demonstrate compatibility of blood groups. You took four test tube labeled O, A, B and AB based on the blood group that each test tube contains. Then you added unknown blood sample into each test tube and no agglutination was observed in each test tube. From this, you can conclude that the unknown blood sample was
   A. Blood group B
   B. Blood group O
   C. Blood group A
   D. Blood group AB

8. Which one of the following diseases increased the likelihood of contracting HIV?
   A. Syphilis
   B. Chancroid
   C. Tuberculosis
   D. Gonorrhoea

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9. Even though pasteurization is a technique use to heat milk, beer and other foods, it cannot preserve food unspoiled for years because it
   A. Cannot kill pathogenic microorganisms
   B. Reduces the nutritional content of the food
   C. Is conducted in containers that are not properly sealed
   D. Uses mild temperature that kills most but not all bacteria
10. Which one of the following statements is correct about the binomial nomenclature?
   A. The genus name is written in full while the species name is abbreviated
   B. Both genus and species names are capitalized and italicized when printed
   C. The first letter of genus name is capitalized while the species name is written in small letters
   D. The genus name is written in English while the species name is written in Latin
11. During gas exchange between the alveoli and blood capillaries, oxygen is absorbed by the blood while carbon dioxide is removed. This opposite direction of gas exchange occurs due to
   A. Difference in the concentration gradient of oxygen and carbon dioxide
   B. Difference in the molecular size of oxygen and carbon dioxide
   C. Thicker layer of the capillaries
   D. The small surface area of the alveoli
12. Identify the correct statement about the effect of exercise on breathing rate.
   A. The vital capacity of the lungs decrease during heavy exercise.
   B. The tidal volume of air at rest is greater than the tidal volume of air during exercise.
   C. Fast and deep breathing are the result of demanding less oxygen
   D. Exercise and getting fit makes the lungs efficient
13. The following is a list of functions of the components of blood
   1. Transport of oxygen
   2. Fight against disease
   3. Cause blood clotting

Which one of the following is a correct order of blood components responsible for the above functions respectively?
A. White blood cells, platelets, red blood cells
B. Platelets, white blood cells, red blood cells
C. Red blood cells, white blood cells, platelets
D. Red blood cells, platelets, white blood cells

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14. What is the cause of the back flow of blood into the atria from the ventricles? It could be damage of the
A. Aorta
B. Pulmonary artery
C. pulmonary vein
D. Valves
15. The capillaries are blood vessels that have thin wall and large surface area since their function is
A. Transport of oxygenated blood
B. Exchange of substance
C. To return blood to the heart
D. Transport of deoxygenated blood
16. During image formation, light is refracted twice before it is focused on the retina. This phenomenon happens at the
A. Aqueous humor and lens
B. Iris and lens
C. Lens and vitrous humor
D. Cornea and lens
4. Three students designed the following experimental settings to understand the actual test of food.
Student 1: Blind folded two students, one of them tightly holding the nose while the other without holding the nose and tasting food
Student 2: Blind folded one student and the other left with open eyes, both holding their nose tightly while tasting the food
Student 3: None of the participants blind folded and only one of them holding her nose tightly while tasting the food
Student 4: None of the participants blind folded and both of them taste the food without holding the nose.
Which student designed the most appropriate experiment?
A. Student 1
B. Student 2
C. Student 4
D. Student 3
5. Short sightedness is corrected by a concave lens because the problem arises due to a
A. Weak eye lens that converges light slightly
B. Powerful eye lens that diverges light
C. Weak eye lens that diverges light
D. Strong eye lens that converges light too soon
6. A patient exhibited symptoms of weight loss, sweating and irritability and the doctor suspected an endocrine malfunction that can be related to the
A. Pancreas
B. Adrenal gland
C. Thyroid gland
D. Gonads
7. Secretions of endocrine glands require receptors on cell membrane to be picked by their targets while exocrine secretions do not. This is because

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A. Exocrine glands secrete hormone directly to the blood stream
B. Endocrine glands are ductless
C. Exocrine glands are controlled by the nervous system
D. Endocrine glands produce protein-based hormones

8. Which of the following diseases is caused by a DNA virus?
A. Cold sore B. AIDS C. Swine flu D. Corona

9. Currently, natural resources are being depleted at increasing rate and waste materials are accumulated and discharged freely into water bodies. What is the best strategy to overcome these problems?
A. Decomposition C. Fossilization
B. Succession D. Recycling

10. Which stage of the nitrogen cycle needs the role of Rhizobium?
A. Decomposition C. Nitrogen fixation
B. Denitrification D. Nitrification

11. A key aspect in the process of recovery of an ecosystem through succession is that
A. Pioneer species always make the environment better
B. Higher organisms colonize the environment first
C. A climax community of lichens and mosses established at the end
D. Pioneer species always tend to be lower animals

12. What are the common plants of the boreal forest?
A. Deciduous trees C. Mosses
B. Conifers D. Epiphytes

13. Which one of the following may NOT be regulate by a plant hormone?
A. Seed dispersal C. Cell elongation
B. Cell division D. Fruit ripening

14. Bushing of a plant results from
A. Cancellation of the positive effect of apical dominance
B. Removal of the inhibitory effect of auxin
C. Decreased production of auxin by side shoots
D. Increased production of auxin by the main shoot

15. Actions such as protection of individual species, reduction of pollution and reduction of deforestation can collectively be taken as methods of
A. Agricultural expansion C. fighting global warming
B. Biodiversity conservation D. fighting soil erosion

16. Which one of the following plants is NOT endemic to Ethiopia?
A. Mango B. Zigba C. Noug D. Enset

17. Which one of the following national parks is characterized by the presence of wildlife such as Elephant, Lion, Cheetah and Giraffe?
A. Bale mountain national park C. Mago national park
B. Awash national park D. Simien mountain national park

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18. In kingdom animalia, which phylum poisons its prey by tentacles that contain stinging cells?
A. Nematoda B. Coelenterata C. Annelida D. Porifera
19. Mosses are grouped under division bryophyte because they
A. Lack true root system and vascular tissue
B. Have true leaves, stems and roots
C. Have flat and broad leaves
D. Are evergreen and photosynthesize all year round
20. If you are asked to draw a pyramid of biomass of an aquatic habitat that contain zooplanktons, herrings, sea lions and phytoplankton, which organisms should be placed at the top of the pyramid?
A. Phytoplankton C. Herrings
B. Zooplanktons D. Sea lions
21. Which one of the following is an adaptation to reduce water loss in plants? Having
A. A single layered epidermis in leaves
B. Thin broad leaves
C. Large and wide stomata
D. Leaves covered with waxy cuticle
22. Why are arctic animals such as seals, walruses and whales large in body size? To
A. Reduce heat loss C. Capture their prey without challenges
B. Move in the water bodies easily D. Increases their surface area
23. The institute of biodiversity conservation is one of the biological research institutes of Ethiopia attempting to conserve the genetic resources of the country. As such, the institute's major research efforts focus on all of the following EXCEPT
A. Microorganism conservation C. Soil conservation
B. Animal conservation D. Plant conservation
24. The importance of stains in microscopic investigations is explained by their effect in
A. Increasing resolution C. Increasing magnification
B. Creating contrast D. Decreasing air bubbles
25. One of the following problems can occur if the liver does NOT properly carry out its digestive functions
A. Increased alkalinity of the intestine
B. Incomplete digestion of polysaccharides
C. Partial absorption of digested foods
D. Partial digestion of fats

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26. Which of the following is the correct order of food substance, enzyme and breakdown product of the digestive process in the stomach respectively?
A. Starch, lipase, glucose
B. Fat, trypsin, glycerol
C. protein, pepsin, amino acid
D. Lipid, amylase, fatty acid

27. Assume you are trying to demonstrate breathing movements using artificial lung and you pulled down the rubber sheet. This action of pulling represents which step of the actual breathing process?
A. Exhalation and contraction of the diaphragm
B. Inhalation and relaxation of the diaphragm
C. Inhalation and contraction of the diaphragm
D. Exhalation and relaxation of the diaphragm

28. When three fatty acid molecules and a glycerol molecule undergo condensation reaction, they form
A. Peptide bond and protein
B. Ester bond and triglyceride
C. glycosidic bond and polysaccharide
D. phosphate bond and phospholipid

29. In water, phospholipids become organized into a bilayer. In this configuration, which part of the bilayer faces towards the water?
A. Hydrocarbon tail
B. Fatty acid tail
C. phosphate head
D. Glycerol head

30. Four students wanted to conduct chemical tests on unknown food particles and used the following chemical and materials
Student A: Test tube and biuret solution
Student B: Test tube, ethanol, water, filter paper
Student C: Test tube, benedict's solution, water bath
Student D: Test tubes and iodine solution
Which of the above students planed a test for lipids?
A. Student B B. Student D C. Student C D. Student A

31. Niacin is used to make the coenzyme
A. FAD B. ADP C. NAD D. AMP

32. The following are reaction types catalyzed by different classes of enzymes
Transferring hydrogen and oxygen atoms
Changing of the molecular form of the substrate
joining of two molecules by the formation of new bonds
Which classes of enzymes catalyze the above reactions according to their consecutive order?
A. Transferases, hydrolases, lyases
B. Oxidoreductases, isomerases, ligases

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C. oxidoreductases, hydrolases, lyases
D. Transferases, isomerases, ligases

33. The hydrogen ion that used to turn the rotor of ATP synthase come from
A. Phosphorylation of ADP
B. Oxidation of NADH and $FADH_2$
C. Hydrolysis of ATP
D. Reduction of NADH and $FADH_2$

34. What triggers fermentation during exercise?
A. Shortage of oxygen
B. Over activity of respiratory enzymes
C. production of ATP beyond the cell demand
D. Presence of excess glucose

35. Which of the following processes explains lactate production during anaerobic respiration in animal cells?
A. Oxidation of pyruvate
pyruvate
B. Decarboxylation of pyruvate
C. phosphorylation of
D. Reduction of pyruvate

36. Which one of the following descriptions holds true for the link reaction of respiration?
A. Pyruvate reacts with coenzyme A to form acetyl coenzyme A
B. Glucose is converted to pyruvate through a chain of reactions
C. Hydrogen ion and ATP are synthesized
D. Reduced NAD and reduced FAD are produced

37. Which one of the following is the neo-Darwinian idea of evolution? Evolutionary changes occur due to
A. Inheritance of acquired characteristics
B. Natural selection through genetic and behavior
C. Use and disuse of anatomical structure
D. Competitions between living organisms

38. What is the significant of *Australopitchecus afarensis* in the study of human evolution? To confirm
A. Evolution of bipedalism before large brains
B. Divergent evolution of *Homo erectus* from *Homo hablis*
C. True common ancestor of chimpanzees and humans
D. Convergent evolution of *Homo neanderthalensis* and *Homo erectus*

39. Instinctive behavior
A. Is common to all member of a species
B. Varies from individual to individual
C. May be modified by new experiences
D. Develops through trial and error

40. Which one of the following statements is an example of latent learning?
A. The random movement of woodlice in response to changes in light intensity and humidity

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B. Knowledge of a mice about the immediate environment of its burrow may help it escape from predator
C. Foraging bees returning form successful trip dance on the honey comb to recruit others to harvest a new
D. Female baboons turns its buttock toward the dominant male to signify readiness to make

41. Which one of the following is correct about the action of apoenzymes and coenzymes?
A. Coenzymes binds with cofactors to form an active enzyme
B. Apoenzymes binds with enzymes to give catalytic activity
C. Both apoenzymes and coenzymes combine with cofactors to form a non-active enzyme
D. Coenzymes bind with enzymes to give catalytic activity

42. Consider the following statement
I. Genetically determined and common to all members of a species
II. The behavior may be modified by new experiences
III. Behavior is fully functional at the first attempt
IV. Generally there is no modification of the behavior
Which numbers describe innate and learned behaviors?
A. Innate: III, IV and learned: I, II
B. Innate: I, II and learned: III, IV
C. Innate: II, III, IV and learned: I
D. Innate: I, III, IV and learned: II

43. Which one of the following is an example of social behavior?
A. Exaggeration of size by fluffing up of feathers
B. Singing of frogs on spring nights
C. Marking a territory with urine
D. Caring for offspring that are not one's own

44. Consider the following reaction
Reactant A + Reactant B + Enzymes → Intermediate → product AB + Enzyme?
At which point are the chemical bonds weakened? At the
A. Intermediate B. Reactant A C. Reactant B D. Product AB

45. How do enzymes bind and act on their substrates? By using the
A. Lock and key model and reducing energy of the substrate
B. Induced fit model and providing extra energy for the substrate
C. Lock and key model and changing themselves into products
D. Induced fit model and putting the substrates under tension

46. Which of the following external factors influence the turnover rate of enzymes by completely damaging their tertiary structure?
A. Competitive inhibitors C. Reversible inhibitors
B. Substrate concentration D. Temperature and PH

47. During baking of Injera, the CO_2 produced by yeast is used to

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A. Kill contaminants that spoil Injera C. Bake Injera with low temperature
B. Give Injera its typical texture D. Increase the shelf life of Injera

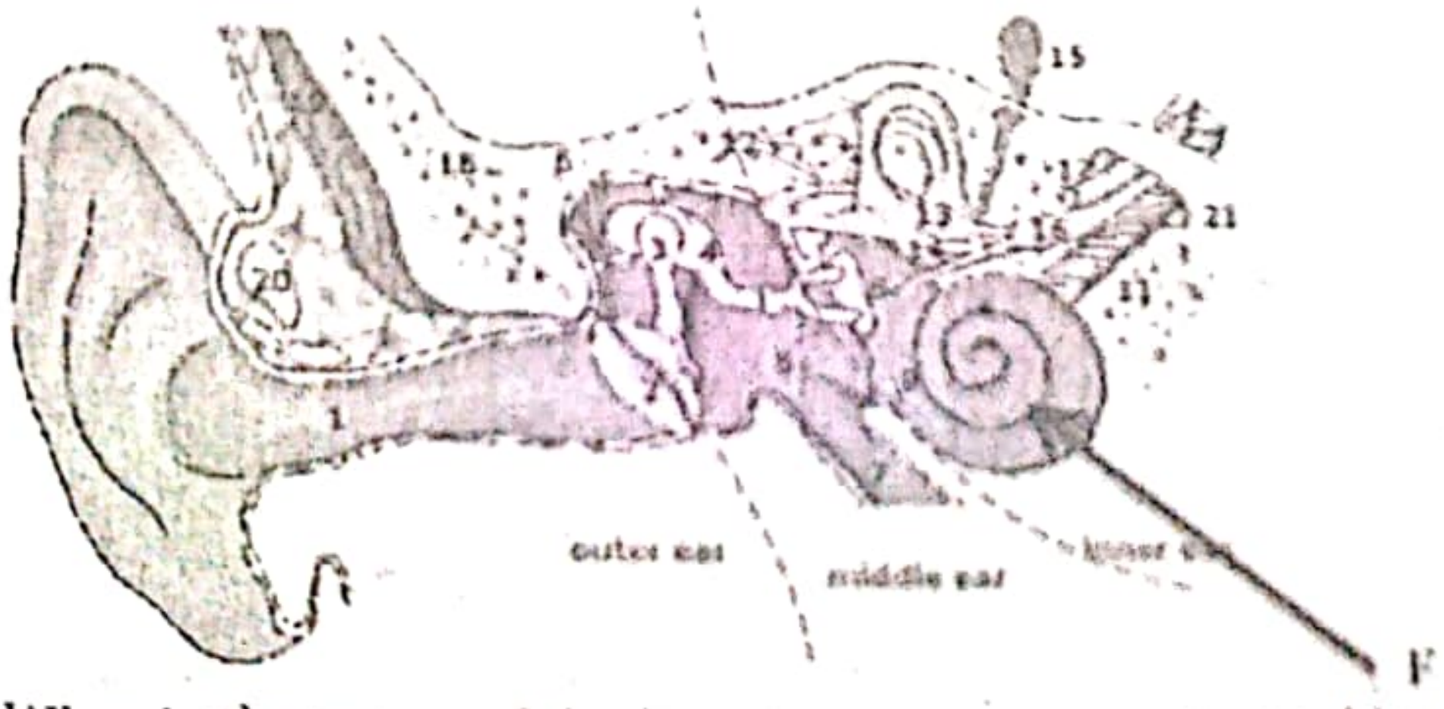
48. The nerve impulse is transmitted from one neuron to another by a neurotransmitter at the
A. Axon B. Cell body C. Synapse D. Myelin sheath

49. What happens if the axon is stimulated? Formation of
A. Negative charge in both side of the axon
B. Negative charge inside the axon
C. Positive charge inside the axon
D. Positive charge outside of the axon

50. Assume you wanted to demonstrate a simple reflex action using the knee jerk reflex. You asked a friend to sit with one crossed over the other. Which one of the following indicates the correct demonstration?
A. Pulling down your friend's leg that is followed by downward movement
B. Hitting your friend on the toes followed by sudden upward movement of the leg
C. Bending up your friend's leg followed by downward movement of the leg
D. Hitting your friend below the knee cap followed by a sudden upward movement

51. A finch with an average length beak was not able to obtain insects out of the cracks in the bark of trees as well as one with a longer beak. It was also not able to crush seed as well as one with a shorter beak. Over time, those with the longer beaks and those with the shorter beaks increased in numbers, while those with average length beaks decreased in number. Which type of selection has been operated in this scenario?
A. Disruptive B. Directional C. Behavioral D. Stabilizing

52. This item is based on the following diagram of the human ear.



What is the name of the human ear part represented by letter 'F' in the above diagram?

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A. Semicircular canal
B. Sacculus
C. Utriculus
D. Cochlea

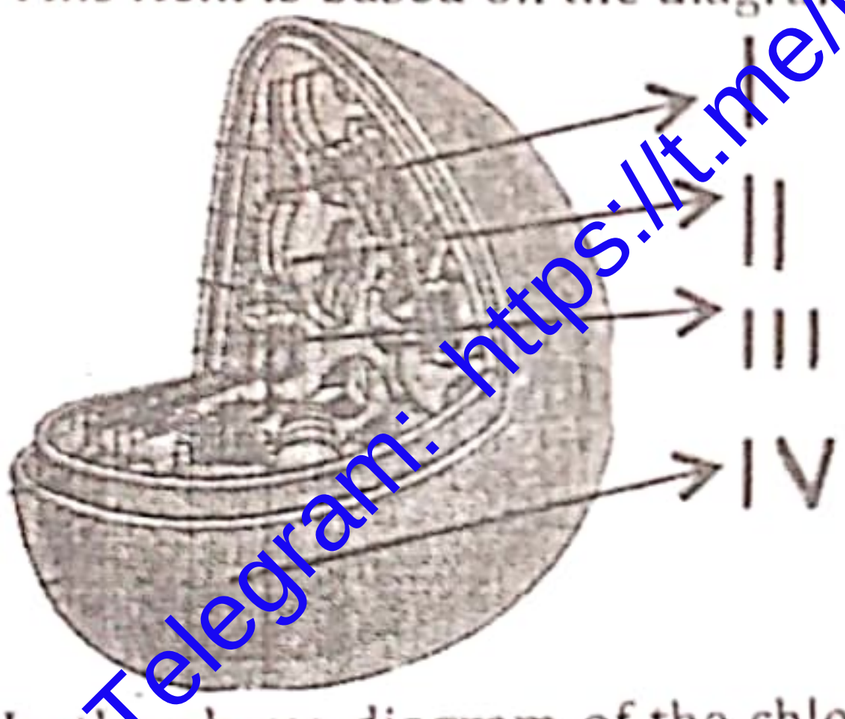
53. The cell theory states that cells
A. Contain different organelles with no specialization
B. Have basically different composition
C. Come from pre-existing cells by division
D. Come form non-living objects

54. Which type of chromosome mutation cause Leukemia?
A. Duplication B. Insertion C. Inversion D. Deletion

55. Which theory of the origin of life states that life has no beginning or ending?
A. Cosmozoan theory
B. Special creationism
C. Eternity theory
D. Biochemical theory

56. Which one of the following is paleontological evidence of evolution?
A. The presence of homologous structures in living organisms
B. Traces of organisms from the remote past
C. Similarities in the DNA base sequences of different organisms
D. Similarities in the development of embryos in different organisms

57. This item is based on the diagram of a chloroplast given below.



In the above diagram of the chloroplast, what are the sites of the light dependent and light independent reactions of photosynthesis respectively?
A. III and IV B. I and II C. II and I D. IV and III

58. A color blind daughter can be born when the
A. Mother is color blind and the father is normal
B. Father is color blind and the mother is normal
C. Mother is carrier and the father is color blind
D. Father is normal and the mother is carrier

59. A farmer cut off several regions of a stem near to the buds of a plant. He removed some of the leaves and dipped the cuts end in some hormones rooting powder and planted the cutting in compost. He kept the cuttings well watered and within a few weeks they developed their own root system and became new independent plants. Which one of the

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following is correct about this experiment? The new plants are most probably
A. Phenotypically and genetypically different form the parent plant and from each other
B. Genetically and phenotypically identical to the parent plant and to each other
C. Genotypically identical but phenotypically different from the parent
D. Phenotypically identical but genetically different from each other

60. During DNA replication
A. DNA helicase transports nucleotides that are used to build the new DNA
B. The base sequence in the DNA will be slightly changed from the parent DNA
C. DNA polymerase assembles new nucleotides into a new strand
D. The parent DNA will become different from the daughter DNA

61. With four different bases to work with, there are 64 possible triplet codes, but only 20 amino acids are used to make all the different proteins. This is because the DNA code is
A. Degenerate code
B. Non-overlapping code
C. Semi-conservative code
D. Universal code

62. If mRNA brings the code "UGA" into the ribosome
A. Translation ceases and protein synthesis terminates
B. tRNA will bring the amino acid methionine
C. ribosome will initiate protein synthesis
D. tRNA will bring "UGA" anticodon

63. breakdown of the thick spongy wall of the uterus at the end of the menstrual cycle is associated with the
A. formation of the corpus lateum
B. maturation of the follicle
D. decrease in progesterone and estrogen concentration
C. increase in oestrogen and progesterone concentration

64. The first incubation period of HIV is characterized by a duration of
A. 20 years before all the symptoms appear in well-nourished people
B. 2 weeks to produce enough antibodies
C. 3 - 12 weeks when mild symptoms are manifested
D. 2 - 3 years for the progress of infection to AIDS

65. Which one of the following is a behavioral method of temperature regulation in homoiotherms?
A. Piloerection B. Licking C. Aestivation D. Vasodilation

66. What will happen if the water contents of the blood it too high?
A. The second coiled tubule of the kidney becomes more permeable
B. Antidiuretic hormone secretion decreases

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C. Osmoreceptors stimulate pituitary gland to release ADH
D. The kidney reabsorbs more water back into the blood.

67. The movement of water from the roots up to the top leaves of tall trees is carried out by
A. Loss of water by transpiration and its replacement by osmosis
B. Active transport in the roots and osmosis in the leaves
C. The low cohesive and adhesive forces of water in the xylem
D. Active transport in both the roots and leaves

68. From the following options, which one contains the steps of writing a scientific report in the correct order?
A. Title → Prediction → Hypothesis → Procedure → Evaluation → Conclusion → Results → Acknowledgement
B. Title → Hypothesis → Predication → Procedure → Results → Conclusion → Evaluation → Acknowledgement
C. Title → Hypothesis → Prediction → Evaluation → Results → Procedure → Conclusion → Acknowledgement
D. Title → Hypothesis → Procedure → Prediction → Results → Conclusion → Evaluation → Acknowledgement

69. What type of force is responsible if a stool solution is placed in a centrifuge spins the tube at high speed and the solid particles are forced to settle down at the bottom?
A. Cohesive force
B. Adhesive force
C. Centripetal force
D. Gravitational force

70. Ten percent of attendants of a wedding ceremony were sick just after having lunch at the ceremony. What should a student do first to prove scientifically that the sickness was actually linked to the food eaten at the ceremony? She should
A. Give drugs to the sick individuals before they contaminate others
B. Take food sample to a laboratory and analyze the presence of toxin
C. Ask herself why ten percent of the attendants were sick
B. Take stool sample of the sick individuals and analyze the presence of toxins

71. Which one of the following importance of water is directly related with its ideal viscosity?
A. Transport medium
B. Habitat for aquatic animals
C. Reactant in photosynthesis
D. Reaction medium

72. Which property of water prevents ponds from drying during the long dry season? High
A. Viscosity property
B. Latent heat of vaporization
C. Transparent ability
D. Surface tension

73. The following table compares prokaryotes and eukaryotes

Feature	Prokaryotes	Eukaryotes

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DNA	X	Linear
Mitochondria	Absent	Y
Ribosomes	Present (70s)	Z
Cell membrane	Present	Present

In the above table X, Y and Z can be respectively stated as
A. Linear, present, present (70s)
B. Circular, present, present (70s)
C. Linear, present, present (80s)
D. circular, present, present (80s)

74. Which one of the following is NOT the function of the plasma membrane in eukaryotic cells?
A. Isolating the cell from the external environment
B. Transferring chemical energy from carbohydrates to ATP
C. Responding to external signals
D. Exchanging materials and information

75. Which one of the following is correct with respect to the structural components of the plasma membrane?
A. Integral protein are found inserted in the lipid bilayer
B. Peripheral protein are located outside the lipid bilayer
C. Channel proteins cover the outer layer of the lipid bilayer
D. Glycoproteins are found on the cytoplasmic side of the cell membrane

76. Suppose three potato cylinders were kept in 25%, 15% and 5% sucrose solution respectively and a fourth cylinder was placed in distilled water. Which of the cylinders will be more flaccid? The cylinder in
A. 5% solution
B. 15% solution
C. Distilled water
D. 25% solution

77. Consider the following table which shows a result of an experiment where red blood cells were placed in three different solutions X, Y and Z.

Solutions	Changes observed on RBCs
X	Shrink
Y	Burst
Z	No change

The three different solutions in which the blood cells were placed in order of X, Y, Z were
A. Isotonic, hypertonic, hypotonic
B. Hypotonic, isotonic, hypertonic
C. Hypotonic, hypertonic, isotonic
D. Hypertonic, hypotonic, isotonic

78. A more comprehensive definition of biodiversity can be given as the
A. Richness, ecological and genetic variability of species
B. Number of individuals of a species in a given area
C. Diversity of plants and animals in a given area
D. Number of individuals of a given animal species

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79. At which stage of the population growth curve is the rate of natality and the tare of mortality are expected to be equal?
A. Lag phase
B. Decline phase
C. Log phase
D. Stationery phase

80. Identify the correct statement about intra-specific competition? It is a competition
A. Between individuals with different adaptation
B. For all types of requirements
C. Of two species inhabiting the same ecological area
D. That leads to local extinction of the species

81. Why *Drosophila melanogaster* are considered ideal for genetic experiments? Because they
A. Contain chromosomes with several genes
B. Have the highest number of chromosomes
C. Breed and keep in large numbers easily
D. Complete their life cycle within one month

82. Which one of the following statements best describes the mechanisms by which meiosis brings about variation? Variation in meiosis is because of
A. Independent assortment and crossing over
B. Frequent spontaneous mutations
C. Variations in the content of the cytoplasm between the cells
D. Reduction division that leads to haploid cell formation

83. In the tropical regions, where there is higher temperature, which one of the following is most likely to take place in the C3 plants? Rubisco binds with
A. CO_2 to produce glucose and water
B. O_2 to produce GP and phosphoglycolate
C. GP to produce lipid
D. H_2O to produce TP

84. During the light independent reactions of photosynthesis
A. Glucose is directly produced from GP
B. RuBP reacts with CO_2 to directly form TP
C. Some of the GP is used to regenerate RuBP
D. GP is reduced to TP by reduced NADP

85. The suitability of the structural arrangement of a chloroplast in relation to its function can indicated by the fact that
A. All pigments are floating around the solution of the stroma
B. All pigments in the photosystem are linked to the electron transport chain
C. Carotenoids are used as a reaction center from which electrons are excited

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D. The different pigments absorb the same wavelength of light

99. Which one of the following events occurs during the day time in the CAM photosynthetic pathway?
A. CO_2 reacts with malate
B. The stomata open to allow CO_2
C. CO_2 enters into Calvin cycle
D. Oxaloacetate is formed

100. How many turns of the light independent reaction cycle are required to make one molecule of triose phosphate?
A. 3
B. 2
C. 6
D. 4

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EDUCATIONAL ASSESSMENT AND EXAMINATIONS SERVICE ETHIOPIAN SECONDARY SCHOOL LEAVING CERTIFICATE BIOLOGY EXAMINATION 2015/ JULY, 2023

1. Which one of the following sentences is correct about the activities of the heart?
   A. The noise of heartbeat one hears through a stethoscope is the sound of valves
   B. As the heart beats blood travels through the arteries with a uniform pressure
   C. Diastole is when the heart muscles contract and force the blood out
   D. Systole is when the heart muscles relax and fills with blood
2. During culturing microorganisms, all of the following activities are important. **EXCEPT,**
   A. Tilting the lid of the petri dish during inoculation
   B. Adding distilled water to a sterilized agar
   C. Sterilizing the inoculating loops
   D. Transferring target microorganisms into an agar medium
3. Which one of the following diseases can be caused by more than one type of microorganisms?
   A. Cholera
   B. Gastroenteritis
   C. Typhoid
   D. Tuberculosis
4. Which one of the following diseases is correctly explained with its causes or symptoms?
   A. Syphilis is caused by *Neisseria gonorrhea*
   B. Gonorrhoea has a burning sensation symptom while urinating
   C. Gonorrhoea is caused by *Treponema pallidum*
   D. Syphilis has sore ulceration on the genitals as a symptom
5. The names of which two taxa are used in binomial nomenclature?
   A. The two lowest units of classification
   B. The two middle units of classification
   C. The highest and the lowest
   D. The two highest units of classification
6. In a dihybrid self- crossing of a plant with genotype DdCc, 3600 offspring were produced. How many of the off-springs will have the ddcc genotype?
   A. 300
   B. 450
   C. 150
   D. 225
7. Identify the correct statement regarding translation in eukaryotic cells.
   A. Helicase enzyme begins to unwind a section of DNA

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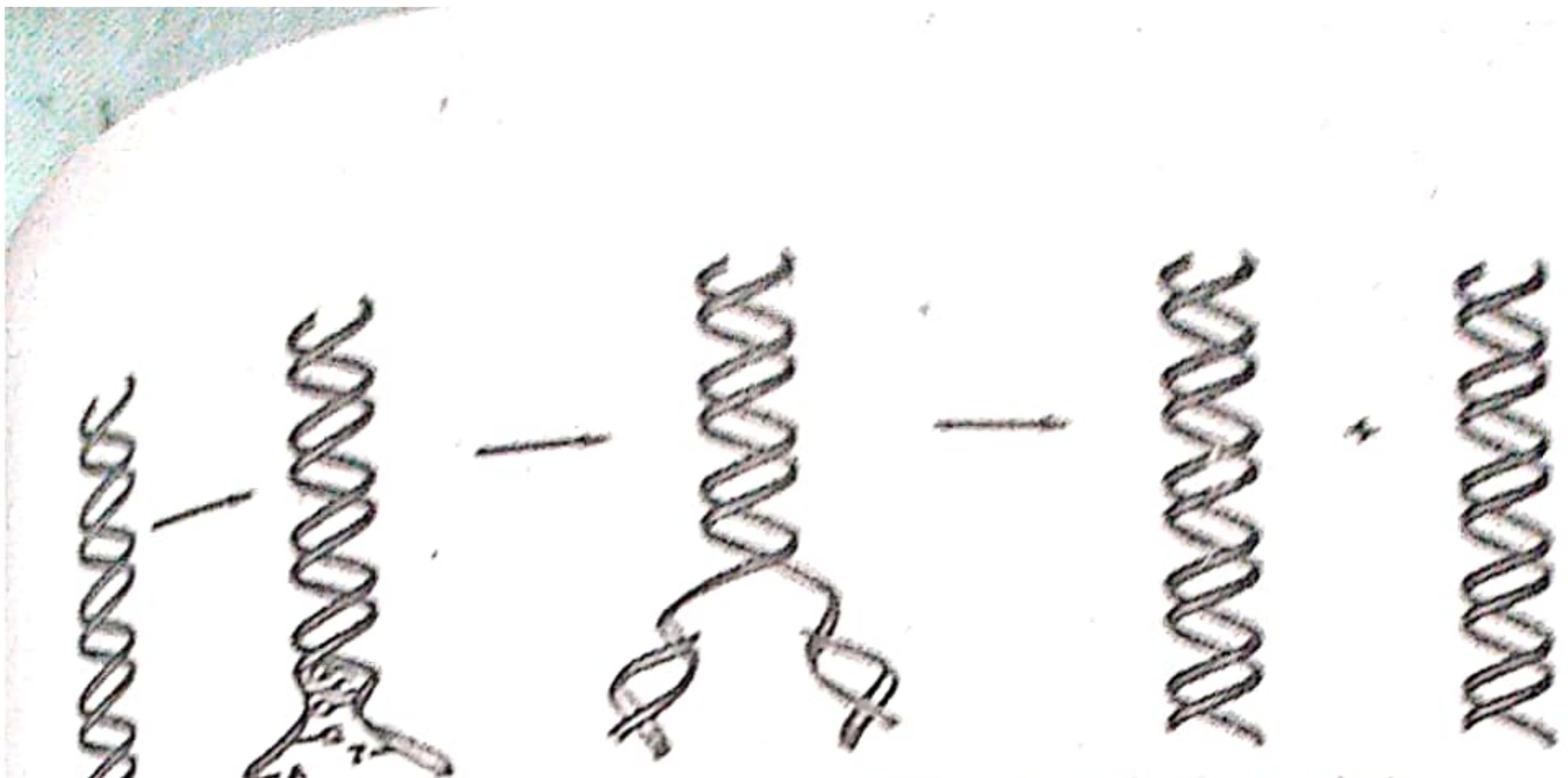

B. Polymerase assembles free RNA nucleotides into a chain
C. First two codons of the mRNA enter the ribosome
D. RNA polymerase moves along the antisense strand

8. Which of the following sentences explains how crossing over brings variation?
   A. In meiosis I, there is splitting up of pairs of homologous chromosomes and line up
   B. In meiosis I, there is exchange of a section of DNA between chromatids
   C. In meiosis II, homologous chromosomes align themselves independent of the other pairs
   D. In meiosis II, chromatids from each chromosome are separated at random

9. Consider the following model of DNA replication
   Which one of the following is true about A, B and C?
   A. A and B are identical with each other but not with C
   B. C is not identical with A and B
   C. B and C are not identical with each other
   D. A, B and C are identical with each other

10. What is the central role of tRNA in translation?
   A. Carrying genetic information from DNA in the nucleus to ribosomes
   B. Recognizing the codon of mRNA and bring the correct amino acid to the ribosomes
   C. Acting as a template for converting a genetic code in to amino acid sequence
   D. Assembling the amino acid in to polypeptide chain

11. Which one of the following sentences explains the traditional use of biotechnology? Production of
   A. Insulin using transgenic bacteria under a laboratory
   B. Dairy products at household level industrial fermenter
   C. Myco- proteins
   D. Vinegar using

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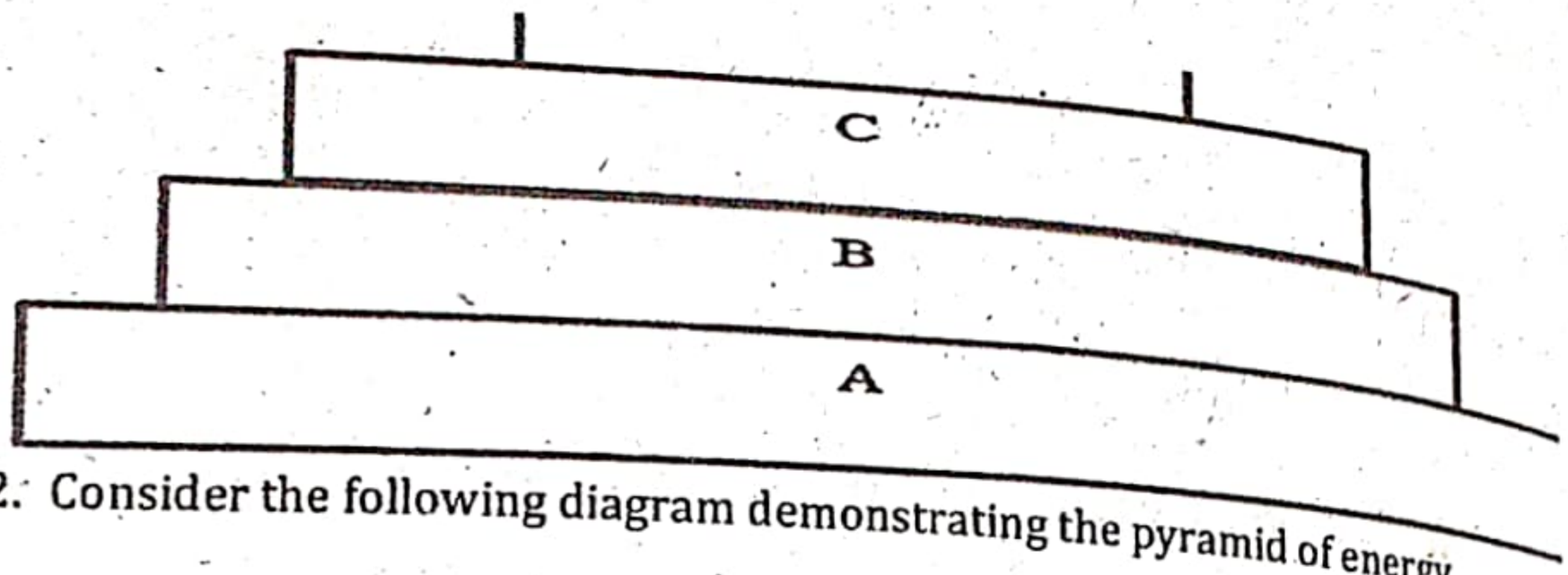

12. Consider the following diagram demonstrating the pyramid of energy

Which of the following is **INCORRECT** about the above pyramid of energy?

A. The organisms of each trophic level use large proportion of the energy to build their body
B. The organisms of the first trophic level convert light energy into chemical energy
C. The energy lost as heat to the surrounding in each trophic level is insignificant
D. The amount of energy transferred through the trophic levels drops with each step up of the pyramid

13. Which one of the following sentences is correct about mitosis? It
A. Is the division of sex cells resulting in identical daughter cells
B. Produces four different daughter cells from a single cell
C. Is the division of somatic cells to make identical daughter cells
D. Produces half number of chromosomes from the original cell

14. Across between pea plants having heterozygous phenotype (Rr) has been made. Which of the following is correct about the offspring according to the Mendelian inheritance? There is genotypic probability of getting,

A. Rr is25% B. Rr is 75% C. RR is 50% D. rr is 25%

15. Which one of the following is correct about DNA? It is
A. Small molecule and coding for only protein
B. Made of phosphate, sugar and bases
C. A single stranded molecule
D. Less stable and degrades quickly

16. Which one of the following phyla is characterized by flexible cartilaginous rod-like structures that run along the dorsal side of the body?
A. Phylum Mollusca
B. Phylum Chordata
C. Phylum Annelida
D. Phylum Arthropoda

17. Which one of the following groups of microorganisms is correctly matched with its phylum?
A. Star fish – Phylum Echinodermata
C. Snail –
Phylum Annelida

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B. Fish – Phylum Arthropoda D. Earthworm – Phylum Mollusca

18. Which one of the following events happens during carbon cycle? Carbon dioxide is,
   A. Transferred from plants to animals by their feeding relationship
   B. Absorbed from the air when microbes decompose plants and animals
   C. Removed from the air by animals during respiration
   D. Return in to the air by plants during photosynthesis

19. Which one of the following is not the function of the liver?
   A. Controlling cholesterol C. Monosaccharide metabolism
   B. Storing glycogen D. Bile formation

20. A zoologist studied 5 vertebrate species (spA, spB, spC, spD & spE) and described their features as follows.
SpA: Smooth moist skin; pass parts of their lives in water and on land
SpB: Streamlined bodies with scales; gills for gaseous exchange
SpC: Feathers over the body and scales on legs; endothermic; horny beak
SpD: Dry skin with scales; ectothermic; lunges for respiration
SpE: Mammary glands; endothermic; higher internal body temperature
Based on the features listed above; spA,spB, spC, spD, and spE can be classified under the vertebrate classes, respectively,
   A. Amphibia, Reptilia, Avea, Mammalia and Pisces
   B. Pisces, Aves, Reptilia, mammalia and Amphibia
   C. Pisces, Amphibia, Aves, Reptilia and Mammalia
   D. Amphibia, Pisces, Aves, Reptilia and Mammalia

21. Which of the following plant adaptation mechanisms helps cactus to adapt in the desert environment?
   A. Spreading its seeds in wider areas C. having broad leaves
   B. Increasing number of stomata in its leaves D. storing water in its tissue

22. What is the function of dermis of the skin? It
   A. Act as protection against heat loss C. Forms a water proof around the body tissue
   B. Protect the entry of pathogens into the body D. Contain blood vessels and sweat glands

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23. Children who play by spinning round and round fast, when they stop, they dizzy. Why?
    A. The otoliths round in the same direction with the head
    B. The optic nerves reports that the children are still moving
    C. The fluid in their semicircular canal keeps on moving after they stop
    D. The sensory hairs vibrate highly in response to movement of their body
24. When does menstruation start in women with no obvious reproductive health problem? It happen when,
    A. The fertilized ovum reaches the uterus and attaches into the thick, spongy lining
    B. The FSH stimulates the development of a follicle in the ovary and the egg ripens
    C. Pregnancy occur and the embryo obtain nourishment and oxygen
    D. Thick spongy lining of the uterus detaches in the absence of pregnancy
25. Which one of the following sentence is correctly matched with function of kidney structures?
    A. Bowman's capsule – act as a filter to produce urine
    B. Cortex – monitors the osmotic pressure of the blood
    C. Glomerulus – concentrates urine and conserves water
    D. Loop Henle – involved in ultrafiltration
26. Which of the following temperature controlling mechanisms in homiotherms is categorized into behavioral method?
    A. Sweating when the temperature increases
    B. Constriction of capillaries when the temperature falls
    C. Shivering as a result of drop in temperature
    D. Seeking shade when the temperature rises
27. Industries in the medicine sector use glucose oxidase in clinistix strips for,
    A. Restoring blood sample to area of heart muscle
    B. Detecting toxic metabolites in the blood
    C. Easy diagnosis of diabetes by examining a urine sample
    D. Reducing the viscosity of mucus
28. In the systematic naming of enzyme, assume an enzyme is designated as EC. 3.4.11.1. Which of the following correctly describes this enzyme?
    A. 11- gives a sub-sub classes to which the enzyme belongs
    B. 1- a reference of enzyme commission
    C. 4 - shows to which main class the enzyme belongs

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D. 3 – indicates a sub class to which the enzyme belongs

29. Which of the following sentences explains the lock-and -key model of enzyme action? The
   A. Attached substrate-the enzyme complex enters in to a transition state.
   B. Model explains enzyme inhibition in a complex manner.
   C. Shape of the substrate is complementary to the active state.
   D. Attachment of substrate to an enzyme produces a conformational change.

30. Which of the following sentence is correct about apo-enzymes? They are
   A. Cofactors that give an enzyme its catalytic activity.
   B. Proteins that combine with cofactors to form active enzymes.
   C. Organic molecules and many are derived from vitamins.
   D. Inorganic molecules that bind loosely with cofactors.

31. How does a competitive inhibitor affect an enzyme catalyzed reaction? By
   A. Denaturing the active site of the enzyme
   B. Producing conformational change of the enzyme
   C. Binding to the allosteric site of the enzyme
   D. Blocking the active site of the enzyme

32. Which one of the following leaf structure is correctly matched with its function?
   A. Palisade mesophyll – the main photosynthetic tissue of the leaf
   B. Waxy cuticle – controls the entry of carbon dioxide in to the leaf
   C. Guard cell – have lots of air space and gas exchange area of the leaf
   D. Spongy mesophyll – helps to prevent water loss from the leaf

33. Which one of the following statements correctly describes the movement of materials in plants?
   A. Passive transport can take place in the phloem
   B. The transport of organic materials is due to transpiration
   C. Water is transported in dead tissue
   D. Active transport can take place in the xylem

34. What will happen if an agriculture extension worker injects gibberellin in to a dwarf plants? The
   A. Leaves will fall
   B. Stem will grow
   C. Roots will elongate
   D. Side shoots will grow fast

35. A biologist collected two types of seeds (SA and SB) from the field and sawed them in moist soils enriched with nutrients. Then, he recorded his observation as follows:

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SA

- The seedling of this seed emerged from the soil curved while pushing up through the soil
- The hypocotyl straightened and carried the cotyledons and plumule above the soil surface

SB

- The plumule of the seedling pushed its way out of the soil, while the cotyledon remained underground
- Unlike what occurred during germination of SA, the plumule of this seed did not form a hook

Which one of the following is correct about the observation of the biologist?

A. Both SA and SB are of monocots
B. SA is of a monocot, while SB is of a dicot
C. Both SA and SB are of dicots
D. SA is of a dicot, while SB is of a monocot

36. Which one of the following statement correctly describes the property of water? It
A. Contracts when it freezes and ice acts as an insulator
B. Takes low energy to change into vapor
C. Allows all light wavelengths to pass through its various depth
D. Requires high amount of energy to heat up

37. An organic molecule which is an important part of structure of chromosomes is,
A. Amino acid
B. Glycogen
C. Amylose
D. Triglycerides

38. In animal cell glycogen is hydrolyzed quickly to release glucose for respiration. This is due to the presence of several,
A. Microfibrils in its side chain
B. α-1,6-glycosidic bonds
C. β-1,4- glycosidic bonds
D. Enzyme that hydrolyze it easily

39. A student has conducted an experiment using different enzymes to assess the enhancement of reaction rates under different conditions. She represents her results as indicated in the following table.

Enzymes used	Cofactors used	Rate enhancement	Optimum temperature in °C
X	Zn++	1.4×10^{17}	37
Y	Cu++	7.7×10^{6}	45

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	No cofactor	$4.6x10^5$	28
M	Co++	$1.7x10^8$	30

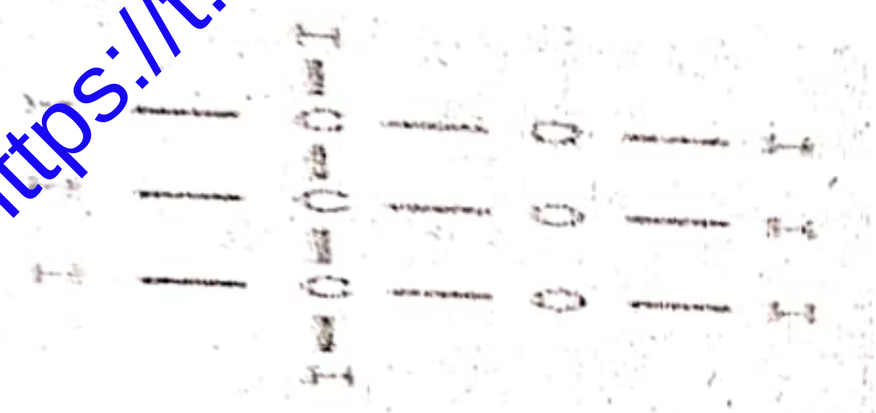
Which of the following is an acceptable conclusion of the experiment?

A. Enzyme 'X' could have enhanced the rate of better at temperature slightly higher than 37°C
B. Enzyme 'M' cannot enhance the rate of reaction further if the temperature exceeds 30°C
C. The rate enhancement for enzyme 'Z' is less because it didn't use a cofactor
D. As temperature increase the rate of chemical reaction decrease

This item is based on the following diagram which depicted structural formula of a biological molecule.

The molecule represented by the above structure is,

A. Cellulose
B. Amylopectin
C. Fatty acid
D. Glycerol



Which of the following is an Ethiopian endemic plant?

A. Noug seed
B. Avocado
C. Bahir zaf
D. Bunna

When a biologist share her research findings with other fellow biologists, she is

A. Making her hypothesis conclusion
C. Drawing her
B. Reporting her result data
D. Analyzing her

Walia ibex is an endemic animal species in Ethiopia which is found in,

A. Nechisar National Park National Park
C. Bale Mountain
B. Omo National Park Mountain National Park
D. Semien

All of the following sentences are correct about pollution. **EXCEPT**,

A. Deforestation increase the amount of CO_2 absorbed from the atmosphere
B. Decomposition of dead bodies increase the amount of CO_2 in the atmosphere

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C. Burning fossil fuels increase pollution due to the release of hydrocarbons
D. Burning of wood increase pollution by casing global dimming which blocks sunlight

45. This item is based on the table below which shows list of molecules

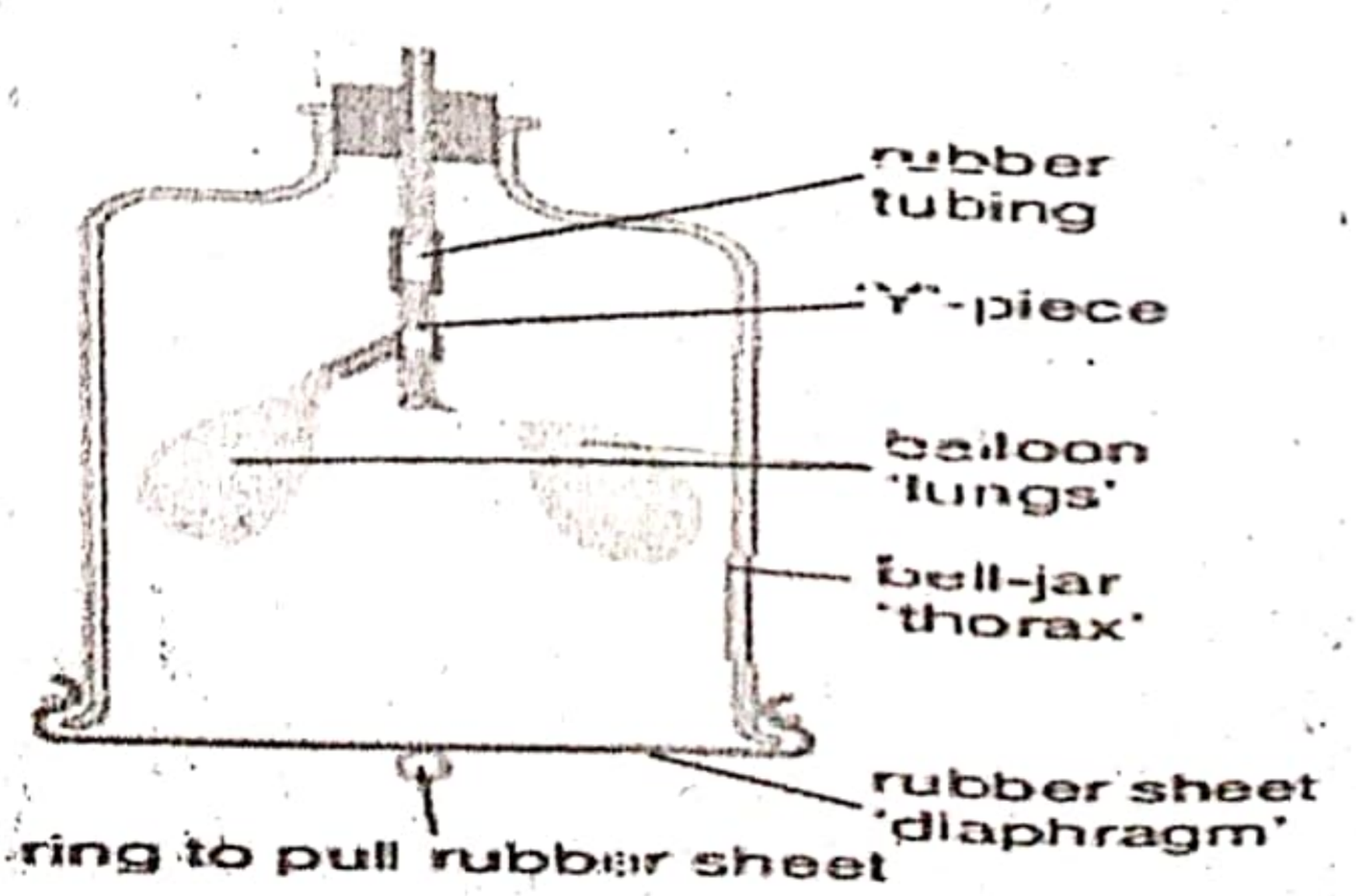
No	Molecules
1.	Calcium carbonate, glucose and methane
2.	Carbon dioxide, linoleic acid and water
3.	Linoleic acid, Glucose and methane
4.	Iron III oxide, glycine and glucose

In the above table, which number holds only organic molecules?
A. 2 B. 3 C. 4 D. 1

46. Which of the following enzyme is correctly matched to the food stuffs it acts on?
A. Pepsin – protein
B. Maltase – lipids
C. Amylase - fats
D. Trypsin - starch

47. Suppose your friends fall suddenly due to heart attack while he is playing football. The victim needs artificial breathing to keep him alive until medical support arrives. Which activity is strictly prohibited in this situation?
A. Remove any obstacle from the mouth that might block the air way
B. Open the air way by removing any obstacles from the mouth
C. Make sure the air way is open and the head is tilted back
D. Give mouth –to- mouth resuscitation if he is conscious

48. Consider the following figure which demonstrates air movements during breathing using the bell-jar model representing the lung.



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Which of the breathing movement happens when air enters into and exit from the balloon lungs?

A. Air enters when the ball- jar thorax contracts
B. Air enters when the rubber sheet is pushed up
C. Air exits when Y-piece is rotated both clockwise and counter clockwise
D. Air exits when the balloon lungs have high pressure than the external environment

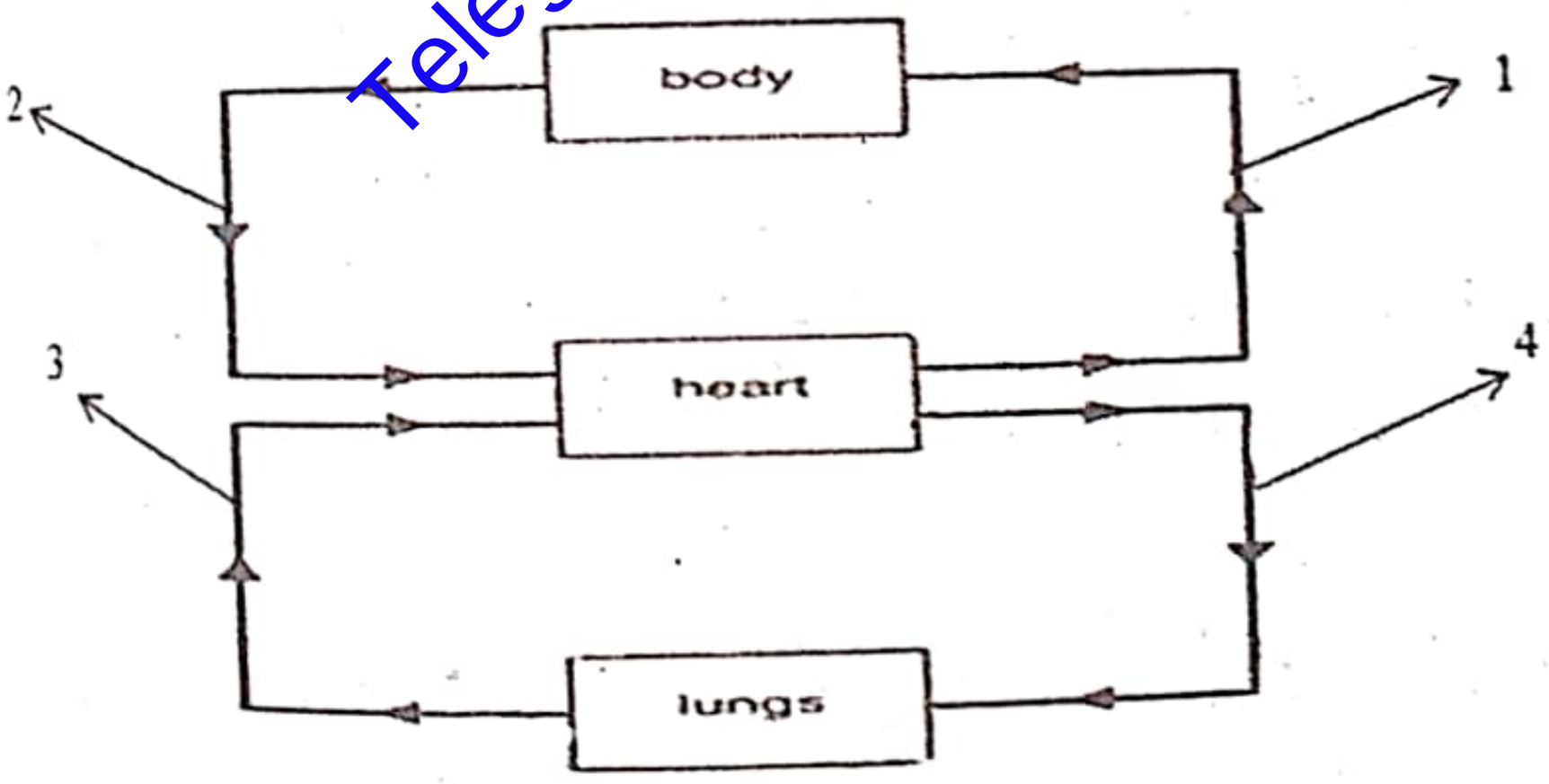
49. Which one of the following structures of the heart is correctly matched with its function?

A. Right atrium – contracts and forces deoxygenated blood into the lungs
B. Left ventricle – contracts and force oxygenated blood around the body
C. Right ventricle – contracts and forces blood into left ventricle
D. Left atrium – contracts and force blood into the right ventricles

50. In the aerobic cellular respiration

A. Krebs cycle take place in the cytoplasm
B. Glycolysis take place in the mitochondria
C. Electro transport chain take place in the cytoplasm
D. Pyruvate enters into the mitochondria for further stages

51. Consider the following diagram showing blood circulation system in the human body



In the above diagram, the part indicated by number 1, 2, 3 and 4 are respectively

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A. Aorta, vena cava, pulmonary vein, pulmonary artery
B. Pulmonary artery, pulmonary vein, vena cava, Aorta
C. Pulmonary vein, Aorta, vena cava, pulmonary artery
D. Vena cava, pulmonary artery, Aorta, pulmonary vein

52. Which one of the following descriptions of proteins is correctly matched with its function in a cell membrane?
A. Peripheral protein – anchor integral proteins to the cytoskeleton
B. Glycoprotein – reduce the fluidity of the membrane
C. Channel proteins – larger molecules can pass through them
D. Carrier proteins – specific ions can pass along them

53. Suppose a hypothetical cell of 15% solute concentration is put in a beaker containing 90% water and 10% solute. In which way will the water move?
A. From the cell to the beaker and out of the cell
B. From the beaker to the cell and into and out of the beaker
C. Both into
D. Both

54. Which of the following is an example of a learned behavior?
A. Nest-building
B. Knee-jerk reflex conditioning
C. weaving a web
D. classical

55. Which one of the following is correct about transport of substances across the cell membrane?
A. Simple diffusion - larger particles transported through channel proteins
B. Facilitated diffusion - particles move from high water potential to lower one
C. Passive transport - particles require energy in the form of ATP to drive the transport
D. Active transport – particles move from low to higher concentration using carrier proteins

56. Which one of the following process of aerobic respiration takes place in the kreb's cycle?
A. Hydrogen atoms carried by reduced NAD and FAD are released and split into protons and electrons
B. Two molecules of ATP are used to phosphorylate each molecule of glucose
C. Two carbon group from acetyl Coenzyme A react with a 4-carbon compound to form citrate

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D fructose 1,6 bis phosphate is split into 2 molecules of the 3-carbon sugar G3P

57. In the course of evolution, which of the following is true about the human brain?

A Modern humans have a brain size between 0.1% and 1% of their body mass

B Australopithecus have a brain size between 1.8% and 2.3% of their body mass

C The pre human brain uses 80% of the resting energy requirement

D Homo – Sapiens brain uses 25% of the resting energy requirement

58. All of the following are example of innate behaviors. EXCEPT.

A Training killer whales at marine parks

B Increasing breathing rate during exercise

C Produce tears when dust gets into the eyes

D. A fly moving its maggot in to more humid environment

59. Which one of the following is an example of insight learning?

A Salivation of a dog when a bell was rung in the absence of food, which was paired with the conditioned stimulus

B Use of boxes after several trials, chimpanzee placed in an enclosed area to obtain bananas placed on the top an enclosure

C. Exhibition of modification of natural behaviors of guide dogs, horses, or zoo animals to desired behaviors

D Lack of response from a lady, which sat in a shop in which a strong deodorant was continuously released from a can

60. Which one of the following an example of territorial behavior of animals?

A A male robin threatening an intruder

B. Fixed action of zebra fish to find a mate

C Penguins huddling to reduce heat loss

D Honey bees nesting in a tree

61. One of the following renowned Ethiopian biologists has developed a vaccine for the rinderpest diseases.

A Professor Sebsebe Demissew

B Professor Yalemtsehay Mokonnen

C. Professor Beyene Petros

D. Professor Tilahun Yilma

62. A grade nine student has planned to study animal cells using a compound light microscope. She prepared a slide of stained human

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cheek cells for this purpose. Which one of the following activities should she first do?

A. Adjust the iris diaphragm until the maximum brightness is coming
B. Set up the microscope with the lowest power objective in place
C. Using course focusing knob to rotate the noise piece for a higher magnification lens
D. Looking through the eyepiece lens to move the objectives down

63. Which one of the following is true about plant cells? They have
A. Vacuole which controls all activities of the cell
B. Lysosomes which contain hydrolytic enzymes
C. Chloroplast which is filled with sap, liquids and minerals
D. Cell wall which gives support and strengthen to the cell

64. A plant cell is placed in a solution which has lower solute concentration than its contents. After a few minutes the cell will become,
A. Turgid B. Flaccid C. Plasmolysed D. Burst

65. Which one of the following is correct about human food digestion?
A. The stomach physically breaks down the ingested food
B. Mouth use protease enzyme to break down proteins
C. Large intestine is the site of lipid hydrolysis
D. The small intestine is site of absorption of digested molecules

66. Sickle cell anaemia is caused by the replacement of glutamic acid by valine at a specific position in the beta chains of haemoglobin. This change is an example of which mutation?
A. Addition B. Substitution C. Inversion D. Deletion

67. Stanley Miller conducted spark – discharge experiment by taking a mixture of gases that he thought to represent the primitive atmosphere of the Earth. All of the following were found in his gas mixture. EXCEPT,
A. NH_3 B. CH_4 C. H_2 D. CO_2

68. Which of the following sentence explain the Darwin theory of evolution?
A. Environment changes, creating a need for organism to change
B. There is a natural variation in features that are heritable
C. New feature acquired in life time are passed through the offspring
D. Their development of new feature in order to survive

69. Comparative anatomy provides evidence of evolution because it,
A. Looks at structural similarity of organisms
B. Has remain of animals and plants from the remote past

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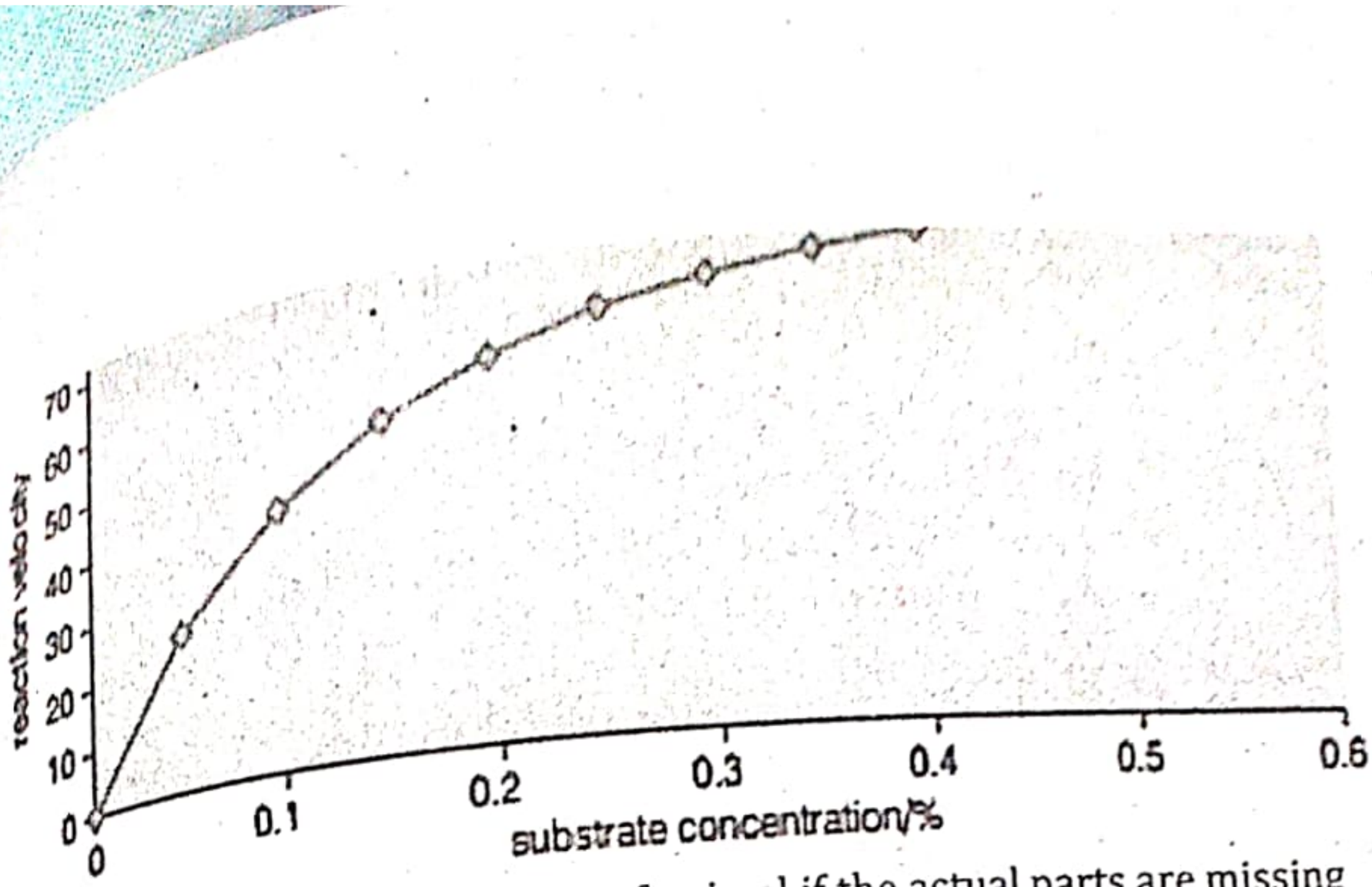

C. Has an impression of animal if the actual parts are missing
D. Studies the development of the embryos of vertebrates

70. About 3 million years ago the Isthmus of Panama separated the shrimp population of the Caribbean Sea and created to population of shrimps, one on either side of the Isthmus. They still look remarkably similar, but they do not inter breed with each other. Which type of isolation is described by this scenario?
    A. Seasonal isolation
    B. Geographical isolation
    C. Behavioral isolation
    D. Temporal

71. In a metabolic path way the end product inhibition occurs when the,
    A. Last product of the path way inhibits the last enzymatic reaction
    B. First product of the path way inhibits the last enzymatic reaction
    C. First product of the path way inhibits the 3rd enzyme of the reaction
    D. Last product of the path way inhibits the first enzymatic reaction

72. The following graph shows the rate of reaction of catalase enzyme at a constant temperature and at different substrate concentration (%). From the above graph we can conclude that catalyze was functional with its minimum turnover rate from substrate concentration of,
    A. 0.4 to 0.5 to 0.4
    B. 0.1 to 0.2
    C. 0.5 to 0.6
    D. 0.3

73. Which one of the following statement is correct about the description of the cell theory?
    A. Scheilden and schwann stated that all cell come from pre-existing cells
    B. Virchow stated that cell is the unit of structure, physiology and organization of living things

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C. Modern cell theory states that cells contain hereditary information that passes from cell to cell

D. Spontaneous generation theory states that all cells have basically different chemical composition

74. A student measured the diameter of onion epidermal cells at a magnification of 100X using a field of view adjusted to 2mm by stage micrometer. What would be the diameter of individual cells if 8 cells fit into the field of view of the microscope at 100x magnification?

A. 250μm B. 200μm C. 150μm D. 300μm

75. Which one of the following characteristics of cells is found only in prokaryotic cell?

A. Division of labour within the cell
long

B. Continuous loop of DNA
complexity

C. 10 to 100 μm

D. Higher level of

76. Which one of the following sentence is true about viruses?

A. AIDS is caused by a DNA virus
caused by an RNA virus

B. Corona virus is a DNA virus
simplex is a DNA virus

C. Swine flu is

D. Herpes

77. The following steps are involved when HIV infects its host cell in the human body.

1. Conversion of the RNA into DNA and its incorporation into its cell's DNA
2. Viral DNA's transcription into viral RNA and production of protein
3. Assemblage of new viral particles from RNA, proteins and reverse transcriptase
4. The binding of gp120 of HIV to CD4 of a T-Lymphocytes
5. Fusion of HIV with the plasma membrane of the host cell
6. Release of its RNA and reverse transcriptase into the cell

The logical order of the steps listed above would be

A. 4, 5, 6, 1, 2, 3
B. 4, 2, 5, 6, 1, 3
C. 4, 6, 5, 2, 1, 3
D. 4, 1, 2, 5, 6, 3

78. Which of the following bacteria produce ammonium ion by forming symbiotic association with peas and beans?

A. Klebssiella B. Azotobacter C. Pseudomonas D. Rhizobuim

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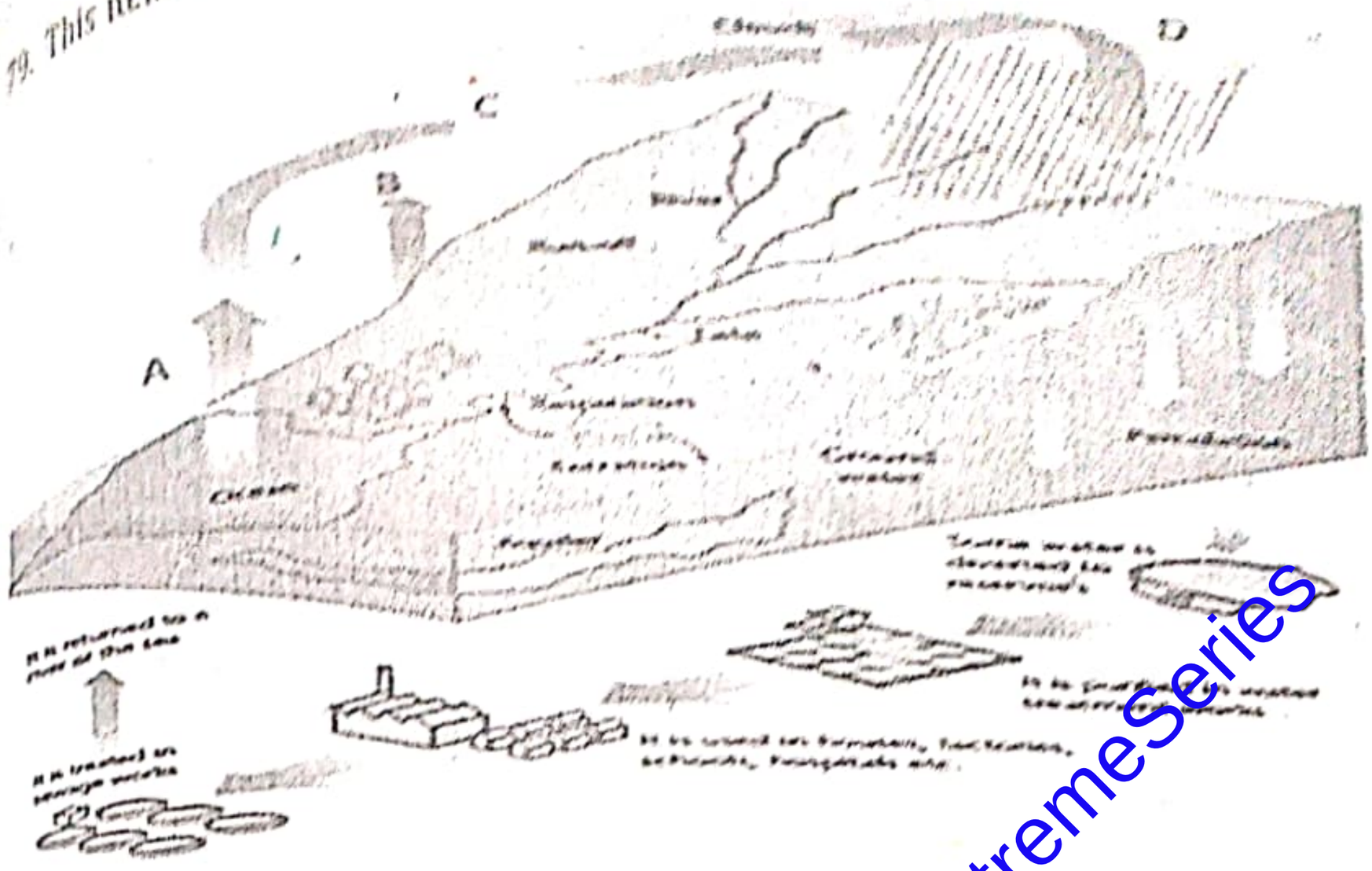
79. This item is based on the following diagram which depicts the water



cycle

In the above diagram the letter A, B, C and D respectively present

A. Evaporation, transpiration, condensation, precipitation
B. Evaporation, precipitation, transpiration, condensation
C. Evaporation, transpiration, precipitation, condensation
D. Evaporation, precipitation, condensation, transpiration

80. Which one of the following comparison is true about primary and secondary succession? Secondary succession has,

A. Reduced species diversity
B. Reduced total biomass of the community
C. Has slower progresses to reach to the climax
D. More stable community

81. Which one of the following sentence describes the tropical rain forest biome? It has

A. Poor and thin soil
B. Sparse – succulent plants animals
C. Frozen and rocky soil
D. Migrating

82. Fresh water biomass like streams and rivers are characterized by the presence of

A. Corals, crabs and planktons
B. Algae, planktons and fish and crabs
C. Corals, crabs and angler fish
D. Angler fish, algae

83. Which one of the following statement correctly describes intra specific and interspecific competition? In intra specific competition the,

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A. Competition is less severe but in interspecific competition it is more
B. Competition is for a new requirement but in interspecific competition the competition is for all requirement
C. Competing individual have similar adaptation but in interspecific competition the adaptation is different
D. Competing individuals belongs to different species but in interspecific competition the individuals are of the same species

84. Which one of the following stages of cell division is correctly described in meiosis II?
A. Prophase II – pairs of homologous chromosomes split up
B. Anaphase II – chromatids from each chromosome separate
C. Metaphase II - chromosomes pair and exchange segments
D. Telophase II – two haploid cells are produce

85. Which one of the following events is an example of a sex influenced trait?
A. Pattern baldness
B. Lactation
C. Retinitis pigmentosa
D. Undescended testicles

86. Which one of the following processes takes place when lactate is produced?
A. DNA and FAD are generated from reduced NAD and FAD
B. ATP is produced through oxidative phosphorylation
C. Protons accumulation created in the inter- membrane space
D. Reduced NAD supplies hydrogen and becomes oxidized itself

87. Which one of the following structures of photosystem is correctly related with its function?
A. Chlorophyll a – a reaction center molecule
B. Antenna complex – light dependent reaction begins
C. Carotenoid – provides the hydrogen ions
D. Thylakoid – absorbs blue and red pigments

88. What happens during the electron transport system of respiration?
A. Four – carbon compound undergoes molecular transformation to generate oxaloacetate
B. Hydrogen atoms carried by FADH and NADH are released and split into protons and electrons
C. Citrate losses a carbon atom to form a five carbon compound and CO_2
D. Five carbon compound is further decarboxylated to form a four carbon compound

89. During light independent reactions of photosynthesis, glucose is synthesized through

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95. Which of the following eye structure correctly matched with its function?

A. Pupil - serves as the eye's protective out layer
B. Iris - controlling the amount of light entering the eye
C. Cornea - provides nutrition to the retina
D. Choroid -bends the light into the eye

96. In the CAM photosynthetic path way

A. Malate is stored in the mesophyll cells during the day
B. Glucose is synthesized in the bundle sheath cell during night
C. ATP and NADPH are produced during the night
D. The C4 stage and the Calvin cycle are separated in time

97. In Gram staining, if some bacteria retain crystal violet after counter staining, then the bacteria is,

A. Gram positive
B. Gram negative
C. Gram intermediate
D. Gram variable

98. Which one of the following sentences explains the mechanism of bacterium to produce diseases?

A. Its hyphae secrets enzyme Which digest substance in the tissue
B. It enters into the living cells and disrupt the metabolic system
C. It attaches with white blood cells and destroy immune system
D. Its genetic material is incorporated in to the cell for instruction

99. Which one of the following sentences explains the role of bacteria in genetic engineering? They are used to

A. Isolate genes from donor DNA
B. Open up a plasmid
C. Cut genes from donor DNA
D. Contain a recombinant plasmid

100. Which one of the following sentences describes a virus?

A. has differentiated organelles
B. is saprophytic respiration
C. is acellular
D. undergoes

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Explanation: All bacteria do have a cell wall but it is not made from cellulose like plant cell walls and instead is made from a substance peptidoglycan, whic i makes it rigid.

Answer: A

3 Explanation: The group of bacteria lived involved in releasing sulphur from proteins of dead matter as hydrogen sulphide is *Desulphovibro* (anaerobic). Sulphur is oxidized to sulphate ions by *Thiobacillus*-aerobic and the genus of bacteria oxidize hydrogen sulphide to sulphate to be taken by plant roots are *Thiobacillus*.

Answer: D

Explanation: *Salonellosis* (Typhoid) is a disease caused by food poisoning bacteria. It is caused by a unique human strain of *Salmonella* called *Salmonella typhi*. The parts of the body infected include intestinal tract and occasionally the blood stream.

Answer: B

4 Explanation: The reproductive cycle of viruses include lytic, lysogenic and chronic release cycle. In lytic cycle, a virus invades a host cell, produce new virus, destroy the host cell and release newly formed viruses. In lysogenic cycle, the virus's DNA becomes incorporated into the DNA of the host cell. The cell divides, the DNA is replicated and each daughter cell gets a copy of the cell's DNA, which now includes the virus DNA.

Answer: B

5 Explanation: Virus infects almost all life forms including humans, animals, plants, fungi and bacteria. Virus lack major organelles such as nucleus, mitochondrion, chloroplast and ribosome because of these, they cannot carry out respiration, photosynthesis, DNA replication, protein synthesis and transport of materials. As a result, all viruses are completely parasitic to all living cell.

Answer: C

6 Explanation: vaccines are made from dead or weakened forms of pathogens to produce immunity (antibody) to dangerous disease. Vaccines need to trigger the immune system without actually causing disease symptom. Vaccines stimulate the immune system.

Answer: A

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7. **Explanation:** During blood transfusion, the donor's red blood cells must be compatible with recipients' plasma.
Anti-A plasma agglutinates type "A" red cells
Anti-B plasma agglutinates type "B" red cells
So these combinations of plasma and red cells are incompatible.
Blood group "O" has no antigens so cannot be agglutinated by blood of any other group. It is called universal donor.

Answer: B

8. **Explanation:** HIV can be transmitted through
- Unprotected sexual intercourse
- From mother to child (baby) in the womb during birth or during breast feeding
- Infected blood

STDs are diseases that can be transmitted from an infected person to a healthy person by unprotected or unsafe sexual contract. The most commonly known STDs are Gonorrhea, chancroids and syphilis. These diseases increase the likelihood of increase HIV. But chancroid, which is caused by *Hermophilus ducreyi* and common problem of men increase the risk of HIV infection more than other STDs due to sores, occurred on genital organ of men.

Answer: B

9. **Explanation:** pasteurization was designed by Louis Pasteur. There are two processes that are used in pasteurization:
10. Holding method: to hold at 60^0C for 30 minutes
11. High temperature method: to heat at 71.6^0C for at least 15 seconds

The process includes heating which is followed by rapid cooling.
- Strictly speaking it does not kill the entire microorganism in the food.

Answer: D

10. **Explanation:** Carl Linnaeus devised the naming system called binomial nomenclature. It consists of two names. The first name is genus name which is followed by species name. Binomial nomenclature follows the following rules:
- The first name is genus name which starts with capital letter
- The second name is species name in which all written in small letter
- The two names are underlined when handwritten or in italics when printed

Answer: C

11. **Explanation:** when the air is breather into the Lungs, oxygen passes into the blood by diffusion along a concentration gradient. At the same

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time carbon dioxide passes out of the blood into the alveoli of the lungs also by diffusion among concentration gradient.

Answer: A

12. Explanation: Exercises affect the breathing system in the following ways:

- More oxygen is consumed during exercise than during rest to produce additional energy by oxidation of more food
- After exercise on tends to breath more rapidly and more deeply
- Exercise increases both the rate of breathing and the depth of breathing
- The heart and the lungs develop bigger and very efficient blood supply

Answer: D

13. Explanation: blood is a complex mixture of cells and liquid that carries a huge range of substance around the body. It consists of

- Plasma: The liquid portion of the blood
  Transport all the blood cells
- Red blood cells: it is greater than any other blood cells and used for carrying oxygen around the body
- White blood cells: are much bigger than red blood cells and fewer red blood cells and they form part of body defense system against microbes.
- Platelets: are small fragment of cells and very important in helping blood to clot at the site of a wound

Answer: C

14. Explanation: valves are structure in the heart which prevents the back flow of blood.

- Tricuspid valve (three parts) found between the right atrium and the right ventricle to prevent the back flow of blood from right ventricle to the right atrium
- Bicuspid (two parts) found between the left atrium and left ventricle to prevent the bake flow of blood
- Semi-Lunar (half-moon) found at the entrance of the arteries from the ventricles.

Answer: D

15. Explanation: the key task of the circulatory system, namely the exchange of materials between tissues and the blood, takes place through the capillaries. These capillaries have certain features that help for a very fast exchange system. One of these features is that they are tiny vessels with walls only one cell thick, which is very thin

Answer: B

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16. **Explanation:** light rays are reflected when they pass from one medium to another. To see things clearly, light from an object must focused on the retina. The focusing of light on the retina is maintained by the bending (refracting) effects of the cornea and the lens

Answer: D

17. **Explanation:** the two sense organs that defect chemical stimuli are the tongue and nose. The chemicals to be tasted and smelled must be in dissolved (solution) form. Taste receptors on the tongue are clusters of cells called taste swellings called papillae; on the upper surface of the tongue since the experiment is designed to understand the actual taste of food, there is no need of noise or vision

Answer: C

18. **Explanation:** a short sighted (near sighted) person can clearly see near objects but cannot properly see distant objects. It is caused due to:
- Long eye balls
- Rigid or strong lens

Images are focused in front of the Retina. It can be corrected by eye glasses made of concave (diverging) lenses

Answer: D

19. **Explanation:** an endocrine malfunction that can be related to symptoms of weight loss, sweating, irritability, lack of energy and feel tired, produce lots of urine and feel thirsty all the time can be related to pancreas.

Pancreas is a small organ located below the stomach it consists of groups of endocrine tissue called Islets of Langerhans which are responsible for secreting the two hormones known as Insulin and Glucagon

Answer: A

20. **Explanation:** endocrine glands are coordinating system of the body which is maintained by chemical messengers called hormones. Hormones act as chemical messages, produced in one part of the body but having an effect some entirely different endocrine glands are ductless glands, where most hormones only affect certain tissues or organs. The hormones are picked up from the blood by receptors in the cell membrane.

Answer: A

21. **Explanation:** RNA virus contains genetic information stored in the form of RNA. Corona and swine flu are caused by RNA virus.

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DNA virus contains genetic information stored in the form of DNA. Example: cold sore is caused by a DNA virus called Herpes simplex virus

Retrovirus is an RNA virus that converts its genetic information from RNA to DNA after it had infected a host. Example AIDS

Answer: A

22. **Explanation:** there is a finite amount of each nutrient in an ecosystem. The same atoms must constantly be re-used over and over again. This happens at the ecosystem and global level. Materials are always being moved from living organisms to another and to the physical environment and subsequently material are lost when they excrete and respire. The waste products of one organism become important nutrient for another.

Answer: D

23. **Explanation:** Rhizobium is a nitrogen fixing bacterium that found in the root nodules of leguminous plants like beans, peas, clover and alfalfa.

Answer: C

24. **Explanation:** Ecological succession is the gradual process by which the structure of a biological community evolves overtime. It is a process of changing form one type of community to another. There are two types of succession; primary and secondary succession.

Primary succession occurs when a habitat is invaded by pioneer organisms such as Lichens, fungi and bacteria.

Answer: A

25. **Explanation:** Boreal forest biome is dominated by cone-bearing evergreen trees such as pines, firs, hemlock and spruce. Plants living in the Taiga (Boreal forest) are adapted for long and cold winters, short summers and nutrient-poor soil. On the cone-bearing evergreen trees, called conifers the waxy needles remain all winter long

Answer: B

26. **Explanation:** growth in plants is influenced by chemical messenger called plant hormone. Auxin hormone stimulates the elongation of the new plant cells, cytokinin stimulates cell division and ethylene causes fruit ripening.

Answer: A

27. **Explanation:** Auxin hormone is involved in apical dominance. Apical dominance is growth concentrated in the terminal bud, allowing it to

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grow taller, thereby increasing its exposure to sunlight. So cancellation of the positive effect of apical dominance results in bushing of plants.

Answer: A

28. **Explanation:** the word biodiversity is used to describe the number, variety and variability of living organisms we must conserve our biodiversity for future generation. Conservation is the use of resources in a sustainable manner to benefit the present and the future generation.

Answer: B

29. **Explanation:** Endemic plants are plants that are found only in one particular area. Therefore mango plant is found in different parts of the world.

Answer: A

30. **Explanation:** Mago national park is found in SNNPR in the area 200KM² and characterized by the presence of wildlife such as Giraffe, Elephants, Lions, Buffalo, Cheetah, Zebra, Leopard, Oxyx, Vulture, Black rhinds

C **Answer:**

31. **Explanation:** Phylum Colentrata (Cidaria) are characterized by soft bodies, a ring of tentacles and stinging cells on their tentacles.

Answer: B

32. **Explanation:** Division Bryophyta is characterized by
    - Are simplest green land plants
    - Do not have true root, leaf and stem
    - Are non-vascular plants
    - Have simple leaves, which are only one cell thick

Answer: A

33. **Explanation:** Biomass refers to the total amount of organic matter. The biomass supported at each trophic level decreases as we go from one trophic level to the next

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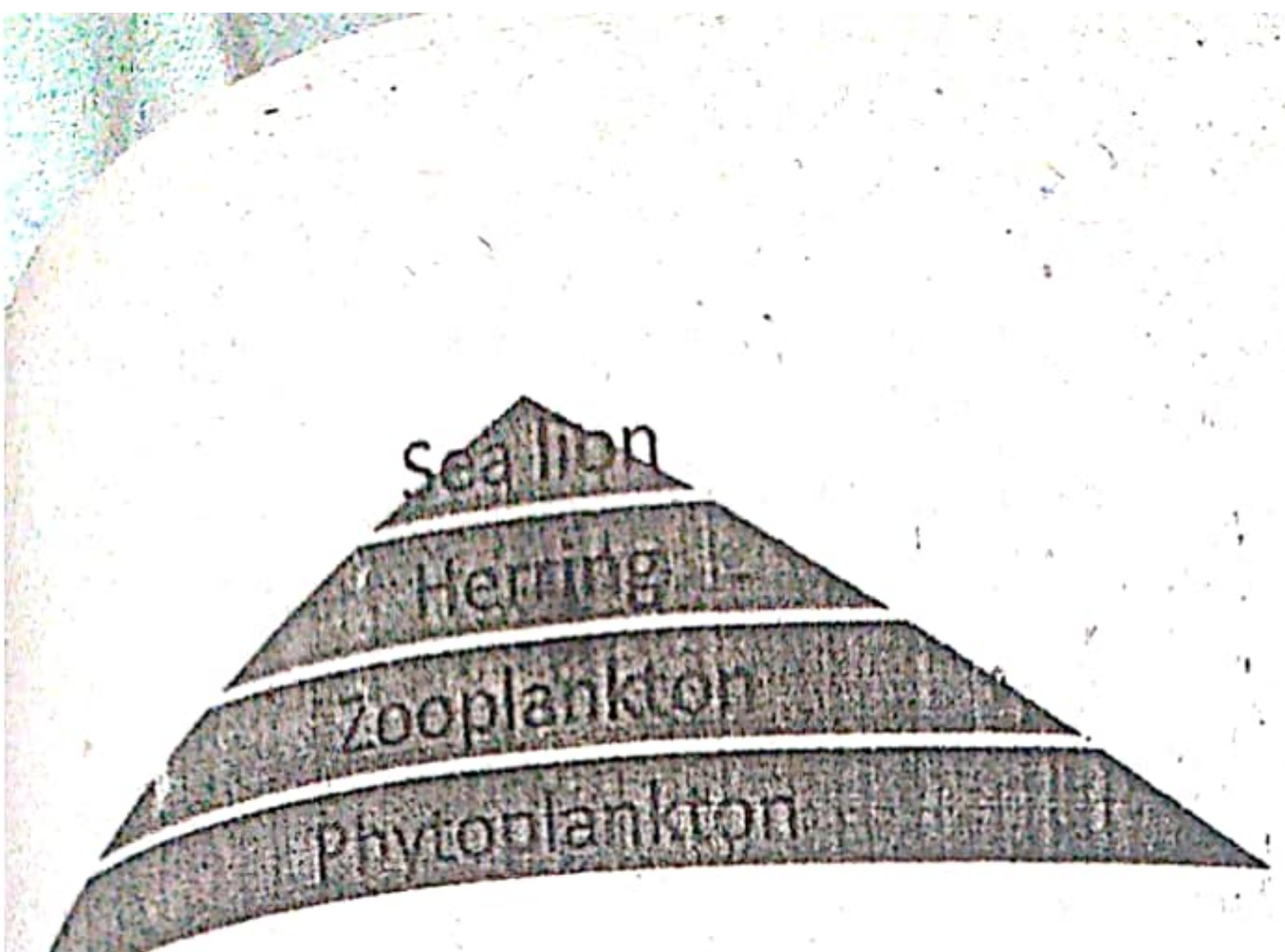

Answer: D

34. **Explanation:** In order to survive within a given habitat an organism must adapt itself to fit into different situations. Adaptations to reduce water loss in plants include;
- Having long roots to absorb more water
- Having thick and fleshy stem store more water (succulents)
- Having reduced number of stomata on the leaf surface to reduce the amount of water loss (transpiration)
- Their leaf surface are covered by waxy cuticles which reduce the loss of water

Answer: D

35. **Explanation:** Animals which live in very cold conditions have an insulating layer of fat under the surface of their skin. This prevents unwanted heat loss. The very thick layer of fat under the skin makes these animals large

Answer: A

36. **Explanation:** IBC is highly dedicated in conserving the genes of Ethiopian endemic plants. Currently the institute is involved in conservation of plants, animals and microorganisms in Ethiopia. In addition, IBC is working on the management of ecosystem

Answer: C

37. **Explanation:** Staining is a process where certain chemicals or dyes are added to the tissues on the slide to make it easier to see particular cells or parts of a cell by creating contrast.

Answer: B

38. **Explanation:** The digestive function of Liver is production (formation) of bile. Bile is a salty substance which has the following roles in digestion.
- Emulsification of fat into fat droplets
- Neutralizing the acidity of chime which is acidic

Answer: D

39. **Explanation:** The stomach is a muscular bag that produces a protease enzyme to digest protein. The main protease made in the stomach is pepsin and rennin. These enzymes convert proteins into amino acids.

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Answer: C

40. **Explanation:** The action of pulling the rubber sheet down represents when the ribcage moves up and out wards. This process occur during inhalation and results in relaxation of the diaphragm

Answer: B

41. **Explanation:** A triglyceride molecule is an ester formed from one -molecule of glycerol and three fatty acid molecules

Answer: B

42. **Explanation:** In water, phospholipids become organized into a bilayer (two layers sandwiched together). In this configuration, the hydrophilic heads face outwards into the water and hydrophobic tails face inwards, away from the water. A hydrophobic (water-hating) region, consisting of the hydrocarbon tails

Answer: A

26. **Explanation:** Reagents- are chemicals which are used to carry out the food tests.
    - The reagent used to test starch in Iodine
    - The reagent used to test simple sugar are Fehling's and Benedicts solution
    - Proteins are tested by Buriet reagent and the reagent used to defect Fats and Oils is Ethanol reagent

Answer: A

27. **Explanation:** Riboflavin is a vitamin used to make the coenzyme Flavin adenine dinucleotide (FAD) and Niacin is a vitamin used to make the coenzyme NAD which is used for oxidation or hydrogen transfer in respiration

Answer: A

28. **Explanation:**
    i. oxidoreductase enzyme transfer of hydrogen and oxygen atoms or electrons form one substrate to another
    ii. Isomerases change of molecular form of the substrate
    iii. Lyases monhydrolytic removal of a group or addition of a group to a substrate
    iv. Ligases joining of two molecules by the formation of real bonds

Answer: B

29. **Explanation:** In cyclic and non-cyclic photophosphorylation, ATP is produced because:
    - There is an accumulation of protons (hydrogen ions) in the interior of thylakoid.
    - This creates a concentration gradient between the thylakoid interior and the stroma of the chloroplast

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Protons move down this concentration gradient through ATP synthase, causing the rotor spin, just as in mitochondria during respiration

Answer: A

31. Explanation: During heavy exercise our body needs energy so much oxygen is required by our cells to carry aerobic respiration. If someone is stranger to such kinds of heavy exercise, it's difficult for him or her to inhale sufficient amount of oxygen needed by our cells to oxidize glucose molecule

Answer: A

31. Explanation: When the amount of oxygen supply is insufficient to the cells of animals they carryout anaerobic respiration which produces lactic acid

Answer: B

32. Explanation: In the link reaction, a molecule of pyruvate reacts with a molecule of coenzyme A (CoA) to form a molecule of acetyl Coenzyme A (Acetyl CoA)

Answer: A

33. Explanation: Neo-Darwinism is revised version of Darwinian Theory of evolution. It combines Darwin's original natural selection theory, genetic theory and ethology (study of animal behavior)

Answer: B

34. Explanation: *Australopithecus afarensis* (Lucy) was discovered by Donald Johansson and Tom Gray in 1974 at Hadar, Ethiopia. The fossil Lucy was significant for showing bipedalism came before large brain

Answer: A

35. Explanation: Instinctive behaviors are the most complex innate behaviors. There is often a fixed action patterns (FAP) for each key stimulus. Once begun, the FAP is not halted until completion. It is common to all members of species.

Answer: A

36. Explanation: Latent learning is a hidden learning when it occurs but it manifests later, without reinforcement, at the time it is needed. A mouse know the burrow protects them from predator and it uses to escape from the predator when needed

Answer: B

37. Explanation: An active enzyme isn't just a single molecule, is made from two molecules. The two parts are the apoenzyme and the cofactor.

i. Apoenzyme is a protein that combines with a cofactor to form an active enzyme. It is inactive on its own

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ii. Cofactor is a small non-protein particle essential for the activity of some enzymes
iii. Coenzymes are organic molecules and many are derived from vitamins. They binds with the to give catalytic activity

Answer: D

38. **Explanation:**

Innate behavior	Learned behavior
Genetically programmed and common to all species member	The behavior is changed or acquired through experience and vary among individual members
The behavior is fully functional	The behavior developed through trial and error or by insight
No (very slightly) modifications of behavior	The behavior may be modified by new experience
Adaptive behavior that retained through natural selection	The behavior may be not adaptive

Answer: D

39. **Explanation:** Social behavior is a set of interaction among individuals of the same species that are usually beneficial to one or more of the individuals. There is cooperation in caring for the offspring; as a consequence, many individuals are caring for offspring that are not their own

Answer: D

40. **Explanation:** The lock and key enzyme action model reduce the activation energy by forming intermediate molecule and then enter the transition to lower the activation energy

Answer: A

41. **Explanation:** Induced fit model suggests that the active site and the substrate aren't naturally complementary in shape, but the binding of substrate molecules produce a conformational change (change in shape) in the active site. This allows the substrate and active site to bind fully the conformational change also puts the substrate molecules under tension.

Answer: D

42. **Explanation:** Temperature - below the optimum temperature, the low level of kinetic energy limits the number of enzyme substrate complexes formed; above the optimum temperature, denaturation of the prevents binding of the substrate.

PH – above and below the optimum PH changes occur in the tertiary structure of the enzyme molecule

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Answer: D

43. **Explanation:** The CO_2 produced by yeast cell during baking Injera is used to give Injera its typical texture.

Answer: B

44. **Explanation:** Synapse is the gap (space) between the terminal branches on the axon of one neuron and the dendrites of another neuron. An impulse is transmitted across the synapse by means of chemical transmitters called neurotransmitters.

Answer: C

45. **Explanation:** when an impulse transmits across the neurons, the axon is stimulated and the positive charge comes inside and the negative charge comes outside the axon membrane. This condition is called Action potential.

Answer: C

46. **Explanation:** The knee-jerk is the simplest type of reflex action because of only two neurons; motor and sensory neurons are involved. This reflex action is not under our conscious control. The correct demonstration of knee jerk reflex is first to ask your friend to sit with one leg crossed over the other and hitting your friend below the knee cap followed by a sudden upward movement

Answer: D

47. **Explanation:** There are three types of natural selections
    - Directional selection is a type of selection in which individuals at one extreme of the range could have a disadvantage whereas those at the other extreme have an advantage
    - Stabilizing selection works in favor of intermediate features and acts against both extremes
    - Disruptive (diversifying) favors individuals at both extreme of a range over those displaying at mean (intermediate) values.

Answer: A

48. **Explanation:** The shell like structure in the inner part of the ear is Cochlea. It consists of hair like sensory cells that can detect the vibration of sound

Answer: D

49. **Explanation:** The important concepts of cell theory includes
    - Cells are the basic units of life
    - Cells are structural and functional unit of life
    - Cells comes from pre-existing cells

Answer: C

50. **Explanation:** Inversion occurs when a chromosome piece rotates 80^0 and rejoins to the same chromosome. One inversion on chromosome 16 can cause Leukemia.

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Insertion describe an increase in the number of genes caused when an equal cross over happens during meiosis
Duplication chromosome duplicates (repeats) itself there by making an additional set of genes available
Deletion a base is "missed out" during duplication

Answer: C

51. Explanation: eternity of life – the theory claims that there is no beginning and no end of life on earth so it is eternal
Cosmozoan theory – asserts that life on earth initially came from somewhere else in the universe
Biochemical evolution - states that the first form of life evolved from chemicals
Special creation theory attributes the origin of life to a divine event that was masterminded by the supernatural being

Answer: C

52. Explanation: Paleontology is the study of ancient life mainly based on fossil evidence. A fossil is a preserved remnant of trace of one living things. It serves as a paleontological evidence of evolution

Answer: B

53. Explanation: There are two phases of photosynthesis. These are light reaction and dark reactions. Light reaction occurs in the grana of the chloroplast and the light independent reactions occur in the stroma of chloroplast

Answer: C

54. Explanation: – linked features in the non-homologous region.
- Are mostly common in males for males inherit only a single X-chromosome that lack corresponding dominant allele on Y chromosome to mask the recessive effect of X- chromosome
- Affected males inherit the allele from their mother for the source of X is mother
- Affected females inherit one allele from each parent, thus father is affected

Answer: C

55. Explanation: There are two types of reproduction. These are sexual and asexual. The type of reproduction used in the experiment is asexual, cutting a part of stem and planted. One of the major characteristics of asexual reproduction is producing offspring phenotypically and genotypically identical to the parent

Answer: B

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56. **Explanation:** The process of making exact copy of DNA is called DNA replication. During replication
- Two copies that each contains one of the original strand and one new strand is produced
- The new DNA molecules are identical to each other and to the original molecule
- The DNA helicase enzymes open the DNA double helix
- DNA polymerases move along each of the DNA strand to add complementary nucleotides to each strand

Answer: C

57. **Explanation:** Since there are 64 possible triplet codes (codons) and only 20 amino acids, a single amino acid can be coded by two or more triplet codes (degenerative codes or codon synonyms)

Answer: A

58. **Explanation:** UGA is a stop codon that terminates the protein synthesis. tRNA could not bring methionine it brings.

Answer: A

59. **Explanation:** An Oestrogen hormone stimulates the uterus to build up a thick, spongy lining with lots of blood vessels ready to support pregnancy. Corpus Lateum secrets a progesterone hormone which keep the uterine wall thick and growth of more blood vessels, ready to receive a fertilized ovum so decrease in progesterone and estrogen concentration is related to breakdown of the thick spongy wall of the uterus

Answer: C

60. **Explanation:** The presence of HIV in the blood can be examined by detecting the antibodies produced by T-helper cell (CD4 lymphocyte). In the early stage of HIV infection, the produced antibody may not yet be sufficient to be detected by antibody-antigen interaction mechanism. The first symptoms appear 3 – 12 weeks after you have become infected this is the initial incubation period of virus

Answer: C

61. **Explanation:** Behavioral method temperature regulation in homoiotherms include:-
- Clothing: wearing suitable clothes depending on weather condition
- Seeking shade or shelter
- Hibernation: a response to cold seasons
- Aestivation: a response to hot seasons

Answer: C

62. **Explanation:** If the water concentration in the blood is too high the amount of ADH produced in less. However, if the concentration of

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water in the blood is less, the pituitary gland releases much ADH into the kidney

Answer: B

63. **Explanation:** Water is taken into a plant through the roots and moves by osmosis to the Xylem tissue. There is no active transport in the Xylem so the transport of water through a plant is as the result of the transpiration stream

Answer: A

64. **Explanation:** Reporting results are means of communication with other scientists by formal publications, such as Nature and science Journals or through informal conversations. Informal publications, the results can be reported via the following major steps. Title, Hypothesis, Predication, Procedures, Results, Conclusion, Recommendations and Acknowledgement respectively.

Answer: A

65. **Explanation:** Centrifuge is used to separate solid particles from the liquid where simple filtration is not adequate for the job. The type of force responsible for the settlement of solid particles down at the bottom is gravitational force.

Answer: D

66. **Explanation:** In scientific experiment, there is a cause and effect relationship. When an event occurs as a direct result of a previous event, there is a cause and effect relationship. In this case it is the food eaten by attendants of wedding ceremony and the result is sickness of the students should take food sample to laboratory analyses the presence of toxin

Answer: B

67. **Explanation:** Viscosity is a measure of how fluid easily flows. It is the measure of how fluid a liquid is. Water has an ideal viscosity for transport medium this property of water is essential for cellular motion and circulatory system

Answer: A

68. **Explanation:** water has a high latent heat of vaporization. Latent heat of vaporization is the energy utilized in converting liquid to a gas at the same temperature water takes in a lot of energy when it turns to vapor. It has the highest latent vaporization of all substances. Thus, water bodies such as ponds do not dry up quickly in hot weather and the organisms in the pond have a better chance of survival

Answer: B

69. **Explanation:** Major difference between prokaryotes and eukaryotes

Feature	Prokaryotic cell	Eukaryotic cell
Size	1 – 10nm	10 – 100nm

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Form	Unicellular	Mainly multi cellular (except many protocista bonded nucleus)
Nucleus	No membrane bounded	Membrane bounded
DNA	In a continuous loop and naked	Linear and associated with histones
Ribosomes	Small ribosome (70s)	Large ribosome(80s)

Answer: D

70. **Explanation:** Plasma membrane has the following major functions
- It isolates and protects a cell from its surrounding
- It regulates (controls) the exchange of substances between the cytoplasm and the environment
- It communicates with other cell (cell signaling)

Answer: B

71. **Explanation:** Fluid mosaic model of plasma membrane consists of the following components

Phospholipid bilayer – it has hydrophilic head and hydrophobic tails

Integral proteins – they span both the phospholipid layers of the membrane. It includes:-

i Peripheral proteins – that span only one of the two phospholipid layers

ii Channel proteins – have ion pores that allow the passage of ions through them

iii. Carbohydrates (glycoproteins and glycolipids)

iv. Cholesterol

Answer: A

72. **Explanation:** If a plant cell is placed in hypertonic solution. It starts to lose its water contents. This results in a loss of turgor pressure and is called plasmolysis. This causes the vacuole to become shrink and the cell flaccid.

Answer: A

73. **Explanation:** Osmosis in living cells can be shown in the following ways
- Isotonic solution – both solutions are the same so there is no water movement across the cell membrane because of osmosis
- In hypertonic solution water passes out the cell or the losses its water content resulting in shrinking of the cytoplasm of the cell
- In hypotonic solution the cell gains water and its swells up

Answer: D

74. **Explanation:** Biodiversity is

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- The measure of wealth of species in a given place or habitat
- The variety of organisms living on a given place include all type of organisms ranging from microbes to large animals
- More accurately in a measure of the number of species breeding in an area at a particular time

But a more comprehensive definition of biodiversity is richness, ecological and genetic variability of species

Answer: A

75. **Explanation:** Most populations develop through four main stages
   a) Lag phase – the number of population remains low or static or increase slowly
   b) Log (logarithmic or exponential) phase – all of the adapted organisms reproduce and increase in number rapidly due to plentiful resources
   c) Stationary phase – here, the carrying capacity is reached and the same numbers are dying as produced in reproduction
   d) Decline phase – number of organisms decline rapidly

Answer: D

76. **Explanation:**
   - Intra specific competition – competition within the same species for the same resources in an ecosystem.
   - Inter specific competition – competition between or among members of different species for the same resources in the same habitat

In this competition two species utilize the same resources in the same way; one will totally outcompetes the other, which will become locally extinct.

Answer: No Answer

77. **Explanation:** Drosophila was selected for the study of genetics for several reasons
   - Short life cycle, easy to handle and large progeny per female insect
   - Easy recognition of male and female flies
   - Easy to make crosses between individuals

Answer: C

78. **Explanation:** The law of independent assortment states that the inheritance of one trait is independent of the inheritance of the other. This means, the alleles of one pair separate independently of the alleles of another pair controlling a different feature vest describe the mechanism by which meiosis brings about variation

Answer: A

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79. **Explanation:** plants in the topics have a problem. It can be very hot and the leaves close their stomata to minimize water loss. When C3 plants do this, the concentration of CO_2 in he leaves falls and the enzyme Rubisco starts to behave in an unusual way. In the low concentration of CO_2, Rubisco binds with oxygen, not CO_2. This means that RuBP is oxidized to one molecule of GP (not two) and a molecule of phosphoglycolate

Answer: B

80. **Explanation:** The main stages of the light-independent reactions are:
- CO_2 reacts with RuBP, the reaction is catalyzed by Rubisco
- Two molecules of GP are formed
- GP is reduced TP by reduced NADP
- Some of the TP formed is used to generate the RuBP while some is used to form glucose and other useful organic compounds

Answer: D

81. **Explanation:** The structure of a chloroplast suited to its function is due to
- The chlorophyll and other photosensitive pigment molecules are arranged in special photosystem that are linked to electron transport chain (ECT)
- The molecules of the photosystems and the ECTs are fixed in the membranes of the thylakoids

Answer: B

82. **Explanation:** The CAM photosynthesis cycle during the day time is as follows;
- The light dependent reactions generate ATP and reduced NADP so that the Calvin cycle can continue
- Malate is released from the vacuoles and is broken down to glycerate, releasing CO_2 for the reaction of the Calvin cycle

Answer: C

83. **Explanation:** In the light independent reactions of photosynthesis three 'turns of the cycle' result in a n output of one molecule of TP. Six turns of the cycle would give am output of two molecules of TP enough to make one molecule of glucose

Answer: A

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ANSWER FOR BIOLOGY HAMILIE, 2015/ JULY 2023

1. A stethoscope is the device used to hear heart beats noise produced by valves
2. B no need of adding distilled water to agar
3. B gastroenteritis can be caused by viruses, bacteria, amoeba and giardia and others.
4. B. burning sensation is the symptom of gonorrhea
5. A in binomial naming the first name is the genus while the second name is species
6. D from the self-cross of (DC, Dc, dC; dc) 225 of the offspring will have genotype of ddcc
7. C entry of mRNA codon in the ribosomes is the correct events
8. B exchange of section of segments (crossing over)at meiosis of prophase-I causes genetic variation
9. D during DNA molecule replicate itself new DNA molecules formed are identical to each other and to the original molecule.
10. B recognizes codon of mRNA and brings respective onion acids and has complementary anticodons
11. B dairy products (yoghurt) for household consumption is traditional use of biotechnology unknowingly used yeast for a long period of time by the people
12. A organism of each tropical level use small proportion of energy
13. C mitosis is a means for division of body cells that results two identical daughter cells
14. D the cross for Rr x Rr will result 25% of rr genotypes
15. B DNA is made of nucleotides with three components (5-C sugar, N - bases and phosphate group
16. B chordates have flexible cartilaginous rod-like structure known as notochord along the dorsal side of the body.
17. A star fish belongs to phylum Echinodermata
18. A carbon transferred from plants to animal by feeding relation ship
19. A different body cells produced phospliphids and cholesterol because of this the liver do not have a chance to control the amount of cholesterol in the body, even if the liver is the main part of the body with high production of cholesterol than other cells
20. D contain sequence of the five vertebrates that matched is described features
21. D storing water in their tissue is one of adaptation mechanism for desert plants
22. D dermis contain sweat glands and blood vessels

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23. C fluid in semi-circular canal keeps moving even though spinning has stopped

24. D menstruation is the result of breaking and discharge of the thickening of uterus if pregnancy has not happened

25. A Bowman's capsules acts as a filter and produce urine

26. D seeking shade is behavioral thermoregulation

27. C glucose oxidase in clinistix strips used for diabetes diagnosis by urine analysis

28. A in the systematic naming of enzymes, an enzyme will have a 'name' such as EC 3.4.11.1. From this the figure 11 indicates sub-sub class of the enzymes

29. D holds the correct sequence of classes for vertebrate species listed above

30. B a protein that combines with a cofactor, to form an active enzyme. The protein is inactive on its own.

31 D. the active site and the substrate aren't naturally complementary in shape, but the binding of substrate molecules produces a conformational change (change in shape) in the active site. This allows the substrate and active site to bind fully.

32. A palisade mesophyll cell is rich in photosynthetic tissues

33. C water is transported in plants through dead tissue known as xylem

34. B gibberellin is one of the plants growth that stimulate growth of stem

35. D monocots undergo hypogeal germination where the cotyledon remains underground (in the soil)

36. D high amount of energy is required to heat up water high specific heat capacity of water

37. A amino acids are building blocks of protein where chromosome contains histone protein

38. B. α-1, 6 glycosidic bonds (having many ends) make it to hydrolyzed easily

39. B if the temperature exceeds beyond the optimum amount it makes the enzymes structures to be denatured

40. D glycerol structure that holds a poly hydroxyl function

41. A Noug seed is an Ethiopian endemic plant

42. B hypothesis is testable by an experiment and after the experiment the researcher now decide whether or not to report the results to other biologists. Someone else might decide to take the work further and try to isolate a particular work

43. D Simien Mountain National Park is the protected area where Walia ibex is found.

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44. B CO_2 can either be used as food by animals, or they die naturally and decay
45. B organic always contain the element carbon.
46. A pepsin is an enzyme that acts on the protein in the stomach.
47. D giving respiration is prohibited for conscious patient
48. D air exit when there is higher pressure in the chest than the external environment
49. B left ventricles contract and force (pump) oxygen rich blood to the body
50. D pyruvate formed at glycolysis and enters to mitochondria for further steps
51. D holds correct sequence parts designated
52. A peripherial protein (extrinsic protein) to the cytoskeleton
53. B the environment is hypotonic than the cell
54. D classical conditioning is learned behavior
55. D active transport occurs from low to high concentration which transfer Ions and medium-sized, non-lipid-soluble particles by requiring energy .
56. C the two-carbon group from acetyl coenzyme A reacts with the four-carbon compound oxaloacetate to form a six-carbon compound called citrate
57. D modern humans have a brain that is between 1.8% and 2.3% of their body mass. The brain of *Homo sapiens* uses 25% of the resting energy requirement, compared with 8% in the great apes.
58. A Innate behaviour is behavior that is present (potentially) at birth or hatching. It does not have to be learned.
59. B Insight learning involves finding solutions to problems that are not based on actual experience (as with trial and error) but on 'trials' occurring mentally.
60. A Territorial behaviour is found in nearly every species of animal, even humans. Possessing territory gives the holder areas to forage for food and so increases the chances of attracting a mate.
61. D Professor Tilahun Yilma developed a vaccine for cattle disease (rinderpest)
62. B At low power there is quite a large space between the objective lens and the specimen. That is way set up the microscope with the lowest power objective in place that is way it preferred
63. A the criss-cross arrangement of cellulose fibers in the cell wall gives it both strength and elasticity. because there are large 'gaps' (on a molecular scale) between the fibers, the cell wall is freely permeable.

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In the hypotonic solution, the cells gain water by osmosis and
66. A swell. However, because of the cell wall, the cell cannot become
much larger. Plant cells in this condition are turgid.
The chemical digestion of food is finally completed in the small
66. D intestine and converts carbohydrates into glucose, protein into
amino acid and fats ito fatty acid and glycerol
substitution of just one base in the sixth triplet of the gene coding
66. A for one of the four polypeptides in the hemoglobin molecule alters
the triplet from GAG to GTG. This results in the amino acid valine
replacing glutamate in the polypeptide chain. The different
hemoglobin molecule formed results in the condition known as
sickle-cell anemia.
67. D Stanely Miller passed electric sparks repeatedly through a mixture
of gases that were thought to represent the primitive atmosphere
of the Earth. These gases were methane (CH4), ammonia (NH3),
water (H2O) and hydrogen (H2).
68. B individuals with advantageous variations of traits survive in greater
numbers and pass on these advantageous variations to their
offspring
69. A the similarity of structure is evidence that the animals have
descended from a common ancestor
70. B natural selection and genetic drift cause the two subpopulation to
diverge, eventually making them incompatible for mating
71. D feedback control (positive or negative) a series of events in which
the last step (end product) controls the first enzyme
72. C at this concentration the enzyme works its minimum turnover rate
and the reaction rate decrease
73. C according to the cell theory the cell contains hereditary information
which is used as instructions for growth, development and
functioning
74. B
75. D bacteria genetic material is circular DNA
76. D DNA viruses – causes, Herpes simplex (causes cold sores)
77. C After HIV has bound to the CD4 receptors on the surface of the T-
helper cell, the following events occur: 1. It fuses with the plasma
membrane and then releases its RNA and reverse transcriptase
enzyme into the cell. 2. The reverse transcriptase converts the RNA
into DNA using building blocks called nucleotides, which are
provided by the cell. 3. The viral DNA becomes incorporated into
the cell's own DNA. 4. The viral DNA is transcribed to viral RNA,
which starts producing viral proteins, including the enzyme
reverse transcriptase. 5. The RNA, proteins and reverse

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transcriptase molecules are assembled by the cell into new lily particles that escape by 'budding' from the cell membrane - this is an example of chronic release. 6. The viruses then infect other T. helper cells.

78. D persuading bacteria like *Rhizobium* to form symbiotic associations with other species of crop plants.

79. A the main stages of the water cycle of evaporation, transpiration, condensation and precipitation

80. D secondary successions to the original climax are usually much quicker than primary successions because:
- the succession is not starting from bare rock/open water
- there is a seed bank of many of the climax plant types available in remaining undamaged plants
- the soil is already present

81. A the tropical rainforest biome is with many of the trees are very tall, the root systems are shallow and trees can easily fall. The shallow root systems grow in shallow, nutrient-poor soils.

82. B freshwater biome a biome consisting of a river, a lake or a pond and contain the species of like algae, Planktons and fish

83. C intra-specific competition - competition between members of the same species and inter-specific competition – competition between members of different species.

84. B Anaphase II chromatids from each chromosome are separated at random (random segregation)

85. A It is the high concentrations of the male sex hormone testosterone that makes the allele dominant in males. Because of this, males need only inherit one pattern baldness allele and they will go bald (because the allele is dominant in males). It doesn't matter whether the second allele is a baldness allele or a non-baldness allele. Females must inherit two before they go bald (because the allele is recessive in females).

86. D hydrogen atoms are transferred to NAD to produce reduced NAD in the glycolysis

87. A the reaction center molecule is always chlorophyll a.

88. B in the inner membrane the hydrogen atoms carried by FADH and NADH are released and split into protons and electrons

89. B Six turns of the cycle would give an output of two molecules of TP - enough to make one molecule of glucose.

90. A CO2 can limits the light independent reaction by influencing the rate of the initial reaction with RuBp

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a test cross requires two elements A/ an organism of dominant phenotype but unknown genotype B/ another organism having a recessive phenotype, which is therefore homozygous recessive

Ethiopia has a high biodiversity at the moment, but the biodiversity is threatened by one case is large scale introduction of exotic plants

93. C Neurons are not continuous 'wires' running about your body. Whenever one neuron ends and another begins there is a gap known as a **synapse**. The electrical impulses that travel along your neurons have to cross these synapses, but an electrical impulse cannot leap the gap. So when an impulse arrives at the end of a neuron, chemicals are released. These chemical transmitters (**neurotransmitters**) cross the synapse and are picked up by special receptor cells in the end of the next neuron.

94. C The reflex jerk of your knee is used by doctors to check that your reflexes are working properly – but it normally works if you stumble and the tendon is stretched. The reflex helps to straighten your leg and stop you falling and then the lower leg to jerk downward and inward

95. B The iris is the coloured part of the eye, but it is not there simply to look pretty. It is made up of muscles that contract or relax to control the size of the pupil and so to control the amount of light reaching the retina.

96. D the desert plants like cacti use the same set of reaction as C4 plants, but they separate the two stages not by carrying them out in different cell, but by carrying them out at different times

97. B Gram negative bacteria are stained pink by Gram's stain whereas Gram positive bacteria is purple

98. B bacteria can releasing toxins that destroy host's cells or inhibit certain metabolic function

99. D the opened up plasmids and the isolated genes are mixed with a DNA ligase enzyme to create recombinant plasmids

100. C viruses are mysterious entities which do not show any sign of life in Free State and are seen to remain as nonliving entities

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CamSca

EDUCATIONAL ASSESSMSNT AND EXAMINATION SERVICE ETHIOPIAN SECONDARY SCHOOL LEVING CERTIFICATE EXAMINATION (ESSLCE) ENGLISH SCHOLASTIC APTITUDE TIKIMTE 2014/2015)

BOOKELET CODE: 075

SECTION ONE : Verba. l Reasining (Question 1-35)

This section which is sub divided in to analogy, synonyms, antonyms classification, reading comprehension, analytical and logical reasoning contains 35 verbal reasoning items altogether. Each subsection is followed by its specific direction.

Classification Items (1-5)

Direction: in question **1-5** a set of four words is given. Three of the words are related in some ways. The remaining word is not related to the rest. Choose the word that does not fit the relation.

1. A. Tiger B. Leopard C. Elephant D. Lion
2. A. Eraser B. Pen C. Pencil D. Book
3. A. Order B. Legal C. Confusion D. Chaos
4. A. Ear B. Tangue C. Heart D. Eye
5. A. Reporter B. Publisher C. Reader D. Writer

Synonyms (6-9)

Direction: Questions 6-9 consists of sentences in which one word is written in bold followed by for other words. Choose the word that is most nearly similar in meaning to the word in bold.

6. The special TV report **disrupted** regular programming due to the day's special events.
   A. Continued C. Complemented
   B. Interrupt D. Satisfied
7. **He dissipated** his inheritance in less than six months and he is now back to where he was
   A. Hoarded B. Saved C. Accumulated D. Wasted
8. If enough members of parliament **dissent,** the new law not be approved
   A. Disagree B. Question C. Support D. Agree
9. During the riot, some of the demonstrators looted shops and supermarkets and took the goods to their home.
   A. shoot B. burned C. wasted D. robbed

Reading Comprehension I (10-14)

Direction: Read the passage carefully and answer questions 10-14. Each question will be based on the information stated or implied in the passage.

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If a person suddenly encounters any terrible danger, the change of nature one undergoes is equally great. Some times fear numbs our senses. Like animals, one stands still, powerless to move a step in fright or to lift a hand in defense of our lives, and sometimes one is seized with panic , and again, act more like the inferior animals than rational beings. On the other hand, frequently in cases of sudden extreme peril, which cannot be escaped by flight, and must be instantly faced, even the most timid men at once as if by miracle, become possessed of the necessary courage, sharp quick apprehension, and swift decision. This is a miracle very common in nature. Main and the inferior animals alike, when confronted with almost certain death'gather resolution from despair' but there can really be no trace of so debilitating a felling in the person fighting, or been;the nerves are steel, there is nothing felt but a wonderful strength and daring. Looking back at certain perilous moments in my own life, I remember them with a kind of joy, not that there was any joyful excitement then, but because they broadened my horizon,lifted me for a time above myself.

Adapted from Reading Comprehension-practice Questions

10. The title that best suits the passage would be:
    A. Courage and panic
    B. The Change of Nature
    C. The Miracle of Confronting Danger
    D. The Will to Fight
11. What is the meaning of the word debilitating?
    A. Fatten B. Strangth C. Debase D. Enfeeble
12. The author feels happy in the recollection of dangers faced and overcome because
    A. he felt elated as he was alive.
    B. these experiences boosted his confidence
    C. they added lifted him above himself for a time.
    D. they brought him a new experience.
13. A man may react to sudden danger in some different ways. Some of these are ;
    A. He may be paraysed with fear, run away, or fight.,
    B. He may be paraysed with fear,seized wiyh panic, or act like an inferior animal.
    C. He my be paraysed with fear, flee panic,seized with panic or act like an inferior animal.
    D. He my flee panic, or fight back, stand still or act like an inferior animal.
14. Explain the phrase 'gather resolution from danger'.
    A. Questions to be enthusiastic and brave the odds during difficulty.
    B. To remain calm and not to lose hope when there is a difficulty.
    C. Find peace in times of difficulty.

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D. Hopelessness makes one determined to face the difficulty.

Reading Comprehension I (15-19)

Direction: read the passage carefully and answer question 15-19. Each question will be based on the information stated or implied in the passage.

Daylilies are beautiful perennial flowers that can brighten up any yard or landscape. They are tolerant of drought and flooding, immune to heat stress, and growb well in full sun or light shade. They are the perfect choice for just about any soil or climate condition.

Different varieties of daylilies can be in bloom from late spring until autumn. Individual flowers last only one day, but each plant produces many buds, and many varieties have more than one flowering period.

Daylilies grow best in

soil that is slightly acidic and they prefer either direct sunshine or light shade. The best time to plant them is in the early fall or early spring, but they are hardy enough to endure planting or transplanting at almost any time of sear. They should be planted 18 to 24 inches apart, and the bulb should be no deeper than one inch below the soil's surface.

Adapted from Reading Comprehension Practice Questions

15. According to the passage, what soil's is best to daylilies /
    A. Moist soil
    B. Any soil
    C. Slightly acidic
    D. Slightly neural \
16. You could infer from this passage that these flowers are called day-lilies because
    A. Charles Day developed the hybrid
    B. the blooms last for one day.
    C. they look like real lilies, but aren't.
    D. they only bloom in the daytime.
17. How deep should daylilies be planted, according to the passage?
    A. any depth work
    B. 18 to 24 inchs
    C. one inch or more
    D. one to 20 inch
18. Daylilies grow best in
    A. sunshine
    B. the evenings
    C. the mornings
    D. darkness
19. which of the following is true according to the passage?
    A. Daylilies can not grow in sunny weather.
    B. Heat can causes stress to daylilies
    C. Daylilies can not stand drought
    D. Daylilies can stand drought

Antonyms (20-22)

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irection: Question 20-22 consist of sentences in which one word is written in bold followed by four other words. Choose the word that is **most nearly opposite** in meaning to the word in **bold.**

20. The officer **exaggerated** the damage caused by the rowdies and, thus made it look more serious.
A. Underestimated C. Condemned
B. Ignored D. Unreported

21. The speaker was unable to **pacify** the crowd.\
A. Challenge B. Antagonize C. Threaten D. Excite

22. His negative attitude to poor people is **deplorable.**
A. Desirable C. Commendable
B. Equitable D. Miserable

Analytical and Logical Reasoning Question (23-30)

Direction: Question 23-30 deals with logical and analytical reasoning. Read the given set of conditions and assumption for each question. Choose the answer that you think is most appropriate among the given alternative.

23. Tamrat's train never bothered him because there were always seats available on it, and he was able to spend his time on the train comfortably reading the newspaper or playing games on his mobile phone. Ever since the train schedule changed, the train has been extremely crowded, and by the time the doors open at his station, there isn't a seat to be found.
A. Tamrat will likely look for a new job closer to home.
B. Any traveler who use the train will complain about the new schedule.
C. Tamrat's train is less comfortable since the train schedule changed.
D. Tamrat would be better off taking the bus to work instead of the train.

24. It is called Restitution when you compensate some one for damaging his or her property in some way. Which of the following situations is the best example of Restitution?
A. pothole in the parking of peter's company caused his flat tyre. He informs his boss and expects that the company should pay for the repair.
B. Peter asks Tom to stay in his apartment when he is out of town. One day Tom arrives to stay and finds that pipe has burst and the apartment is filled with water. . He calls the plumber to repair the pipe and pays for the repair.
C. Tom borrows his friend's that camera. Tom fails to zipper the case, and the camera falls on the ground, and the lens shatters. When Tom returns the camera, he tells his friend that he will pay for the damage.

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D. Tom borrows his friend's car and returns the car with an empty petrol tank. He apologizes and tells his friend that he will fill the tank tomorrow.

25. A four-person crew is painting Mr. Omod's house. Michael is painting the front of the house. Roza is in the alley behind the house painting the back. Jemal is painting the window frames on the north side. Shawul is on the south. If Michael switches places with Jamal, and Jemal then switches places with Shawul?
A. on the south side of the house
B. in front of the house
C. on the north side of the house
D. in the alley behind the house

26. Four friends in the sixth grade were sharing a pizza. They decided that the oldest friend would get the extra piece. Hawi is two months older than Geremew, who is three months younger than Nardos. Kemal is one month older than Geremew. Who should get the extra piece of pizza?
A. Kemal B. Geremew C. Nardos D. Hawi

27. If you are a fitness walker, you don't need to goto a health gym. You also don't need any fitness equipment for workout. All you need is a pair of comfortable athletic shoes.
Which of the following statements does this paragraph support?
A. Poorly designed shoes can cause foot injuries
B. Fitness walking is an effective and convenant form of exercise
C. Walking outdoors provides more health benefits than walking indoors
D. Fitness walking is better than weight lifting

28. Sara lives in Adama. Her younger cousin Martha lives in a small town in the eastern part of Ethiopia. Martha has visited Sara several times during the past five years. In the same period of time, Sara has visited Martha only once.
A. Martha wants to move to another town.
B. Sara is older than Martha.
D. Sara thinks small towns are boring.
D. Martha likes Sara better than Sara likes Martha.

29. Solomon introduced Sara as the mother of his sister's brother. How is Sara related to Solomon?
A. Sister B. Mother C. Niece D. Aunt

30. Mrs. Genet wishes to renovate her cottage. She hires the services of a plumber, a carpenter, a painter, an electrician, and an interior decorator. The renovation is to be completed in a period of one weekie . Monday to Friday. Every worker will be taking one complete day to do this job, Mrs. Genet will allow just one person to work per day.

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- The painter can do this work only after the plumber and the carpenter have completed their job.
- The interior decorator has to complete his job before that of the electrician.
- The carpenter can not work on Monday or Tuesday.

In case the painter works on Thursday, which among the following alternative is possible ?

A. The plumber and the painter work on consecutive days.
B. The interior decorator dose his work after the painter.
C. The electrician works on Friday.
D. The electrician works on Tuesday.

Analogy (31-35)

Direction: To answer the items ***31-35***, read each of the following words carefully and choose the word that correctly completes the analogy.

31. EXERCISE is to GYM as EATING is to:_____
A. Restaurant B. Drink C. Dieting D. Food

32. The EVENING is to MORNING as DINNER is to:_____
A. Brunch B.Lunch C. Breakfast D. Supper

33. GUN is to BULLET as a BOW is to:_____
A. Sky B. Fly C. Wood D. Arrow

34. PEN is to POET as NEEDLE is to _____
A. Tailor B. Sewing C. Button D. Thread

35. PLAY is to ACTOR as CONCERT is to:_____
A. Persuasion B. Piano C. Musician D. Symphony

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EDUCATIONAL ASSESSMENT AND EXAMINATION SERVICE (EAES)
ETHIOPIAN SECONDARY SCHOOL LEAVING CERTIFICATE EXAMINATION (ESSLCE) SCHOLASTIC APTITUDE TEST
Hamlie, 2015 E.C/ July, 2023 G..C

Section One: Verbal Reasoning

Direction: This section contains **35** verbal reasoning items. It is subdivided into analogy, synonyms, antonym, classification, reading comprehension, analytical and logical reasoning and sentence completion. Each subsection has its own specific direction.

Antonyms

Direction: For Questions **1-3**, choose the letter of the word or phrase which is an opposite in meaning to the words in bold type.

1. **Devaluation**
   A. Conversion
   B. Appreciation
   C. Reduction
   D. Speculation
2. **Prospect**
   A. Retrospect
   B. Traditional
   C. Ideal
   D. Modernist
3. **Distraction**
   A. Concentration
   B. Recovery
   C. Interference
   D. Wandering

Synonyms

Direction: For questions 4-7, choose the letter of the word or phrase that is nearly similar in meaning with the words in bold type.

4. **Abundance** of water resources characterize Ethiopia that it is well styled as the water tower of Africa.
   A. Drainage
   B. Plenitude
   C. confluence
   D. Irrigation
5. Some African countries suffer from **regular** drought and famine.
   A. Frequent
   B. constant
   C. Unpredictable
   D. Perpetual
6. The **invention** of the atomic bomb was regrettable for Albert Einstein himself
   A. Exploration
   B. Discovery
   C. Extinction
   D. Explosion
7. The Committee does not know why the company has **revoked** the contract of employment.
   A. Endorsed
   B. Cancelled
   C. Revised
   D. Approved

Analogy

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Direction: For questions **8-12**, read the pair of words in **block letters** which are followed by four alternatives **A-D**. Choose the letter which best expresses a relationship similar to, that of the original and **blacken** the letter of your choice on the separate answer sheet provided.

8. RAINBOW: SIPECTRA; Constellation: _____
   A. Oasis
   B. Star
   C. Blue
   D. Glacier

9. EYES: GOGGLES; Foreheads: _____
   A. Belt
   B. Helmet
   C. Gloves
   D. Necklace

10. WILDFIRE: DEFORESTATION; Drought: _____
   A. Thunderbolt
   B. Flooding
   C. Landslide
   D. Desertification

11. KING :CROWN; Archpriest: _____
   A. Screeches
   B. Headgear
   C. Whistles
   D. Horn

12. AMBITIOUSNESS: BEHAVIORAL; Elegance_____
   A. Modesty
   B. Physical
   C. Faith
   D. Wisdom

Reading Comprehension

Direction: Read the passages carefully and answer questions **13 -22**. Each question is based on the information stated or implied in the passage. Then **blacken** the letter of your choice on the separate answer sheet provided.

Reading Passage I

Over the last fifty decades, the Western media had associated Ethiopia with images of war, disease and mass starvation due to man-made and natural disasters. And yet, Ethiopia's history and culture are far greater than the sum total of such tragic events. Some of these enduring legacies are its literary traditions, religious, antiquity and political sovereignty.

Firstly, unlike many other African countries, Ethiopia has not only its own alphabet but it also has a rich literary tradition including translations from ancient Greek, Arabic and Coptic sources into **indigenous** languages. Secondly, the Ethiopian multi-cultural identity is intertwined with a uniquely African Christianity that traces its ancestry directly to pre-colonial traditions, rather than arriving via Europe. Thirdly, the country's exceptional status must underscore its political sovereignty. Thus, in Africa's colonial history, Ethiopia was much less a victim than an astute player in the imperial game, able to wring territorial concessions out of European powers like fascist Italy.

13. According to the passage, two of the legacies which characterize Ethiopia are

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A. Tourism and manufacturing industries
B. Import and export trade
C. Literary and sovereignty
D. Mechanization and industrialization

14. What was the attitude of Western media towards Ethiopia's traditions?
A. Destructive
B. Objective
C. Constructive
D. Mysterious

15. In which of the following countries were fewer victims observed in African colonial history?
A. Great Britain
B. Imperial Egypt
C. Ethiopia
D. Italy

16. The word 'indigenous' could be used interchangeably with_____.
A. Foreign
B. Local
C. Exotic
D. Borrowed

17. What does the 'imperial game' in the 8th line of paragraph 2 refer to?
A. Decolonization of Africa
B. European colonial ambition
C. Liberation movement
D. Diplomacy

Reading Passage II

1. Two **contemporary** concerns that require attention in any development initiative are said to be social and human development. Social and human development necessarily requires a unified approach of integrating the economic and social components in planning, policies and programmes for people's betterment. The challenge appears to be one of simultaneously integrating cross-sectoral and regional developmental needs as well as encouraging participatory or bottom-up approaches. The issues of environment, empowerment, habitat, hunger and unemployment have become the burning issues of the day which require public and institutional attention along with resource allocations.

2. **We** need to ensure that development initiatives are geared towards meeting the needs of the present generation without compromising on the expectations of future generations. Fulfilment of basic needs of mankind and improving the living standard of the people should be the primary objective of development. Achievements which fulfill such requirements via the expansion of infrastructure and mega projects for the good of society would definitely transform the economic sector which in turn increases the per capita income of the people..

18. According to the writer, development initiatives should prioritize _____and_____
A. Poverty and prosperity

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B. Social and human development
C. War and peace
D. Migration and resource allocation.

19. The word 'Contemporary' in the first line denotes
A. Controversial
B. Foreseeable future
C. Glorious past
D. Concurrent time

20. Which one of the following strategies is **NOT** a democratic aspect of development?
A. Integrating social and economic parameters
B. Empowerment of the people
C. Top-down approach
D. Respect for cultural diversity

21. To which of the following does 'such' as used in paragraph 2 line 5 refer?
A. development initiatives
B. expectations of future generation
C. expansion of infrastructure
D. fulfillment of mankind's basic needs

22. What does the word '**We**' in paragraph 2 refer to?
A. The student population
B. Future generations
C. Development planners and policy-makers
D. The people

Analytical and Logical Reasoning Questions

Direction: The following statements are followed by two conclusions numbered I and II. Read each statement carefully and decide the conclusion that can be drawn from the premises. Then, **blacken** the letter of your choice on the separate answer sheet provided.

23. **Statement:** Abraham Lincoln..who preserved the Union with his vigorous executive action during the national emergency is a legendary American hero. The abolition of slavery and the national unity of America are a tribute to his leadership.

Which one of the following conclusions could be drawn from this premise?
A. The determination of leaders salvages a country from balkanization.
B. The example of Abraham Lincoln is not applicable to Africa.
C. Slavery was abolished before the civil war
D. American slaves were liberated before the civil war

24. **Statement:** Ethiopia is the third populous country in Africa next to Nigeria and Egypt in that order.

What is the implication of this statement in terms of human wealth and economic growth?

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A. The population of Ethiopia is as large as that of Egypt
B. The population of Nigeria is as large as that of Egypt
C. The population of Nigeria is twice as large as that of Egypt and Ethiopia
D. The population of Ethiopia is far less than that of Nigeria and Egypt.

25. **Statement:** The company has been repeatedly challenged by its staff who are demanding pay rise in the face of the soaring cost of living. The Labor Union is threatening; to call off a strike against the company in the near future.

Scenario (I) The company would sack leaders of the Labor Union from their work

Scenario (II): The Company would make a reasonable salary increment to resolve the conflict

Scenario (III): It all depends on the contractual agreement of the signatories

Scenario (IV): Most of the workers may cancel their agreement Which one of the following scenarios is sound enough?

A. Only scenario II
B. Only scenario III
C. Scenario II and III are more likely
D. Only scenario IV would

26. The gap between natural resources and population growth is one the pressing problems of Afro-Asian countries. They are consequently under socio-economic pressure.

Conclusions: **(I)** These countries should adopt population check mechanisms

(II) These countries should mobilize their natural resources.

(III) These countries should encourage cross-border migration

(IV) These countries should adhere to traditional farming

Which one of the conclusions would logically follow from the above observation?

A. Both III and IV would follow
B. Only I would follow
C. Both I and II are feasible
D. Conclusions I, II and IV would follow

27. **Statement:** The war between Russia and Ukraine which were once the integral parts of USSR has claimed millions of resources and thousands of lives

Conclusion I: Balkanization of countries leads to socio-political and economic crisis.

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Conclusion II: The interference of Western countries in the internal affairs of others should be tolerated.
Conclusion III: The East-West rivalry is no longer threat to world peace
The implication of this destructive war for other countries is stated under

A. Conclusion I
B. Conclusion II
C. Both conclusions I & II
D. Not give

28. Paradoxically, some candidates with great distinction in the entrance examinations can hardly survive the freshman year.
   A. All leading students graduate with great distinction
   B. Average learners can hardly cope with the demands of higher education
   C. Such students are not necessarily intelligent but are opportunists.
   D. All high-achievers in the entrance exam are guaranteed graduation

29. The evaluation of student performance is nothing more than enforcing academic justice.
   A. The evaluation of student performance is a personal practice
   B. Student evaluation is a matter of subjective judgment
   C. Non-academic requirements should not interfere in the evaluation of student performance
   D. No justice could be done to student evaluation

30. Diplomats enjoy a special privilege known as diplomatic immunity. By implication
   A. All men are born equal
   B. International law is not discriminatory
   C. This privilege is unjustifiable
   D. Diplomats are not ordinary individuals

Sentence Completion

Direction: Question **31-35** is incomplete. Choose the most appropriate word that best completes each sentence and blacken the letter of your choice on the separate answer sheet provided.

31. The investors who import commodities from abroad _____ their national currency into hard currency.
   A. save
   B. convert
   C. hoard
   D. deposit

32. The _____ of power causes considerable constraints on the operation of manufacturing industries.
   A. supply
   B. disruption
   C. discharge
   D. erosion

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33. The great game parks in Africa have been responsible for _____ preserving animals from extinction.

A. pandemic
B. endemic
C. wild
D.domestic

34. Only some of Magellan's crew members were privileged to gaze at the_____ view of the Pacific Ocean.

A. telescopic
B. panoramic
C. provincial
D. aerial

35. The_____ of modernization and national unity are the guiding principles of all countries.

A. agenda
B. ideals
C. memories
D. slogans

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ENGLISH SCHOLASTIC APTITUDE TEST (SAT) ANSWERS AND BRIEF EXPLANATION TO THE 2014/2015 EXAMINATION

BOOKLET CODE :075

Classification

1. C. Elephant is a grass eating animal where as the Others are flesh eaters.
2. D. Eraser, pen and pencil are writing materials except 'book' – materials prepared to be read
3. ***Has no answer,*** because order and legal on one side confusion and chase on the other side are the same
4. C. Ear, Tangue and Eyes are sense organs, it is heart different from the others.
5. C. Reporter,Publisher and writer 679repare the material to be read by the reader
6. B. In this context the word **disrupted** is used to mean interrupted
7. D. ' Dissipated` means waste time or money foolishly.
8. A. ' Dissent` is expiration of disagreement
9. D. ' loot`- privet property taken away unlawfully and by force.
10. C. `The miracle of conforting Danger`can be the title best saits the passage.
11. D. Debilitating means make a person weak and the answer is Enfeeble`- make weak,faint or with out energy.
12. D. Facing and over coming danger of the auther brought him new experience.
13. D. He may flee panic, or fight back, stand still or act like an inferior animal.
14. C. `Gather resolution from danger `means find solution from problem.
15. C. Paragraph 3 shows daylilies grow best in soil that is slightly acidic,
16. B. Paragraph 2 states individual daylily flower lasts only one day
17. ***There is no answer*** given in the choice . But in paragraph 3 it says daylilies

    shoudlbe no deeper than one inch below the soil's surface.
18. A. in paragraph 3 it has been stated that daylilies prefer direct sun light
19. D. paragraph1- Daylilies are tolerant of drought.
20. A. Exaggerated means make something seen larger which is the opposite of underestimate form too low estimation.
21. B. Pacify means quieten end violence whose opposite is antiagonize means to make some body angery with you.
22. C. Deplorable is one filled with sorrow or regrate but commendable is deserving praise and approval

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23. C. the situation has been changed since the train schedule changed and it has become less comfortable
24. D. Asking for apology and filling the tank with petrol is the best example of restitution or compensation.
25. B. Michael-front of the house
Rosa – back of the house
Jemal –window frames on the north side
Shawul – on the south side
Switch] Michael window fromes on the north side
Jemal – front of the house, and
Switch] Shawul- front of the house(answer)
Jemal – southern side of the windows.
26. C. Hawi 2 months older than Geremew.
Geremew 3 months younger than Nardos.
Kemal 1 month older than Geremew
So Nardos is the oldest to get the extra piece of Pizza.
27. B. Fitness walking is an effective and Convenient form of exercise which you can do by your self with out going to gym and fitness equipment
28. B. ` the expression` her younger cousin Martha` shows Marttha is younger than Sara.
29. B. Sara is Solomon's mother, Sara is Solomon's Sister brother's mother. This means Solomon has a sister and a brother, and sara is their mother.
30. C. the electrician works on Friday

Mondy	Tuesday	Wednesday	Thursday	Friday
↓	↓	↓	↓	↓
Interior decorator	plumber	Carpenter	painter	electrician

Analogy

31. A. Gym- short form for Symnasium is where exercise is made like restaurant
is where some one gets food to eat,
32. C. Dinner is eaten in in the evening as breakfast is eaten in the morning.
33. D.` Arrow` is shout from abow as `bullet` is fired from a gun
34. A. Needle is used by a tailor like pen in is used by a poet.
35. C. An actor plays a drama on the stage as musician shows concert.

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Antonyms:-words with opposite meaning

1. A. Devolution means Change into a new, lower fixed Price -is the opposite of appreciation means rise in Value eg. of business sham or goods.
2. A. prospect means look forward to, something expected or hoped for is the opposite of retrospect means looking back at past events.
3. A. Distraction means draw away somebody's attention from-is the opposite of Concentration means focus One's attention on.

Synonyms:-words with similar meaning

4. B. Abundance- Great, plenty is nearly similar to plenitude , fulness or much.
5. A Regular - Coming or happening again and again is nearly Similar to frequent-be often happen.
6. B. Invention- Creating or designing something is nearly similar to discover- find out
7. B. Revoked-canceled is nearly similar to cancel - Something will not be done, will not take place.

Analogy

8. B. Rainbow. - group of Colors of Spectrum found in the Sky as Constellation is group of stars in the sky.
9. B. Goggles - large round spectacles with hoods to Protect eyes from wind, dust, water etc as helmet protective of forehead or head Covering worn by fire men, mines etc.
10. D. Wild fire-Spread of five easily and damage the forest Causes deforestation as drought is Caused by desertification .
11. B. Crown is worn on head of a king as headgear is worn on
12. B. Elegance - One who has grace is physical as ambitiousness is behavioral.

Reading Comprehension

13. C. According to the passage in paragraph 1 last Sentence, two of the legacies which characterize Ethiopia are literary tradition and sovereignty.
14. A. Western media created image or destruction due 14 to man made and natural disasters.
15. C. In paragraph 2 Stated as Ethiopia was much less Victim of African colony.
16. B. The word 'indigenous' means native, belonging naturally i.e. local
17. B. In paragraph 2 'imperial games' refer to European Colonial ambition.

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18. B.. Development initiatives give attention to social and human development.
19. D. The word 'Contemporary' refers to existing time means concurrent time.
20. C. From the given Strategies the one which is not a democratic aspect of development is top down approach but instead it is bottom-up approach.
21. D. In paragraph 2 'such' refers to fulfillment of mankind's basic needs.
22. A. The pronoun 'we' in paragraph 2 refers to development Planners and policy-makers

Analytical and Logical Reasoning Questions

23. D. The American slaves were librated before the Civil war Answer:- D because of Abrham Lincoln's determination.
24. D. The idea given in the premise indicates the population of Ethiopia is for less than that of Nigeria and Egypt.
25. A. Scenario II ie The company would make a reasonable salary increment to resolve the Conflict Sounds enough.
26. C. Afro- Asian Countries should adopt population Check mechanisms and mobilize their natural resources to be free of pressing economic Problems.
27. A. The destructive war between Russia and Ukraine which has claimed millions of resources and thousands of lives has implication to other Countries that balkanization of Countries leads to Socio Political and economic crises.
28. C. The Conclusion given under 'c' Can be reached based on the given statement.
29. C. The Statement in the question leads to what has been given in Choice 'c'.
30. D. Diplomats enjoy a special privilege known as diplomatic immunity means they are not ordinary individual
31. B. Convert-means change from one Currency to the others.
32. A. Supply- means give or provide wanted need
33. B . wild animals which are only found in that country
34. B. Panoramic-means uninterrupted view, Constantly Changing.
35. A. Agenda-things to be done.

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BOOKLET CODE: 075

SECTION TWO: Quantitative Reasoning (Questions 1-25)

DIRECTIONS: Solve each of the problems given in questions 1-25 below and select the best answer from the given alternatives and blacken the letter of your choice on the separate answer sheet.

1. A student earns 13 Birr per hour working for the computer center's help desk and 9.50 birr per hour working at the copy center. Because of her course load, she limits her work to 25 hours per week. She wants to earn at least 275 Birr each week. Which of the following system of linear inequalities represent this information?

A. $\begin{cases} 13x + 9.5y > 275 \\ x + y \leq 25 \\ x \geq 0,\ y \geq 0 \end{cases}$

B. $\begin{cases} 13x + 9.5y < 275 \\ x + y \leq 25 \\ x \geq 0,\ y \geq 0 \end{cases}$

C. $\begin{cases} 13x + 9.5y \geq 275 \\ x + y \leq 25 \\ x \geq 0,\ y \geq 0 \end{cases}$

D. $\begin{cases} 13x + 9.5y \leq 275 \\ x + y \geq 25 \\ x \geq 0,\ y \geq 0 \end{cases}$

2. Consider the following system of linear inequalities:

$$x - y \leq 3$$
$$3x - 3y \geq d$$

If the solution set of the system is empty, which of the following numbers could be the value of d?

A. 15 B. 8 C. 7 D. 5

3. A Pediatrician uses the model $h = 3.3a + 28.6$ to estimate the height h of a boy, in inches, in terms of the boy's age a, in years, between the age of 2 and 5. Based on the model, what is the estimated increase, in inches, of the boy's height each year?

A. 4.3 B. 3.3 C. 28.6 D. 14.3

4. The red, green and black lollies in a packet are in the ratio 2:3:5. How many green and black lollies are there, if there are 10 red lollies?

A. 30 green lollies and 50 black lollies

B. 25 green lollies and 15 black lollies

C. 15 green lollies and 25 black lollies

D. 6 green lollies and 15 black lollies

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5. At a restaurant the bill came to 82 Birr. If you leave 119.72 Birr, what percent tip is that?

A. 46

C. 38

B. 31.5

D. 64

6. Suppose a population of a town is given by the model

$$p(t) = 17250\left(\frac{1}{2}\right)^{\frac{t}{10}}$$

, where t denotes the number of years since 2000. Which of the following statements is true?

A. The population is growing by 50% every ten years.

B. The population is halved every ten years.

C. The population doubles every ten years.

D. The population in 2000 was 8625.

7. Meseret begins a saving program in which she will save Birr 1000 the first year, and each subsequent year will save Birr 200 more than she did the previous year. How much will she save during the eighth year?

A. Birr 1360

C. Birr 2400

B. Birr 2600

D. Birr 2200

8. A ball bounces one-half the distance from which it falls. How high does it bounce up on the fifth rebound if the ball is dropped from a height of 20 m?

A. 1.25 m

C. 2.5 m

B. 0.625 m

D. 5 m

9. A contest offers a total of 18 prizes. The first prize is worth Birr 10,000, and each consecutive prize is worth Birr 500 less than the next higher prize. What is the total value of the prizes?

A. Birr 105,000

C. Birr 113,500

B. Birr 108,000

D. Birr 103,500

10. Suppose a chain letter is started by sending a letter to five people and the letter requires each person who receives the letter to send it on to five additional people. Assume that no person receives more than one letter. How many letters are sent all together if the chain reaches the eighth stage?

A. 390,625

C. 480,280

B. 490,625

D. 488,280

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B. 83.71

D. 78.25

16. The total amount of bonus paid by the firm during the whole period is approximately what percent of the total amount of salary paid during this period?

A. 0.58

C. 0.60

B. 0.62

D. 0.72

17. What is the approximate ratio between the total expenditure on interest on loan for all the years and the total expenditure on fuel and transport for all the years?

A. 10:39

C. 10:29

B. 10:36

D. 10:26

18. Two six sided dice are thrown together. What is the probability that the sum of the two numbers on top is 10?

A. 1/12 B. 5/6 C. 1/2 D. 1/6

19. Nine identical balls numbered 1 through 9 are placed in a bag. A ball is randomly drawn and replaced. Then, the second ball is randomly drawn. What is the probability that the first ball drawn and the second ball drawn are both even numbered?

A. 25/81 B. 20/81 C. 16/81 D. 8/81

20. Mesfin bought a laptop valued Birr 7700. One year later the value of the laptop is decreased by 2/7. What is the new value of the laptop?

A. 9900 Birr

C. 4400 Birr

B. 5500 Birr

D. 2200 Birr

21. Hamza spent 22 minutes on his English homework and 25.08 minutes on his Mathematics homework. What percent more times Hamza spent on his Mathematics homework?

A. 28 B. 87 C. 12 D. 14

22. Let $x \neq -2, 3, -\frac{4}{7}$. Then $(\frac{2}{x+2} + \frac{5}{x-3}) \div (\frac{7x+4}{2x-6})$ equals:

A. $\frac{7x+4}{x+2}$

C. $\frac{7x-1}{(x+2)(7x+4)}$

B. $\frac{3}{x-3}$

D. $\frac{2}{x+2}$

23. When the polynomial $5x^4 - 2x^3 + x^2 - x + 1$ is divided by $x^2 - 1$, the remainder is:

A. $\frac{x^2-1}{-3x+7}$

C. $-3x + 7$

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B. $5x^2 - 2x. + 6$

D. $\frac{-3x+7}{x^2-1}$

24. Given that x and y are negative numbers.

Quantity A	Quantity B
$x + y$	$x - y$

Which of the following is true about the relationship between the quantiles A and B?

A. Quantity B is greater than Quantity A.

B. Quantity A is greater than Quantity B.

C. The relationship cannot be determined.

D. The two quantities are equal.

25. Given that $2^{1.5894} = 3$.

Quantity A	Quantity A
$2^{4.327}$	11

How are the quantiles A and B related?

A. Quantity B is greater than Quantity A.

B. Quantity A is greater than Quantity B.

C. The relationship cannot be determined.

D. The two quantities are equal.

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SECTION TWO: Quantitative Reasoning (Questions 36-60)

DIRECTIONS: Solve each of the problems given in questions 36-60 below and select the best answer from the given alternatives and blacken the letter of your choice on the separate answer sheet.

36. Suppose $\{a_n\}_{n=1}^{\infty}$ is an arithmetic sequence and $a_5 + a_8 = 41$ and $a_8 - a_5 = 9$. Which one of the following is the sequence?

A. 4, 7, 10, 13, 16, ...
B. 2, 5, 8, 11, 14, ...
C. 3, 6, 9, 12, 15, ...
D. 4, 7, 10, 13, ...

37. Which one of the following is the 100^{th} term of a geometric sequence $\{G_n\}_{n=1}^{\infty}$, where $G_n \div G_{n-1} = -1$ and $G_3 = 5$?

A. -5
B. 1
C. -1
D. 5

38. If the n^{th} term of the arithmetic sequence $\{a_n\}_{n=1}^{\infty}$ is given by $a_n = 3n - 2$, then which one of the following is the sum of the first seven terms of the sequence?

A. 19
B. 70
C. 28
D. 72

39. Which one of the following is the sum of the first six terms of a geometric sequence $\{3(\frac{1}{2})^{n-1}\}_{n=1}^{\infty}$?

A. 189/32
B. 45/8
C. 381/64
D. 93/16

40. Consider the system of linear equations $x - 2y = 3$ and $-3x + 2y = 1$. Then, which one of the following is true about the system?

A. The system has infinitely many solutions.
B. The system has no solution.

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C. The system has unique solution.
D. The system has two different solutions.

41. Suppose you want to select a 3-persons committee from a group of 7 teachers. How many ways are there to do this?
A. 35
C. 120
B. 210
D. 60

42. A box contains 12 pieces of paper, each of which has the name of the 12 students in a class consisting of 7 male and 5 female students, all with different names. What is the probability of a randomly selected piece of paper has a female student name?
A. 5/7
C. 7/5
B. 5/12
D. 7/12

43. According to surveys in an area, 25% of the trees planted did not grow in the year 2011 E.C. and 35 % of the trees planted did not grow in the year 2014 E.C.. What was the percentage increase in the fraction of trees planted that did not grow from the year 2011 E.C. to the year 2014 E.C.?
A. 40 %
C. 30 %
B. 60 %
D. 50 %

44. A lake was 30 m deep. Two months later, its depth decreased by 20%. If the depth of the lake increased from where it was after two months by 15% in three months' time, what was the cumulative percentage change in the depth of the lake during the five months?
A. 5%
C. 10%
B. 6%
D. 8%

45. Which one of the following is the solution set of the system of linear equations given by $x + y = 1$ and $3x - y = 7$?
A. {(2, -1)}
C. {(-2, 1)}
B. {(-1, 2)}
D. {(2, 1)}

46. Which one of the following ordered pair of numbers is **not** in the solution of the system of linear inequalities by $x + y \geq 1$ and $x - 2y \leq 2$?
A. (2, 1)
C. (2, -1)
B. (1, 2)
D. (-1, 2)

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47. Given the system of linear inequalities $\begin{cases} ax + by \geq c \\ ax + by \leq c \end{cases}$ for arbitrary constants a, b and c in $\mathbb{R}$ with $c \neq 0$, then which one of the following is true about the solution set of the system?

A. $\{(x, y): ax + by \leq c\}$

B. $\{(x, y): ax + by \geq c\}$

C. $\{(x, y): ax + by \neq c\}$

D. $\{(x, y): ax + by = c\}$

48. A woman spent one fifth of her salary for clothing, two fifth of her salary for different bills and 100 Birr for transport. After covering all these costs she is left with 4,500 Birr. What was the salary she received?

A. 12,500 Birr

B. 11,500 Birr

C. 10,500 Birr

D. 13,500 Birr

49. During the sunshine, an upright building 16 meters tall is casting a shadow of 12 meters long. At the same time, a nearby upright building is casting a shadow of 18 meters long. If the length of the shadows are proportional to the height of the buildings, what is the height, in meters, of the taller building?

A. 17

B. 32

C. 24

D. 36

50. Which one of the following is true for every natural number n?

A. $6^n - 1 \leq 6^{n-1}$

B. $(n+1)^3 \geq n^3 + 1$

C. $n^2 \geq n!$

D. $(n+1)^2 \leq n^2 - 2n + 1$

Directions:- Questions 51 - 55 are based on the data given in the table below. Answer each of the questions based on the given data.

Student enrollment at in a college
Distribution of enrollment by class and gender

Total enrolment: 4,800

Class	Male	Female	Total
Freshman (1st Year)	1,000	800	1,800
Sophomore (2nd Year)	750	450	1,200
Junior (3rd Year)	620	380	1,000
Senior (4th Year)	480	320	800
Total	2,850	1,950	4,800

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Percent of total enrollment majoring in selected academic areas

Area	Percent
Social Science	30 %
Natural Science	50 %
Other Science	20 %

51. What is the ratio of the number of male freshman social science students to the number of female sophomore natural science students?

A. 4 to 3

C. 3 to 4

B. 9 to 20

D. 20 to 9

52. If the total enrollment is 20 percent greater than it was five years ago, what was the total enrollment five years ago?

A. 4,600

C. 4,000

B. 3,600

D. 3,000

53. How many students are either junior or male or both?

A. 2,850

C. 3,230

B. 1,000

D. 620

54. In which enrollment do the female students have highest percentage?

A. Junior

C. Senior

B. Freshman

D. Sophomore

55. What is the number of male junior social students?

A. 620

C. 500

B. 310

D. 186

56. If your total income is 12,000 Birr per month and you can't spend more than 60% on housing, what is the maximum amount of your total income that can be spent on housing?

A. 5,600 Birr

C. 7,200 Birr

B. 9,000 Birr

D. 6,000 Birr

57. Suppose that the perimeter of a rectangular farm is 80 meter. If the farm is y meter long and x meter wide, then which one of the following functions of x will model the total area of the farm in square meter?

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A. $g(x) = x^2 - 40x$

B. $f(x) = -x^2 + 40x$

C. $k(x) = 2x^2 + 80x$

D. $h(x) = -2x^2 + 80x$

58. Which one of the following is the coefficient of x^3y in the expansion of the algebraic expression $(x + y)^4$?

A. 6

B. 2

C. 3

D. 4

59. Which one of the following is the degree of the polynomial function $f(x) = (2x + 7)^5(3x - 1)^2$?

A. 5

B. 6

C. 2

D. 7

60. Which one of the following is necessarily true about any real numbers x and y?

A. $(x + y)^2 \leq (x - y)^2$

B. $(x - y)^2 \leq (x + y)^2$

C. $|x + y| = |-x - y|$

D. $|x + y| \geq |x - y|$

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SECTION TWO: Quantitative Reasoning (Questions 36-60)

DIRECTIONS: Solve each of the problems given in questions 1-25 below and select the best answer from the given alternatives and blacken the letter of your choice on the separate answer sheet.

1. For x hours by 13 Birr/hr and for y hrs by 9.50 birr/hr; and
She wants to earn at least 275 Birr each week: $13x + 9.5y \geq 275$
She limits her work to 25 hours per week: $x + y \leq 25$
Number of hours is non-negative: $x \geq 0, y \geq 0$

Answer: C

2. The given system $\begin{matrix} x - y \leq 3 \\ 3x - 3y \geq d \end{matrix}$ can equivalently be put as:

$$x - y \leq 3$$
$$x - y \geq \frac{d}{3}$$

Since $x - y = 3$ and $x - y = \frac{d}{3}$ are parallel lines the above system has no solution if $\frac{d}{3} > 3$, i.e., $d > 9$.

Answer: A

3. The given model $h = 3.3a + 28.6$ is linear with slope $m = 3.3$. The slope is the estimated increase, in inches, of the boy's height each year.

Answer: B

4. If r, g and b are number of red, green and black lollies respectively and x is the common factor, and then given number of red, green and black lollies in the ratio 2:3:5; and $r = 10$ red lollies

$$10 = r = 2x \Longrightarrow x = 5$$

Thus, $g = 3x = 15$ and $b = 5x = 25$.

Answer: C

5. If a bill came to 82 Birr and if you leave 119.72 Birr, then the percent of the tip is:

$$\frac{\text{Total Paid} - \text{Bill amount}}{\text{Bill amount}} \times 100\% = \frac{119.72 - 82}{82} \times 100\% = 46\%$$

Answer: A

6. If the rate of change of the population of the model

$$p(t) = 17250\left(\frac{1}{2}\right)^{\frac{t}{10}}$$

, in every ten years is r then

$$1 - r = \frac{1}{2} \Longrightarrow r = \frac{1}{2}$$

Thus, the population is halved every ten years.

Answer: B

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7. Saving Birr 1000 on first year, and saving each subsequent year Birr 200 more than the previous year is arithmetic with $a_1 = 1000$ and common difference $d = 200$ so that the save during the eighth year is: $a_8 = a_1 + 7d$ = Birr (1000+7(200)) = Birr 2400.

Answer: C

8. If a ball dropped from a height of h, bounces one-half the previous distance from which it falls each time, then after the n^{th} rebound it rises the height:

$$h_n = r^n h$$

so that for

$$r = \frac{1}{2}, \text{ and } h = 20 \text{ m}$$

, after the fifth rebound (i.e., $n = 5$) it rises

$$h_5 = (\frac{1}{2})^5(20 \text{ m}) = 0.625 \text{ m}$$

Answer: B

9. The first prize is worth Birr 10,000, and each consecutive prize is worth Birr 500 less than the next higher prize so that this is arithmetic with $a_1 = 10{,}000$ and d = Birr -500.

Thus, the total value of the 18 prizes is S_{18} so that:

$$S_n = \frac{n}{2}(2a_1 + (n-1)d)$$

$$\Rightarrow S_{18} = \text{Birr } \frac{18}{2}(2(10{,}000)+(18-1)(-500)) = \text{Birr } 103{,}000$$

Answer: D

10. First stage sent to $G_1 = 5$ people, then each of these five people sends to five people so that on the second stage it reaches for $G_2 = 25$ people and so on gives $G_3 = 125$, $G_4 = 625$...which is geometric with common ratio $r = 5$ so that by

$$S_n = \frac{G_1(1-r^n)}{1-r}$$

, at the eighth stage it reaches for

$$\Rightarrow S_8 = \frac{5(1-5^8)}{1-5} = 488{,}280 \text{people}$$

Answer: D

11. The first line passing through (x, y) and (-1, 2) has a slope of 3 so that it has an equation: $y = 3x + 5$, and

The second line passing through (x, y) and (2, -1) has a slope of 1 so that it has an equation: $y = x - 3$

Hence, $y = 3x + 5 = x - 3 \Rightarrow x = -4$ and $y = -7$

Answer: A

12. Let x and y be the amounts invested at a profit rate of 9.5% and 8% respectively. Then

Total investment Birr 20,000 $\Rightarrow x + y = 20{,}000$.....(1)

Total yearly interest of Birr 1690 $\Rightarrow 9.5\%x + 8\%y = 1690$

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$\Rightarrow 9.5x + 8y = 169{,}000$ .....(2)

Solving (1) and (2) simultaneously:

$$\begin{cases} x + y = 20{,}000 \\ 9.5x + 8y = 169{,}000 \end{cases} \Rightarrow x = 6000 \text{ and } y = 14{,}000$$

Therefore, he should invest Birr 6000 at 9.5%; Birr 14,000 at 8%.

Answer: D

Questions 13 - 17 are based on the table below, which gives the expenditure of a firm (in millions of Birr) per annum over the given years.

Year	Items of Expenditure				
	Salary	Fuel and Transport	Bonuses	Interest on Loan	Taxes
2006	314	92	3	24	85
2007	258	110	2.5	33	71
2008	732	131	3.4	28	112
2009	450	104	2.6	31	98
2010	370	125	3.8	42	93

First build a table containing each year's total expenditure and total budget for each item of expenditure.

Year	Items of Expenditure					Total Expenditure in a year
	Salary	Fuel and Transport	Bonuses	Interest on Loan	Taxes	
2006	314	92	3	24	85	**518**
2007	258	110	2.5	33	71	**474.5**
2008	732	131	3.4	28	112	**1006.4**
2009	450	104	2.6	31	98	**685.6**
2010	370	125	3.8	42	93	**633.8**
Total	**2124**	**562**	**15.3**	**158**	**459**	

13. The total expenditure of the firm in the year 2007 is 474.5 million.

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Answer: C

14. The average amount of tax per year which the firm had to pay during the whole period is

$$\frac{459}{5} = 91.8$$

Answer: B

15. Total expenditure on all these items in year 2006 was 518 million, and items in year 2010 was 633.8 million

Thus, percent of 518 of that of 633.8 is:

$$\frac{518}{633.8} \times 100\% = 81.73\%$$

Answer: A

16. For the whole period, total amount of bonus paid is 15.3 million and that of salary is 2124 million.

Thus, percent of 15.3 of that of 2124 is:

$$\frac{15.3}{2124} \times 100\% = 0.72\%$$

Answer: D

17. Total expenditure for all the years for interest on loan is 158 million and that of fuel and transport is 562 million. Thus, for all the years, the ratio between the total expenditure on interest on loan and the total expenditure on fuel and transport is: 158:562

Dividing by 15.8, the approximate ratio is 10:36

Answer: B

18. In throwing two six sided dies, the sample space S is with

$$n(S) = 6 \times 6 = 36$$

The event E of sum of the two numbers on top is 10 is

$$E = \{(4, 6), (5, 5), (6, 4)\} \text{ with } n(E) = 3$$

Thus, the probability that the sum of the two numbers on top is 10 is

$$p(E) = \frac{n(E)}{n(S)} = \frac{3}{36} = \frac{1}{12}$$

Answer: A

19. Since the sample space S has $n(S) = 9$ and the event of getting even E= $\{2, 4, 6, 8\}$ is with $n(E) = 4$; we have $p(E) = \frac{4}{9}$ for the first draw which is also a probability of the second draw. Thus, since replacement is done the events are independent events so that the probability that the first ball drawn and the second ball drawn are both even numbered is:

$$p(2 \text{ evens}) = p(E) \times p(E) = \frac{4}{9} \times \frac{4}{9} = \frac{16}{81}$$

Answer: C

20. If Birr 7700 is reduced by 2/7, then the new value is:

$$\text{Birr } 7700 - \frac{2}{7}(\text{Birr } 7700) = \text{Birr } 7700\left(1 - \frac{2}{7}\right)$$

$$= \text{Birr } \frac{5}{7} \times 7700 = \text{Birr } 5500$$

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Answer: B

21. The percent of 22 that 25.08 is more than 22 is:

$$\frac{25.08-22}{22} \times 100\% = 14\%$$

Answer: D

22. For $x \neq -2, 3, \frac{4}{7}$:

$$\left(\frac{2}{x+2}+\frac{5}{x-3}\right)\div\left(\frac{7x+4}{2x-6}\right) = \left(\frac{2}{x+2}+\frac{5}{x-3}\right)\times\left(\frac{2x-6}{7x+4}\right)$$

$$= \frac{7x+4}{(x+2)(x-3)} \times \frac{2(x-3)}{7x+4} = \frac{2}{x+2}$$

Answer: D

23. The remainder is a ***polynomial*** with ***lesser degree*** than that of the divisor so that the only possible answer is $-3x + 7$.

Answer: C

24. Given that x and y are negative numbers.

Quantity A	Quantity B
$x + y$	$x - y$

Then, $x = x \Rightarrow x - y > x + y$ since $y < 0$.

Therefore, Quantity B is greater than Quantity A.

Answer: A

25. Given that $2^{1.5894} = 3$, and

Quantity A	Quantity A
$2^{4.327}$	11

Then, $2^{1.5894} = 3 \Rightarrow 2^{1.5894} \times 4 = 3 \times 4$

$$\Rightarrow 2^{1.5894} \times 2^2 = 12$$

$$\Rightarrow 2^{1.5894+2} = 12$$

$$\Rightarrow 2^{3.5894} = 12$$

Thus, $11 < 12 = 2^{3.5894} < 2^{4.327} \Rightarrow 11 < 2^{4.327}$

Therefore, Quantity A is greater than Quantity

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36. Since a_5 is listed in each choice with different values solve for it.

Subtract down to the system: $\begin{cases} a_5 + a_8 = 41 \\ \underline{a_8 - a_5 = 9} \end{cases}$

$\Rightarrow 2a_5 = 32 \Rightarrow a_5 = 16.$

Thus, A is the possible answer.

Answer: A

37. A geometric sequence $\{G_n\}_{n=1}^{\infty}$, where $G_n \div G_{n-1} = -1$ means the common ratio is -1, and it is alternating sequence. And 3 is odd and given $G_3 = 5$ implies $G_n = 5$ if n is odd and $G_n = -5$ if n is even.

Thus, $G_{100} = -5$.

Answer: A

38. Use the formula $S_n = \frac{n}{2}(a_1 + a_n)$ for $n = 7$. Thus, $a_n = 3n - 2$ gives

$$S_7 = \frac{7}{2}(a_1 + a_7) = \frac{7}{2}([3(1) - 2] + [3(7) - 2]) = 70.$$

Answer: B

39. You better list the terms and add using properties of Σ rather than using the formula $S_n = \frac{G_1(1-r^n)}{1-r}$ for $r \neq 1$.

$$S_6 = \sum_{n=1}^{6} 3\left(\frac{1}{2}\right)^{n-1} = 3\left(\left(\frac{1}{2}\right)^{1-1} + \left(\frac{1}{2}\right)^{2-1} + \ldots + \left(\frac{1}{2}\right)^{6-1}\right)$$

$$= 3\left(1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32}\right) = 3\left(\frac{63}{32}\right) = \frac{189}{32}$$

Answer: A

40. Let a system S be given by $\begin{cases} ax + by = c \\ dx + ey = f \end{cases}$. Then,

1) If $ae - bd \neq 0$, then S has unique solution, and
2) Let $ae - bd = 0$. Then,
   (i) S has infinitely many solutions if $ce - bf = 0$ and $af - cd = 0$,
   (ii) S has no solution if $ce - bf \neq 0$ and $af - cd \neq 0$.

Hence, for $\begin{cases} x - 2y = 3 \\ -3x + 2y = 1 \end{cases}$; $ae - bd = 1(2) - (-2)(-3) = -4 \neq 0$ so that the system has unique solution.

Answer: C

41. Selection is combination. So the number of ways to select a 3-persons committee from a group of 7 teachers is:

$$C(7, 3) = \frac{7!}{3!(7-3)!} = 35.$$

Answer: A

42. Out of 7 male and 5 female, the probability of a randomly selected is a female is given by:

$$P(F) = \frac{n(F)}{n(M)+n(F)} = \frac{5}{7+5} = \frac{5}{12}.$$

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Answer: B

43. According to surveys in an area, 25% of the trees planted did not grow in the year 2011 E.C. and 35 % of the trees planted did not grow in the year 2014 E.C.. Then, the percentage increase in the fraction of trees planted that did not grow from the year 2011 E.C. to the year 2014 E.C. is: $\frac{p(2014\text{ E.C.})-p(2011\text{ E.C.})}{p(2011\text{ E.C.})} \times 100\% = \frac{35\%-25\%}{25\%} \times 100\% = 40\ \%.$

Answer: A

44. A lake was $d_o = 30$ m deep. Two months later, its depth decreased by 20% so that its depth became

$$d_1 = 30\text{m} - 20\% \times 30\text{m} = 24\text{m}.$$

And then, if the depth of the lake increased from where it was after two months by 15% in three months' time, then its depth became

$$d_2 = d_1 + 15\% d_1 = 24\text{m} + 15\% \times 24\text{m} = 27.6\text{m}.$$

Thus, the cumulative percentage change in the depth of the lake during the five months is:

$$\frac{d_2-d_o}{d_o} \times 100\% = \frac{27.6\text{m}-30\text{m}}{30\text{m}} \times 100\% = -8\ \%.$$

The negative sign shows that, the depth has decreased.

Answer: D

45. Given a system of linear equations $\begin{cases} x + y = 1 \\ 3x - y = 7 \end{cases}$

$$\Rightarrow 4x = 8 \Rightarrow x = 2 \text{ so that } x + y = 1 \Rightarrow y = -1$$

Therefore, the solution set is $\{(2, -1)\}$.

Answer: A

46. All ordered pairs (2, 1), (1, 2), (2, -1), and (-1, 2) satisfy the inequality $x + y \geq 1$ but only (2, 1), (1, 2), and (-1, 2) satisfy the inequality $x - 2y \leq 2$. Thus, the ordered pair (2, -1) is not in the solution of the given system of linear inequalities.

Answer: C

47. The system of linear inequalities $\begin{cases} ax + by \geq c \\ ax + by \leq c \end{cases}$ for arbitrary constants a, b and c in $\mathbb{R}$ with $c \neq 0$ the intersection of the relations $ax + by \geq c$ and $ax + by \leq c$ is the line $ax + by = c$; and the answer is D if and only if a and b cannot be 0 at a time.

Answer: Faulty Item

48. Given information, a woman spent one fifth of her salary for clothing, two fifth of her salary for different bills and 100 Birr for transport. After covering all these costs she is left with 4,500 Birr. Let her total salary be x in Birr. Then,

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Amount for clothing is $\frac{x}{5}$, amount for different bills $\frac{2x}{5}$, 100 Birr for transport, and left with 4,500 Birr so that:

$$\frac{x}{5} + \frac{2x}{5} + 100 \text{ Birr} + 4{,}500 \text{ Birr} = x$$

$$\Rightarrow \frac{2}{5}x = 4{,}600 \text{ Birr} \Rightarrow x = 11{,}500 \text{ Birr}$$

Therefore, her salary is 11,500 Birr.

Answer: B

49. Given information: During the sunshine, an upright building $h_1 = 16$ meters tall is casting a shadow of $h_{1s} = 12$ meters long. At the same time, a nearby upright building of height h_2 is casting a shadow of $h_{2s} = 18$ meters long.

If the length of the shadows are proportional to the height of the buildings, then

$$\frac{h_2}{h_1} = \frac{h_{2s}}{h_{1s}} \Rightarrow \frac{h_2}{16\text{ m}} = \frac{18\text{ m}}{12\text{ m}} \Rightarrow h_2 = 24\text{ m}$$

Therefore, the height of the taller building is 24 m.

Answer: C

50. Since $n \in \mathbb{N}$, i.e., $n > 0$ and $(n+1)^3 = n^3 + 3n^2 + 3n + 1$

$n^3 + 3n^2 + 3n + 1 \geq n^3 + 1$ because $3n^2 + 3n > 0$

Therefore, B is true.

But, $6^n - 1 \leq 625$, and $n^2 \geq n!$ are false for $n \geq 4$ so that A and C are False. Since $n \in \mathbb{N}$ and $n + 1 \geq n - 1$ we have that

$$(n+1)^2 \geq (n-1)^2 = n^2 - 2n + 1$$

, which implies that D is False.

Answer: B

Directions: Questions **51 – 55** are based on the data given in the table below. Answer each of the questions based on the given data.

Student enrollment at in a college
Distribution of enrollment by class and gender

Total enrolment: 4,800

Class	Male	Female	Total
Freshman (1st Year)	1,000	800	1,800
Sophomore (2nd Year)	750	450	1,200
Junior (3rd Year)	620	380	1,000
Senior (4th Year)	480	320	800
Total	2,850	1,950	4,800

Percent of total enrollment majoring in selected academic areas

Area	Percent

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Social Science	30 %
Natural Science	50 %
Other Science	20 %

51. The ratio of the number of male freshman social science students to the number of female sophomore natural science students is $\frac{30\%(1000)}{50\%(450)} = \frac{4}{3}$, i.e., 4 to 3

Answer: A

52. If the total enrollment is 20 percent greater than it was five years ago, and the total enrollment five years ago is x, then
$x + 20\%x = 4{,}800 \Rightarrow 1.2x = 4{,}800 \Rightarrow x = 4{,}000.$
Therefore, the total enrollment five years ago is 4,000

Answer: C

53. There are 1,000 junior students, 2,850 − 620 = 2,230 non-junior male students so that there are 1,000 + 2,230 = 3,230 junior or male or both students.

Answer: C

54. Table below shows percentage of females in each year.

Class	Female	Total	Percentage
Freshman (1st Year)	800	1800	$\frac{800}{1800} \times 100\% \approx 44.4\%$
Sophomore (2nd Year)	450	1,200	$\frac{450}{1200} \times 100\% = 37.5\%$
Junior (3rd Year)	380	1,000	$\frac{380}{1000} \times 100\% = 38\%$
Senior (4th Year)	320	800	$\frac{320}{800} \times 100\% = 40\%$

As shown in the first cell of the fourth column female students have highest percentage in freshman enrollment.

Answer: B

55. The number of male junior social students is $30\%(620) = 186$.

Answer: D

56. If your total income is 12,000 Birr per month and you can't spend more than 60%
, i.e., $\leq 60\%(12{,}000 \text{ Birr}) \Rightarrow \leq 7{,}200$ Birr
, on housing, then the maximum amount of your total income that can be spent on housing is 7,200 Birr.

Answer: C

57. Suppose that the perimeter of a rectangular farm is 80 meter. If the farm is y meter long and x meter wide, then $2x + 2y = 80$ so that $y = 40 - x$. And the area is $A(x, y) = xy$ so that using the relation $y = 40 - x$, the area is given as:

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$$A(x) = x(40-x) = -x^2 + 40x.$$

Answer: B

58. In the expansion of $(x+y)^n$ the coefficient of $x^{n-i}y^i$ is $C(n,i)$.

Thus, for $n = 4$ and $i = 1$:

$$C(4,1) = \frac{4!}{1!(4-1)!} = 4.$$

Answer: D

59. For the polynomial $f(x) = (2x+7)^5(3x-1)^2$ the leading term is $(2x)^5(3x)^2 = 288x^7$ so that degree of f is 7.

Answer: D

60. By property and definition of absolute value $|z| = |-z|$ for every real number z so that for $z = x + y$ for any real numbers x and y

$$|z| = |-z| \Rightarrow |x+y| = |-(x+y)| = |-x-y|.$$

Therefore, C is necessarily true.

But, $x = 1, y = 1 \Rightarrow (1+1)^2 \leq (1-1)^2 \Rightarrow 4 \leq 0 \Rightarrow$ A is false,

$x = -1, y = 1 \Rightarrow (-1-1)^2 \leq (-1+1)^2 \Rightarrow 4 \leq 0 \Rightarrow$ B is false.

$x = 1, y = -1 \Rightarrow |1+(-1)| \geq |1-(-1)| \Rightarrow 0 \geq 2 \Rightarrow$ D is false.

Answer: C

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